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SCIENCE MEDICINES HEALTH

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Committee for Medicinal Products for Human Use (CHMP)

Assessment report

Tecentriq

International non-proprietary name: atezolizumab

Procedure No. EMEA/H/C/004143/II/0019

Note

Variation assessment report as adopted by the CHMP with all information of a commercially confidential nature deleted.



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List of abbreviations

ADA	anti-drug antibody, also called anti-therapeutic antibody
ADR	adverse drug reaction
AE	adverse event
AESI	adverse event of special interest
Atezo	Atezolizumab
CCOD	clinical cut-off date
CD47	Cluster of Differentiation 47
CE	Carboplatin + Etoposide
CI	confidence interval
CMC	chemistry, manufacturing, and control
Cmin	minimum serum or plasma concentration
CNS	central nervous system
CR	complete response
CSR	Clinical Study Report
DOR	Duration of response
ECOG PS	Eastern Cooperative Oncology Group Performance status
eCRF	electronic Case Report Form
EMA	European Medicines Agency
EORTC QLQ-C30	European Organization for the Research and Treatment of Cancer Quality of Life Questionnaire Core 30
EORTC QLQ-LC13	European Organization for the Research and Treatment of Cancer Quality of Life Questionnaire Lung Cancer Module
ES-SCLC	extensive-stage small cell lung cancer
EU	European Union
FNA	fine needle aspiration
HR	hazard ratio
HRQoL	Health-Related Quality of Life
iDMC	Independent Data Monitoring Committee
IgG1	immunoglobulin G1
IHC	immunohistochemistry
IL-10	Interleukin 10
ITT	intent-to-treat

IV	intravenous
IxRS	Interactive Voice/Web Response System
KM	Kaplan-Meier
MAA	Marketing Authorisation Application
mAb	monoclonal antibody
NSCLC	non-small cell lung cancer
ORR	objective response rate
OS	overall survival
PBO	placebo
PCI	prophylactic cranial irradiation
PD	progressive disease
PD-1	programmed death-1
PD-L1	programmed death-ligand 1
PFS	progression-free survival
PK	Pharmacokinetic
PopPK	population pharmacokinetics
PR	partial response
PRO	patient-reported outcome
q3w	every three weeks
RECIST	Response Evaluation Criteria for Solid Tumors
RMP	Risk Management Plan
SAE	serious adverse event
SAP	Statistical Analysis Plan
SBP	Summary of Biopharmaceutics
SCE	Summary of Clinical Efficacy
SCP	Summary of Clinical Pharmacology
SCS	Summary of Clinical Safety
SmPC	Summary of Product Characteristics
TGF- β	transforming growth factor beta
Ttd	time to deterioration
UC	urothelial carcinoma

1. Background information on the procedure

1.1. Type II variation

Pursuant to Article 16 of Commission Regulation (EC) No 1234/2008, Roche Registration GmbH submitted to the European Medicines Agency on 15 October 2018 an application for a variation.

The following variation was requested:

Variation requested		Type	Annexes affected
C.I.6.a	C.I.6.a - Change(s) to therapeutic indication(s) - Addition of a new therapeutic indication or modification of an approved one	Type II	I and IIIB

Extension of Indication to include Tecentriq, in combination with nab-paclitaxel and carboplatin, indicated for the first-line treatment of adult patients with metastatic non-squamous non-small cell lung cancer (NSCLC) who do not have EGFR mutant or ALK-positive NSCLC; as a consequence, sections 4.1, 4.2, 4.8 and 5.1 of the SmPC are updated. The Package Leaflet is updated in accordance. RMP version 9.0 has been submitted.

The requested variation proposed amendments to the Summary of Product Characteristics and Package Leaflet and to the Risk Management Plan (RMP).

Information on paediatric requirements

Pursuant to Article 8 of Regulation (EC) No 1901/2006, the application included an EMA Decision P/0220/2015 on the agreement of a paediatric investigation plan (PIP).

At the time of submission of the application, the PIP P/0220/2015 was not yet completed as some measures were deferred.

Information relating to orphan market exclusivity

Similarity

Pursuant to Article 8 of Regulation (EC) No. 141/2000 and Article 3 of Commission Regulation (EC) No 847/2000, the applicant did not submit a critical report addressing the possible similarity with authorised orphan medicinal products because there is no authorised orphan medicinal product for a condition related to the proposed indication.

Scientific advice

The applicant did not seek Scientific Advice at the CHMP.

1.2. Steps taken for the assessment of the product

The Rapporteur and Co-Rapporteur appointed by the CHMP were:

Rapporteur: Sinan B. Sarac Co-Rapporteur: Jan Mueller-Berghaus

Timetable	Actual dates
Submission date	15 October 2018
Start of procedure:	3 November 2018
CHMP Co-Rapporteur Assessment Report	20 December 2018
CHMP Rapporteur Assessment Report	17 December 2018
PRAC Rapporteur Assessment Report	4 January 2019
PRAC members comments	9 January 2019
Updated PRAC Rapporteur Assessment Report	10 January 2019
PRAC Outcome	17 January 2019
CHMP members comments	21 January 2019
Updated CHMP Rapporteur(s) (Joint) Assessment Report	24 January 2019
Request for supplementary information (RSI)	31 January 2019
CHMP Rapporteur Assessment Report	6 May 2019
CHMP members comments	20 May 2019
Updated CHMP Rapporteur Assessment Report	23 May 2019
Request for supplementary information (RSI)	29 May 2019
Restart of procedure	26 June 2019
CHMP Rapporteur Assessment Report	9 July 2019
CHMP members comments	15 July 2019
Updated CHMP Rapporteur Assessment Report	18 July 2019
Opinion	25 July 2019

2. Scientific discussion

2.1. Problem statement

2.1.1. Disease or condition

The claimed indication is for Tecentriq in combination with nab paclitaxel and carboplatin, for the first line treatment of adult patients with metastatic non-squamous non-small cell lung cancer (NSCLC) who do not have EGFR mutant or ALK positive NSCLC.

2.1.2. Epidemiology

Lung cancer remains the leading cause of cancer death worldwide, estimated to be responsible for nearly one in five cancer deaths globally (1.76 million deaths, 18.4% of the total; Globocan 2018). There were estimated to be 2.1 million new cases in 2018 (11.6% of the total), 58% of which occurred in less developed regions. This disease is the most common cancer in men worldwide (1.4 million, 15% of the total cancers in men) and accounts for the highest absolute number of cancer deaths globally (1.2 million deaths, 24% of cancer deaths in men). Non-small cell lung cancer (NSCLC) is the predominant subtype, accounting for approximately 85% of all cases. NSCLC can be divided into two major histologic types: non-squamous and squamous cell carcinoma. Non-squamous histology accounts for more than half of all

NSCLC, whereas squamous histology accounts for approximately 30% (Brambilla et al, 2014 and Schrump DS et al. NSCLC; Principles and Practice of Oncology. 9th Edition. 2011).

2.1.3. Clinical presentation, diagnosis and stage/prognosis

More than half of the patients are diagnosed at an advanced stage of disease, which directly contributes to poor survival, as expressed by an untreated median OS of 4 months and a metastatic 5-year survival rate of <5% (Lindsey A. et al, 2016). Poor prognostic factors for survival in patients with NSCLC include advanced stage of disease at the time of initial diagnosis, poor performance status (PS), and a history of unintentional weight loss. More than half of the patients with NSCLC are diagnosed with distant metastatic disease, which directly contributes to poor survival prospects.

2.1.4. Management

For patients who have metastatic NSCLC without an oncogenic driver mutation (such as EGFR mutations or ALK translocation), the standard of care is platinum-based chemotherapy, partnered with agents such as taxanes (paclitaxel, docetaxel), vinorelbine, gemcitabine, and pemetrexed, with or without bevacizumab.

Over the past 4 years, immune checkpoint inhibitors, such as PD-1/PD-L1 blocking antibodies, have emerged as effective alternatives to chemotherapy for many tumour types. In 2L+ NSCLC, PD-1 inhibitors (nivolumab, pembrolizumab) and PD-L1 inhibitors (atezolizumab) demonstrated superiority over docetaxel as monotherapy. Subsequently, it was also shown that pembrolizumab monotherapy is effective in the first line setting for patients who express high levels of PD-L1. Pembrolizumab is currently approved for 1L treatment of NSCLC patients whose tumours express high levels of PD-L1 ($\geq 50\%$ tumour proportion score). Atezolizumab has also been authorised for the treatment of 1L metastatic non-squamous NSCLC when used in combination with bevacizumab, paclitaxel and carboplatin. In the authorised indications, patients with EGFR or ALK positive tumour mutations should also have received targeted therapy before treatment with PD-1/PD-L1 inhibitors in combination with chemotherapy.

About the product

Atezolizumab is an Fc-engineered humanized immunoglobulin (IgG1) monoclonal antibody (MAb) targeting the programmed death-ligand 1 (PD-L1). Binding of atezolizumab to PD-L1 inhibits the interaction of the PD-1 and B7.1 receptors. Both of these interactions are reported to provide inhibitory signals to T cells.

Tecentriq is currently authorised in the following indications:

- as monotherapy for the treatment of adult patients with locally advanced or metastatic urothelial carcinoma (UC):
 - after prior platinum-containing chemotherapy, or
 - who are considered cisplatin ineligible, and whose tumours have a PD-L1 expression $\geq 5\%$.
- in combination with bevacizumab, paclitaxel and carboplatin, for the first-line treatment of adult patients with metastatic non-squamous non-small cell lung cancer (NSCLC). In patients with EGFR mutant or ALK-positive NSCLC, Tecentriq, in combination with bevacizumab, paclitaxel and carboplatin, is indicated only after failure of appropriate targeted therapies.
- as monotherapy for the treatment of adult patients with locally advanced or metastatic NSCLC after prior chemotherapy. Patients with EGFR mutant or ALK-positive NSCLC should also have received targeted therapies before receiving Tecentriq.

The current application is supported by the results from the primary analysis of IMpower130 (Study GO29537; n=723), a Phase III trial designed to evaluate the safety and efficacy of atezolizumab-carboplatin-nab-paclitaxel (Atezo-CnP) versus treatment with carboplatin-nab-paclitaxel (CnP) in chemotherapy-naïve patients with Stage IV non-squamous NSCLC.

The following new indication is recommended:

Tecentriq, in combination with nab paclitaxel and carboplatin, is indicated for the first line treatment of adult patients with metastatic non-squamous NSCLC who do not have EGFR mutant or ALK positive NSCLC.

During the induction phase, the recommended dose of Tecentriq is 1,200 mg administered by intravenous infusion, followed by nab-paclitaxel and carboplatin every three weeks for four or six cycles. For each 21-day cycle, Tecentriq, nab-paclitaxel, and carboplatin are administered on day 1. In addition, nab-paclitaxel is administered on days 8 and 15.

The induction phase is followed by a maintenance phase without chemotherapy in which 1,200 mg Tecentriq is administered by intravenous infusion every three weeks.

For first line NSCLC, in patients given Tecentriq in combination with carboplatin and nab-paclitaxel, it is recommended that patients are treated with Tecentriq until disease progression or unmanageable toxicity. Treatment beyond disease progression may be considered at the discretion of the physician (see section 5.1).

2.2. Non-clinical aspects

No new non-clinical data have been submitted in this application, which was considered acceptable by the CHMP.

2.2.1. Ecotoxicity/environmental risk assessment

Atezolizumab is an IgG1 monoclonal antibody produced by recombinant DNA technology, a protein with a molecular mass of ~150 kDa. As an unaltered protein, being extensively degraded in the patient's body by regular proteolytic mechanisms before excretion, atezolizumab is unlikely to result in a significant environmental exposure. Atezolizumab is expected to biodegrade in the environment and does not pose a significant risk to the environment. Thus, according to the "Guideline on the Environmental Risk Assessment of Medicinal Products for Human Use" (EMA/CHMP/SWP/4447/00), atezolizumab is exempt from the submission of Environmental Risk Assessment studies as the product and excipients do not pose a significant risk to the environment.

2.2.2. Discussion and conclusion on non-clinical aspects

No new nonclinical data has been provided to support this application. The applicant did not submit studies for the ERA. According to the relevant guideline, in the case of products containing proteins as active pharmaceutical ingredient(s), this is acceptable.

2.3. Clinical aspects

2.3.1. Introduction

GCP

The Clinical trials were performed in accordance with GCP as claimed by the applicant>

The applicant has provided a statement to the effect that clinical trials conducted outside the community were carried out in accordance with the ethical standards of Directive 2001/20/EC.

- Tabular overview of clinical studies

Table 1: Summary of studies contributing to efficacy evaluation

Study No.	Study Design	Population	No. of Patients Enrolled	Dose, Route, and Regimen	Primary Efficacy Endpoint	Analysis Timing (Data Cutoff Date)
Pivotal Study						
GO29537 (IMpower130)	Phase III, global, open-label, multicenter, randomized study	Patients with 1L chemotherapy-naïve non-squamous NSCLC. Patients were stratified by sex (male vs. female), presence of liver metastases at baseline (yes vs. no), and PD-L1 tumor expression by IHC (TC3 and any IC vs. TC0/1/2 and IC2/3 vs. TC0/1/2 and IC0/1).	Total: n=723 ^a Arm A (Atezo+CnP): n=483 Arm B (CnP): n=240	Atezolizumab 1200 mg IV q3w Carboplatin AUC of 6 mg/mL/min IV q3w Nab-paclitaxel 100 mg/m ² IV qw	OS and investigator-assessed PFS per RECIST v1.1	Primary analysis After 357 OS events in the ITT-WT population across both arms (15 Mar 2018)

1L=first-line; Atezo=atezolizumab; AUC=area under the concentration-time curve; CnP=carboplatin+nab-paclitaxel; IC=tumor-infiltrating immune cell; IHC=immunohistochemistry; ITT=intent-to-treat; IV=intravenous; NSCLC=non-small cell lung cancer; OS=overall survival; PD-L1=programmed death-ligand 1; PFS=progression-free survival; q3w=every 3 weeks; qw=once a week; RECIST=Response Evaluation Criteria in Solid Tumors; TC=tumor cell; WT=wild type.

^a Reflects the ITT population included in this summary. A total of 724 patients were randomized; however, 1 patient was excluded from the ITT population because the death date was prior to the randomization date.

Source: IMpower130 Primary CSR ([Figure 3](#) and [Table 17](#)).

2.3.2. Pharmacokinetics

The application includes key results of the atezolizumab clinical pharmacology program and new data from Study IMpower130 (Study GO29537). The clinical pharmacology properties of atezolizumab has been characterized in previous studies when atezolizumab was administered as monotherapy in patients predominantly with mUC and NSCLC (see Tecentriq EPAR). The new data investigates Atezolizumab when administered in combination with carboplatin and nab-paclitaxel in stage IV non-squamous NSCLC.

Phase III Study GO29537; IMpower130

Pharmacokinetics of atezolizumab, carboplatin and nab-paclitaxel and immunogenicity were investigated in IMpower130, a randomized, Phase III, multicenter, open-label study designed to evaluate the safety and efficacy of atezolizumab in combination with carboplatin + nab-paclitaxel (atezo+CnP) compared with carboplatin + nab-paclitaxel in chemotherapy naïve patients with Stage IV non squamous NSCLC.

Patients were randomized (2:1) to one of the following treatment regimens:

- Arm A: atezolizumab+carboplatin+nab paclitaxel (atezo+CnP) for four or six 21 day cycles during induction; followed by atezolizumab maintenance
- Arm B: carboplatin+nab paclitaxel (CnP) for four or six 21 day cycles during induction; followed by best supportive care or pemetrexed maintenance

Atezolizumab was administered at 1200 mg IV every three weeks (q3w), carboplatin at AUC 6 mg/mL/min IV q3w on Day 1 and nab-paclitaxel (100 mg/m²) IV was administered on Days 1, 8 and 15.

The PK objectives included in study IMpower130 were:

- To evaluate the maximum observed serum concentration (C_{max}) of atezolizumab after infusion on Cycle 1 Day 1 (C1D1) and minimum observed serum concentration

- (C_{min}) of atezolizumab prior to infusion at selected cycles, at treatment discontinuation, and at 120 days (± 30 days) after the last dose of atezolizumab for patients randomized to Atezo+CnP
- To evaluate the plasma concentrations of carboplatin (Atezo+CnP and CnP arms)
- To evaluate the plasma concentrations of nab-paclitaxel reported as total paclitaxel (Atezo+CnP and CnP arms)

The immunogenicity objective was:

- To evaluate the incidence and titers of anti-drug antibody (ADAs) response to atezolizumab and potential correlation with pharmacokinetic (PK), pharmacodynamic, safety, and efficacy parameters

Pharmacokinetic Analysis Population

The PK-evaluable population was defined as all patients who received any dose of atezolizumab, carboplatin, or nab-paclitaxel and who had evaluable PK samples.

Pharmacokinetic Analysis

A total of 465 patients had evaluable atezolizumab PK data. Atezolizumab PK + immunogenicity samples were collected according to table below:

Table 2: Schedule of PK, biomarker, and anti-therapeutic antibody assessments

Study Visit	Time	Arm A	Arm B
Cycle 1, Day 1	Pre-dose (same day as treatment administration)	MPDL3280A ATA MPDL3280A pharmacokinetics TBNK Biomarkers ^a Carboplatin pharmacokinetics Nab-paclitaxel pharmacokinetics Optional RCR blood (DNA extraction) ^b	TBNK Biomarkers ^a Carboplatin pharmacokinetics Nab-paclitaxel pharmacokinetics Optional RCR blood (DNA extraction) ^b
	90 min (± 10 min) after end of MPDL3280A infusion	MPDL3280A pharmacokinetics	
	5–10 min before the end of nab-paclitaxel infusion	Nab-paclitaxel pharmacokinetics	Nab-paclitaxel pharmacokinetics
	1 hr after end of nab-paclitaxel infusion	Nab-paclitaxel pharmacokinetics	Nab-paclitaxel pharmacokinetics
	5–10 min before the end of carboplatin infusion	Carboplatin pharmacokinetics	Carboplatin pharmacokinetics
	1 hr after end of carboplatin infusion	Carboplatin pharmacokinetics	Carboplatin pharmacokinetics
Cycle 2, Day 1 (± 2 days)	Pre-dose (same day as treatment administration)	MPDL3280A ATA MPDL3280A pharmacokinetics TBNK Biomarkers ^c	TBNK Biomarkers ^c
Cycle 3, Day 1 (± 2 days)	Pre-dose (same day as treatment administration)	MPDL3280A ATA MPDL3280A pharmacokinetics TBNK Biomarkers ^c Carboplatin pharmacokinetics Nab-paclitaxel pharmacokinetics	TBNK Biomarkers ^c Carboplatin pharmacokinetics Nab-paclitaxel pharmacokinetics
	30 min (± 10 min) after end of MPDL3280A infusion	MPDL3280A pharmacokinetics	
	5–10 min before the end of nab-paclitaxel infusion	Nab-paclitaxel pharmacokinetics	Nab-paclitaxel pharmacokinetics
	1 hr after end of nab-paclitaxel infusion	Nab-paclitaxel pharmacokinetics	Nab-paclitaxel pharmacokinetics
	5–10 min before the end of carboplatin infusion	Carboplatin pharmacokinetics	Carboplatin pharmacokinetics
	1 hr after end of carboplatin infusion	Carboplatin pharmacokinetics	Carboplatin pharmacokinetics
Cycles 4, 8 and 16, Day 1 (± 2 days)	Pre-dose (same day as treatment administration)	MPDL3280A ATA MPDL3280A pharmacokinetics TBNK Biomarkers ^c	TBNK Biomarkers ^c
After Cycle 16, every eighth cycle, Day 1 (± 2 days)	Pre-dose (same day as treatment administration)	MPDL3280A ATA MPDL3280A pharmacokinetics Biomarkers ^c	Biomarkers ^c
At time of fresh biopsy (on-treatment or at progression, including during follow-up)	At visit	TBNK ^d Biomarkers ^{a,c}	TBNK Biomarkers ^a

Treatment discontinuation visit	At visit	MPDL3280A ATA MPDL3280A pharmacokinetics TBNK Biomarkers ^a	TBTK Biomarkers ^a
120 ± 30 days after last dose of MPDL3280A	At visit	MPDL3280A ATA MPDL3280A pharmacokinetics TBNK Biomarkers ^a	

ATA=anti-therapeutic antibody; TBTK=T, B, and NK cells.

Notes: Serum pharmacokinetic samples for MPDL3280A; plasma pharmacokinetic samples for carboplatin, and nab-paclitaxel.

Twenty patients in each treatment arm will undergo the additional PK assessments for carboplatin and nab-paclitaxel.

^a Plasma, serum, whole blood for biomarkers.

^b The optional RCR blood sample (for DNA extraction) requires an additional informed consent and can be collected at any time during the course of the study.

^c Plasma and serum for biomarkers.

^d TBTK and biomarker samples at the time of biopsy do not have to be collected if the biopsy visit occurs ≤ 3 days before or after another protocol-defined sample collection timepoint and this period does not overlap with a dosing with MPDL3280A or other therapy.

Atezolizumab serum concentration data (Cmin and Cmax) were tabulated and summarized.

A validated ELISA with a lower limit of quantification (LLOQ) of 0.06 µg/mL was used to quantify atezolizumab in human serum. A validated inductively coupled plasma mass spectrometry with a LLOQ of 100 ng/mL and a validated liquid chromatography tandem mass spectrometry with a LLOQ of 2 ng/mL were used to determine carboplatin and paclitaxel concentrations, respectively, in the plasma. The concentrations of atezolizumab, carboplatin and paclitaxel were summarized using descriptive statistics including means, medians, ranges, and SD.

Immunogenicity Analysis

Baseline prevalence and post-baseline incidence of ADA against atezolizumab were summarized. Analyses of pharmacokinetics, efficacy, and safety by atezolizumab ADA status were conducted to explore the potential impact of immunogenicity, as appropriate.

PK results from IMpower130

Atezolizumab pharmacokinetics:

Descriptive statistics of serum concentrations of atezolizumab, Cmax (30 minutes following the end of the atezolizumab infusion) and Cmin (pre-dose) following the administration of Atezolizumab 1200 mg every 3 weeks (q3w) IV in combination with carboplatin and nab-Paclitaxel are summarized in Table 3.

Table 3: Summary Statistics for Atezolizumab C_{max} and C_{min} Following Multiple IV Doses of Atezolizumab 1200 mg, Administered Every 3 Weeks in Combination with Carboplatin and nab-Paclitaxel

Treatment	Visit ¹	Nominal Time from First Dose (day)	N	AM (µg/mL)	AM SD (µg/mL)	GM (µg/mL)	GM %CV	Min (µg/mL)	Median (µg/mL)	Max (µg/mL)
Atezo + CnP	C1D1	0	446	NR	NR	NE	NR	NR	0	0
	C1D1	0.0625	437	392	114	344	136	0.03	377	943
	C1D21	21	416	70.9	35.1	61.7	84.5	0.03	67	380
	C2D21	42	384	111	52.2	97.1	76.5	0.03	106	610
	C3D1	42.04	356	454	170	422	50.4	1.41	441	2340
	C3D21	63	352	134	57.8	122	47.5	23.7	129	460
	C7D21	147	257	218	93.7	199	45.3	24.3	204	596

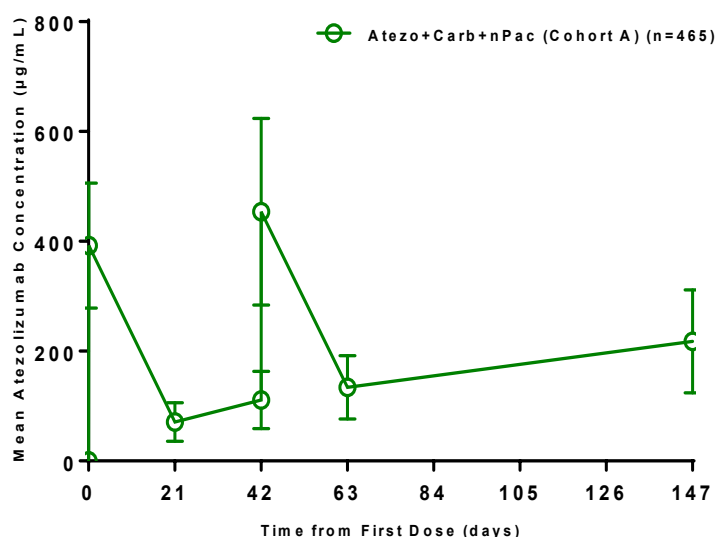
AM=Arithmetic Mean; Atezo + CnP=atezolizumab in combination with carboplatin and nab-paclitaxel; C_{max}=maximum serum concentration; C_{min}= minimum serum concentration; CV=coefficient of variation; GM=Geometric Mean; Max=maximum; Min=minimum; SD=standard deviation; NE=Not Evaluable; NR=Not Reportable.

¹Visit is denoted by Cycle abbreviated by "C" and Day abbreviated by "D". For example, C1D1 corresponds to Cycle 1, Day 1. Predose Cycle 1 is C1D1, 0 days. C_{max} is C1D1 30 minutes post end of infusion. Predose Cycle 2 is C1D21, predose Cycle 3 is C2D21, C_{max} is C3D1 30 minutes post end of infusion etc.

Data source: CSR [IMpower130](#), Table 30

A total of 465 patients (out of 483 total enrolled patients) had evaluable atezolizumab PK. Mean $\pm$ standard deviation (SD) serum atezolizumab concentrations over time are shown in Figure 1.

The steady state based on atezolizumab C_{min} was reached by approximately Cycle 4-8. The geometric mean C_{min} of atezolizumab was 61.7 µg/mL at the end of Cycle1 and 199 µg/mL at the end of Cycle 7, respectively.

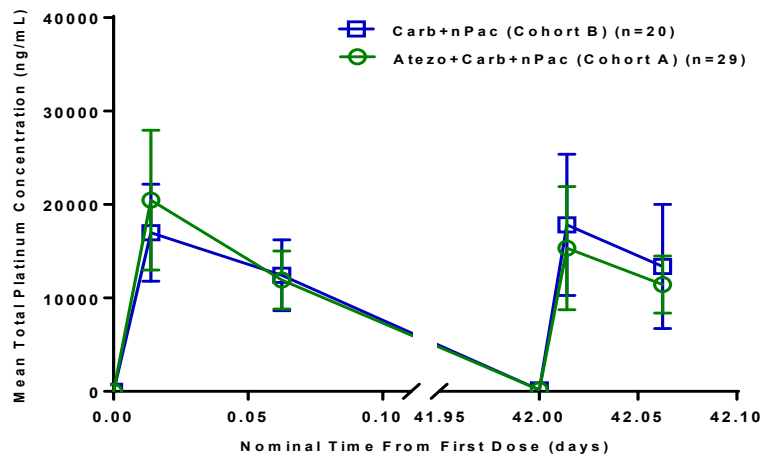


Atezo + Carb + nPac=atezolizumab in combination with carboplatin and nab-Paclitaxel. Data source: CSR [IMpower130](#), Figure 17

Figure 1: Mean ($\pm$ SD) Plot of Atezolizumab Serum Concentrations versus Time Following Multiple IV Doses of Atezolizumab 1200 mg, Administered Every 3 Weeks in Combination with Carboplatin and nab-Paclitaxel

Carboplatin Pharmacokinetics

A total of 49 patients had evaluable carboplatin PK (29 from the Atezo+CnP arm and 20 from the CnP arm). The mean plasma carboplatin concentrations over time are presented in Figure 2.

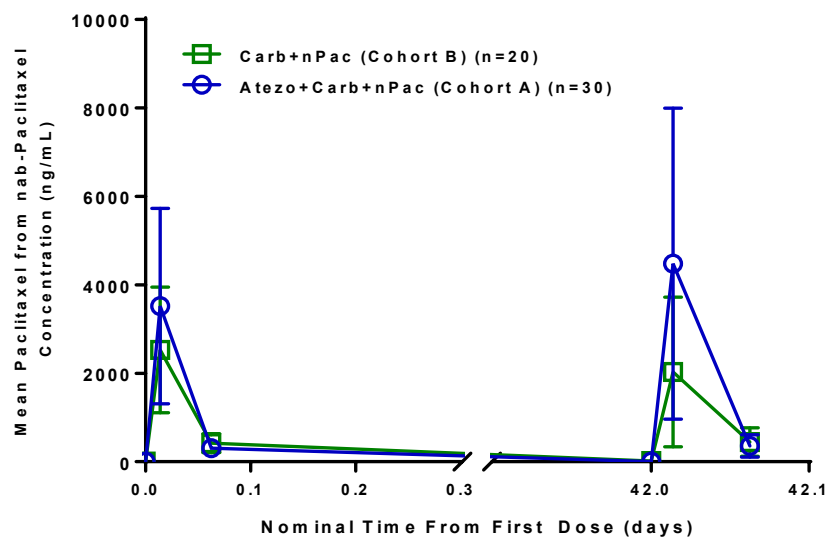


Data source: CSR [IMpower130](#), Figure 18

Figure 2: Mean (±SD) Plot of Carboplatin Concentrations versus Time Following Multiple IV Doses of Carboplatin AUC6 mg/mL/min, Administered Every 3 Weeks in Combination With nab-Paclitaxel, With or Without Atezolizumab (PK-Evaluable Population)

Nab-Paclitaxel Pharmacokinetics

A total of 50 patients had evaluable paclitaxel PK (30 from the Atezo+CnP arm and 20 from the CnP arm). The mean plasma paclitaxel concentrations over time are presented in Figure 3.



Data source: CSR [IMpower130](#), Figure 19

Figure 3: Mean (±SD) Plot of Paclitaxel Concentrations versus Time Following Multiple IV Doses of nab-Paclitaxel 100 mg/m², Administered on Day 1, 8, and 15 of a 3 Week Cycle, in Combination With Carboplatin, With or Without Atezolizumab (PK-Evaluable Population)

Pharmacokinetics by Treatment-Emergent ADA Status

The baseline prevalence of atezolizumab ADA was 3.1% in the Atezo+CnP arm and 4.8% in the CnP arm (cross-over patients). The incidence of post-baseline treatment-emergent ADA was 22.4% for the

Atezo+CnP arm and 23.5% for the CnP arm (crossover patients). These atezolizumab ADA-positive rates were within the ADA-positive rate ranges (16.7%-54.5%) observed in atezolizumab monotherapy studies.

Atezolizumab concentrations up to Cycle 7 by treatment emergent ADA status for Atezo + CnP arm are summarized in Figure 4 and shown in Table 4. The geometric mean C_{min} for Cycle 7 Day 21 was 194 $\mu\text{g/mL}$ (48.2% CV) and 201 $\mu\text{g/mL}$ (44.6% CV) for ADA-positive (n=54) and ADA-negative (n=203) patients, respectively.

Table 4: Summary Statistics for Atezolizumab C_{max} and C_{min} Following Multiple IV Doses of Atezolizumab 1200 mg Given Every 3 Weeks in Combination with Carboplatin and nab-Paclitaxel by Atezolizumab Anti-Drug Antibody Status (PK-Evaluable Population)

ADA Status	Visit ¹	Nominal Time From First Dose (day)	N	AM ($\mu\text{g/mL}$)	AM SD ($\mu\text{g/mL}$)	GM ($\mu\text{g/mL}$)	GM %CV	Min ($\mu\text{g/mL}$)	Median ($\mu\text{g/mL}$)	Max ($\mu\text{g/mL}$)
Negative	C1D1	0	330	0	0	NE	NE	0	0	0
	C1D1	0.0625	326	391	121	330	172	0.03	378	943
	C1D21	21	322	74	36.4	67.2	48.4	3.43	69.6	380
	C2D21	42	297	116	53.1	105	48.9	5.72	110	610
	C3D1	42.04	276	460	174	432	38.8	61.8	448	2340
	C3D21	63	273	141	58.3	130	43.5	34.6	133	460
	C7D21	147	203	218	91.5	201	44.6	24.3	206	596
Positive	C1D1	0	97	0	0	NE	NE	0	0	0
	C1D1	0.0625	92	395	96.8	384	24.5	218	375	641
	C1D21	21	94	60.2	27.9	45.9	195	0.03	57.9	124
	C2D21	42	87	93.4	45	73.5	153	0.03	93.5	212
	C3D1	42.04	80	435	153	389	82.7	1.41	432	997
	C3D21	63	79	112	50.4	99.9	53.9	23.7	105	268
	C7D21	147	54	215	102	194	48.2	73.5	195	508

ADA=anti-drug antibodies, AM=arithmetic mean, SD=standard deviation, CV=coefficient of variation, GM=geometric mean, NE=not evaluable, BLQ=below the limit of quantification

¹Visit is denoted by Cycle abbreviated by "C" and Day abbreviated by "D". For example, C1D1 corresponds to Cycle 1, Day 1. Predose Cycle 1 is C1D1, 0 days. C_{max} is C1D1 30 minutes post end of infusion. Predose Cycle 2 is C1D21, predose Cycle 3 is C2D21, C_{max} is C3D1 30 minutes post end of infusion etc.

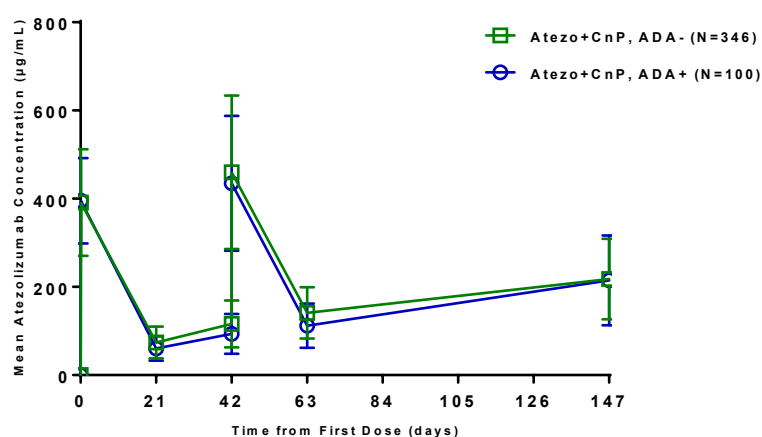


Figure 4: Mean ($\pm$ SD) Plot of Atezolizumab Concentrations versus Time Following Multiple IV Doses of Atezolizumab 1200mg Given Every 3 Weeks in Combination with Carboplatin and nab-Paclitaxel by ADA Status (PK-Evaluable Population)

2.3.3. PK/PD modelling

Base Population Pharmacokinetic Model

The popPK of atezolizumab was first assessed based on Phase I data from two clinical studies (the “Phase I popPK Model”): Study PCD4989g and Study JO28944 (see Tecentriq EPAR).

For the Phase I popPK Model, serum PK of atezolizumab across the two studies PCD4989g and JO28944 (dose range: 1-20 mg/kg q3w, including the fixed 1200 mg dose of atezolizumab q3w) was described by a linear two compartment disposition model with first-order elimination. The estimated typical population CL was 0.200 L/day and typical V1 was 3.28 L for a male patient, 77 kg with 40 g/L of albumin, tumor burden of 63 mm and no ADA. The typical Vss was 6.9 L and terminal t1/2 was 27 days.

Final Population Pharmacokinetic Model

In the final popPK model, body weight, albumin, tumor burden, and ADA were statistically significant covariates for CL; body weight, and albumin were statistically significant covariates for V1, and body weight, albumin and gender were statistically significant covariates for V1, and gender was statistically significant for V2. The results from the popPK analysis suggest the unexplained IIV is moderate for CL (i.e., 29%), V1 (i.e., 18%), and V2 (i.e., 34%).

In treatment-emergent ADA positive patients, CL is estimated to be 16% higher than in ADA negative patients. In females, volume of distribution would be 13% and 27% lower than in males for V1 and V2, respectively. No covariate induced more than 27% change from the typical PK model parameter for extreme values.

The Phase I popPK Model was subjected to an external validation by prediction-corrected visual predictive check (pcVPC) for UC, NSCLC, and ES-SCLC separately, using PK data collected in IMvigor210 and IMvigor211 for UC, data collected in BIRCH, POPLAR, FIR, OAK, IMpower150 and IMpower130 for NSCLC and data collected in IMpower133 for ES-SCLC. The results of pcVPC by study indicated that Phase I popPK Model was adequate to describe atezolizumab PK data in studies mentioned above and suggested a consistent atezolizumab PK in patients with NSCLC, SCLC and UC.

External Validation- IMpower130

The Phase I popPK Model was used to derive the individual PK estimates based on atezolizumab observed concentration-time profiles in IMpower130.

The prediction-corrected visual predictive check suggested that the median, 95th, and 5th percentiles of observed Cmax and Cmin were generally well within the prediction intervals of the Phase I popPK Model. Covariate effects (i.e., BWT, gender, ADA status, baseline albumin levels and tumor burden) in the IMpower130 data were generally consistent with those identified in the popPK Model. Both cycle 1 and steady-state exposure metrics were similar to those estimated in other NSCLC studies (RDR 1067735 and 1074540) for atezolizumab in monotherapy or in combination with carboplatin + paclitaxel ± bevacizumab (IMpower150), which suggested that the atezolizumab PK was not impacted by the co-administration chemotherapy (carboplatin + nab-paclitaxel).

There appeared to be a trend toward under-prediction of atezolizumab exposure data (troughs) after long-term administration (7th, 15th, 23rd+ dose), with observations above the prediction intervals by the Phase I popPK Model. This observation is consistent with the results of a post-hoc analysis reported in the USPI, where atezolizumab clearance was found to decrease over time, with a mean maximal reduction % (coefficient of variation [CV%]) from the baseline CL value of approximately 17.1% (40.6%).

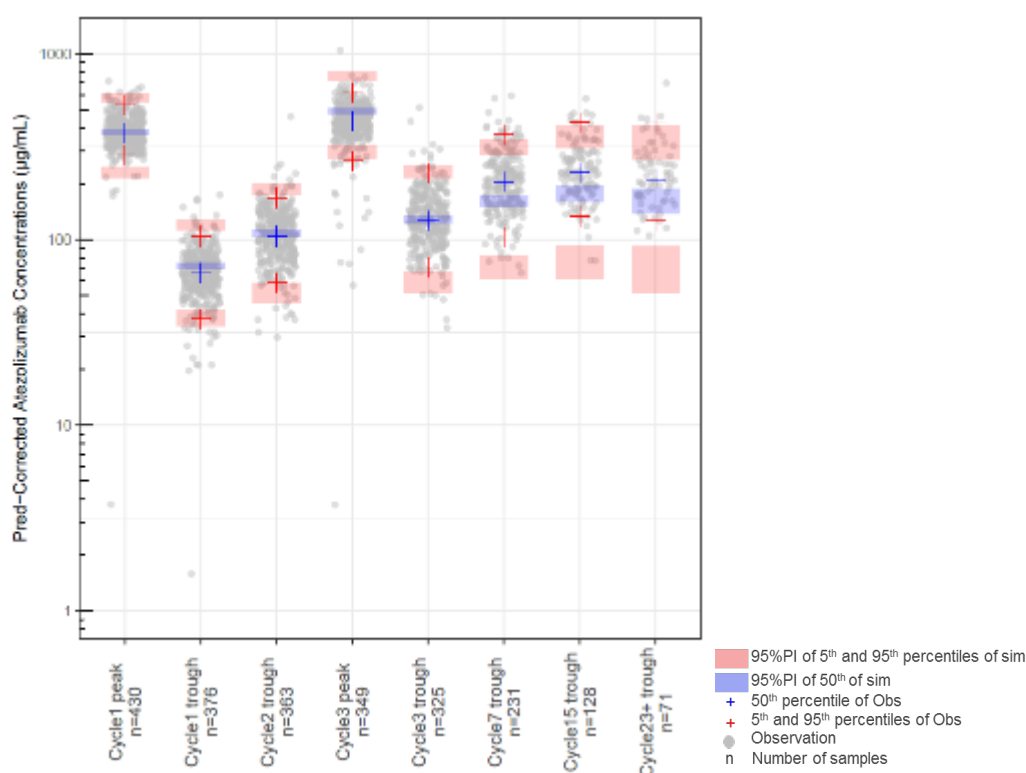
The geometric mean accumulation ratio based on AUC was 1.81-fold. Cmin and Cmax accumulated 2.04- and 1.41-fold (geometric means), respectively.

Table 5: Summary Statistics (Geometric Mean [Geometric Mean CV%]) of Atezolizumab Exposure Metrics at Cycle 1 and Steady-State in Study IMpower130 Predicted using popPK Model

IMpower130 (N=465)			
Cmax (µg/mL)	Cmin (µg/mL)	AUC (µg.day/mL)	t _{1/2} beta (day)
382 [19.2]	72.6 [30.6]	2862 [19.9]	20.2 [5.1]
Cmax _{ss} (µg/mL)	Cmin _{ss} (µg/mL)	AUC _{ss} (µg.day/mL)	Accumulation ratio
537 [22.8]	148 [45.9]	5171 [31.5]	1.81 [15.0]

N=Number of patients; C_{max,ss}=C_{max} at steady-state; C_{min,ss}=C_{min} at steady-state; AUC_{ss}=AUC at steady-state; Accumulation ratio is derived as the ratio between AUC at cycle 1 and AUC_{ss}; CV%=coefficient of variation.

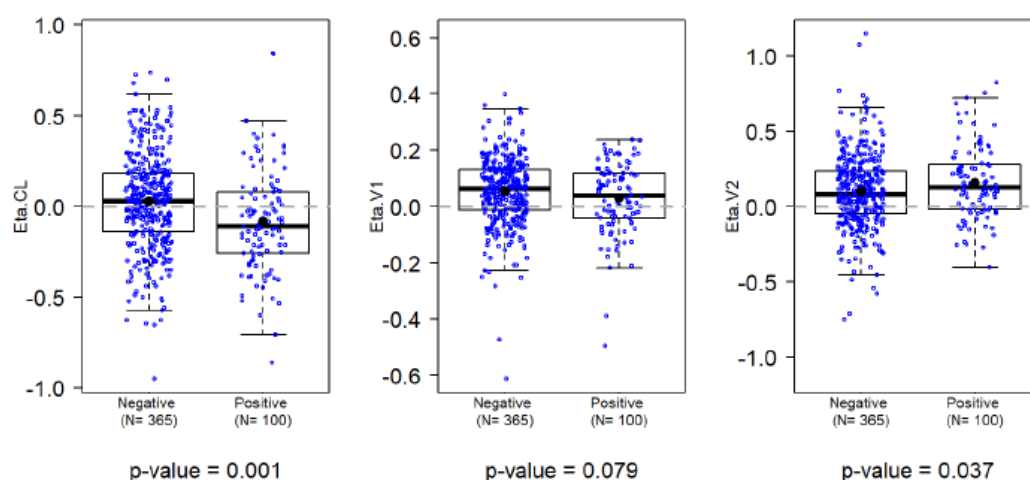
Source: IMpower130 PopPK report Tables 6 and 7.



CI=confidence interval; popPK=population pharmacokinetics; VPC=visual predictive check. Source: [IMpower130 PopPK report](#) 1089066, Figure 2

Figure 5: Pred-Corrected Visual Predictive Check Atezolizumab Concentration

Exploratory analyses of the relationships between individual CL, V1 and V2 patient-level random effects and covariates (continuous, categorical, and including body weight, albumin, tumor burden, gender, and ADA) indicated that covariate effects in IMpower130 data were generally consistent with those identified in the Phase 1 popPK Model. In addition, these exploratory analyses did not reveal any unexpected covariate effect not identified in the Phase 1 popPK Model. Finally, random effects of parameters were not influenced by PD-L1 expression (IC/TC status).



ADA = Anti-drug antibodies; Eta.CL = random effect on clearance; Eta.V1 = random effect on central volume of distribution; Eta.V2 = random effect on peripheral volume of distribution; N = number of patients

Figure 6: Exploratory analyses of the relationship between random effects and ADA on IMpower130 data

Negative eta CL in positive ADA patients compared to negative ADA ($p < 0.001$) indicates less effect in IMpower130 compared to Phase 1 popPK Model (Figure 6).

2.3.4. Pharmacodynamics

Exposure - response (ER) analyses

No new ER analyses have been submitted with this application.

Immunogenicity in IMpower130

Demographic and baseline characteristics were generally balanced between post baseline treatment emergent atezolizumab ADA-positive and ADA negative patients in the Atezo+CnP arm of ADA-evaluable WT population. Some imbalance for Atezo + CnP patients was seen in baseline SLD (sum of longest diameters) between ADA positive and ADA negative patients, with medians of 87 mm and 71 mm respectively.

Further information on the impact of ADA on efficacy and safety is further described in the clinical efficacy and safety sections respectively.

Dose rationale

2.3.5. Discussion on clinical pharmacology

The clinical pharmacology of atezolizumab has been characterized in previously submitted studies with atezolizumab administered as monotherapy. The PK of atezolizumab is linear over a dose range of 1 to 20 mg/kg of atezolizumab, including the fixed 1200 mg dose of atezolizumab used in the IMPower130 study in patients with metastatic NSCLC in combination with carboplatin and nab-paclitaxel.

The pharmacokinetics of atezolizumab, carboplatin and nab-paclitaxel were investigated in IMPower130 and an external validation of the Phase 1 Pop PK model was conducted with the PK data from IMpower130. The PK of atezolizumab in combination with carboplatin and nab-paclitaxel was similar the PK of atezolizumab when given as monotherapy and there was no evidence of drug-drug interactions. Covariates were investigated in the Pop PK model and found to be comparable to previous findings. Significant covariates were body weight, gender, ADA status, albumin level and tumour burden.

The recommended dose of atezolizumab is 1200 mg q3w via intravenous infusion. IMpower130 evaluated atezolizumab at this dose level and schedule in combination with carboplatin and nab paclitaxel in chemotherapy-naïve patients with Stage IV non-squamous NSCLC.

The original rationale for the recommended dose was based on data from nonclinical studies and clinical data from Phase I Study GO27831 (PCD4989g). Atezolizumab 1200 mg q3w as a single agent showed clinical efficacy in NSCLC and mUC and patients receiving atezolizumab 1200 mg q3w had a safety profile that was well tolerated. In IMpower130, the PK of atezolizumab 1200 mg q3w in combination with carboplatin and nab paclitaxel were similar to atezolizumab 1200 mg q3w as monotherapy. There is no evidence of a PK drug-drug interaction with the co administration of atezolizumab with carboplatin and/or nab paclitaxel. Treatment-emergent ADA did not appear to have a clinically relevant impact on pharmacokinetics, efficacy, or safety.

Collectively, atezolizumab 1200 mg q3w administered with carboplatin and nab paclitaxel provided a favorable benefit-risk profile which supports the recommended dosing regimen of atezolizumab in combination with carboplatin and nab-paclitaxel for treatment of chemotherapy-naïve patients with Stage IV non-squamous NSCLC.

The proportion of patients who developed treatment-emergent ADAs for atezolizumab in IMpower130 were 22.4%, which is comparable to historical data for atezolizumab (ranges from 16.7% to 54.5%). Even though the MAH has submitted data to support a valid and sensitive NAb assay, no data regarding NAb have been included. Treatment emergent ADAs did not appear to have a clinically meaningful impact on PK, efficacy, and safety, however, the number of ADA positive subjects is limited and the MAH is recommended to propose how an assessment of the effect of atezolizumab ADAs and NAb on PK parameters and efficacy endpoints including OS, PFS and ORR in NSCLC studies could be conducted. These will be investigated as part of a post-authorisation commitment.

ADAs have previously in the Pop PK model been found to increase atezolizumab clearance with 16 %, but no impact of ADAs on clearance was identified in IMPower130. However, the numerical differences in clearance estimated between the IMpower130 study and the PCD4989g+JO28944 studies are small. In the IMpower130, clearance is higher in ADA+ positive as also previously identified in other atezolizumab studies. The difference in clearance between ADA+ and ADA- is slightly smaller compared to the difference identified in the final Pop PK model (11.2 % vs 15.9% respectively). The Applicant has not identified any apparent cause of this small difference. This is acceptable as clinical consequences are unlikely. The impact of ADA on PK – presumably predominately clearance – will be investigated further as part of a post-approval commitment.

Exposure-efficacy (ORR, OS) and exposure-safety (adverse events of special interest [AESI], adverse events of Grade 3 to 5 [AEG35]) analyses were performed using Phase I, II, and III trials in mixed solid tumour, NSCLC, and mUC patients when atezolizumab was administered as monotherapy. No clinically meaningful ER relationships were identified when patient prognostic factors were considered in the analysis. Since combination administration poses additional confounding factors, the Sponsor obtained agreement (meeting minutes with EMA) with regulatory agencies that no further ER analysis are required for atezolizumab administered in combination settings.

2.3.6. Conclusions on clinical pharmacology

The clinical pharmacology of atezolizumab has previously been characterized and the PK investigations in IMPower130 overall confirm previous findings. The PK of atezolizumab is similar when administered in combination with carboplatin and nab-paclitaxel without evidence of drug-drug interactions. No unexpected covariates have been identified and the proposed dose of 1200 mg q3w seems to be appropriate and is endorsed.

The number of patients with ADAs is about 22 %, which is consistent with previous findings and ADAs do not appear to impact PK. However, data regarding neutralising antibodies (NABs) are lacking, the MAH is recommended to further investigate it post authorisation.

2.4. Clinical efficacy

2.4.1. Dose response study

No dose response study was submitted. Please refer to the clinical pharmacology assessment.

2.4.2. Main study

IMpower 130 (GO29537): Randomised phase III trial in chemotherapy-naïve patients with metastatic non-squamous NSCLC, in combination with nab paclitaxel and carboplatin

Table 6: Study design features of IMpower130

Study Design Feature	Arm A: Atezo+CnP	Arm B: CnP
Line of therapy	First-line	
Sample size	2:1 based on 715 total patients planned	
PD-L1 status	All-comers population stratified by PD-L1 expression by IHC (TC3 and any IC vs. TC0/1/2 and IC2/3 vs. TC0/1/2 and IC0/1)	
Dose/regimen (as applicable per arm)	Atezolizumab 1200 mg IV q3w Carboplatin AUC of 6 mg/mL/min IV q3w ^a Nab-paclitaxel 100 mg/m ² IV on Days 1, 8, and 15 of each 21-day cycle	
Treatment duration	<u>Induction</u> Atezo+CnP for four or six cycles ^b or until loss of clinical benefit <u>Maintenance</u> Atezolizumab until loss of clinical benefit	<u>Induction</u> CnP for four or six cycles ^b or until PD per RECIST v1.1 <u>Maintenance</u> BSC or pemetrexed until PD per RECIST v1.1
Response assessment	Response assessed by investigator using RECIST v1.1 and modified RECIST ^c Tumor assessments every 6 weeks up to Week 48, then every 9 weeks until PD or loss of clinical benefit, withdrawal, death, or study end	
Follow-up assessment	Survival follow-up every 3 months until death, loss to follow-up, or study end	

Atezo=atezolizumab; AUC=area under the concentration-time curve; BSC=best supportive care; CnP=carboplatin+nab-paclitaxel; IC=tumor-infiltrating immune cell; IHC=immunohistochemistry; IV=intravenous; PD=progressive disease; PD-L1=programmed death–ligand 1; q3w=every 3 weeks; RECIST=Response Evaluation Criteria in Solid Tumors; TC=tumor cell.

^a Refer to [Protocol v7 Section 4.3.2.2](#) for details on dose calculations of carboplatin.

^b One cycle was 21 days. The number of planned cycles of induction treatment (four or six) was at the discretion of the investigator and was determined and documented prior to randomization.

^c Tumor assessments using modified RECIST were performed only for patients in Arm A (Atezo+CnP).

Methods

Study participants

IMpower130 recruited patients with histologically or cytologically confirmed Stage IV non-squamous NSCLC according to the American Joint Committee on Cancer (AJCC) 7th edition, regardless of the PD-L1

expression level in tumour tissue. No prior treatment for Stage IV non-squamous NSCLC was allowed, except targeted therapy for *EGFR*-mutant and/or *ALK*-positive patients (see below for details).

IMpower130 included standard eligibility criteria for Stage IV NSCLC studies.

Key efficacy-related inclusion criteria were as follows:

- Male or female, 18 years of age or older
- ECOG performance status of 0 or 1
- Histologically or cytologically confirmed, Stage IV non-squamous NSCLC
 - Patients with tumors of mixed histology (i.e., squamous and non-squamous) were eligible if the major histological component appeared to be non-squamous.
- No prior treatment for Stage IV non-squamous NSCLC
 - Patients with a sensitizing mutation in the *EGFR* gene had to have experienced disease progression (during or after) or intolerance to treatment with one or more *EGFR* tyrosine kinase inhibitors for the treatment of *EGFR*-mutant NSCLC.
 - Patients with an *ALK* fusion oncogene must have experienced disease progression (during or after) or intolerance to treatment with one or more *ALK* inhibitors for the treatment of NSCLC in patients having an *ALK* fusion oncogene.
 - Patients with unknown *EGFR* and *ALK* status were required to have test results at screening. *ALK* and/or *EGFR* could be assessed locally or at a central laboratory.
 - Patients who received prior neo-adjuvant, adjuvant chemotherapy, radiotherapy, or chemoradiotherapy with curative intent for non-metastatic disease had to have experienced a treatment-free interval of at least 6 months from randomization since the last chemotherapy, radiotherapy, or chemoradiotherapy cycle.
- Known PD-L1 tumor status as determined by an IHC assay performed by a central laboratory on previously obtained archival tumor tissue or tissue obtained from a biopsy at screening
- Measurable disease, as defined by RECIST v1.1
- Previously irradiated lesions could only be considered as measurable disease if disease progression had been unequivocally documented at that site since radiation and the previously irradiated lesion was not the only site of disease.

Key efficacy-related exclusion criteria included the following:

- Active or untreated CNS metastases as determined by computed tomography (CT) or magnetic resonance imaging during screening and prior radiographic assessments
- Spinal cord compression not definitively treated with surgery and/or radiation or previously diagnosed and treated spinal cord compression without evidence that the disease had been clinically stable for >2 weeks prior to randomization
- History of autoimmune disease, including, but not limited to, myasthenia gravis, myositis, autoimmune hepatitis, systemic lupus erythematosus, rheumatoid arthritis, inflammatory bowel disease, vascular thrombosis associated with anti-phospholipids syndrome, Wegener's granulomatosis, Sjögren's syndrome, Guillain-Barré syndrome, multiple sclerosis, vasculitis, or glomerulonephritis
- Leptomeningeal disease

- Malignancies other than NSCLC within 5 years prior to randomization, with the exception of those with a negligible risk of metastasis or death treated with expected curative outcome
- Known tumor PD-L1 expression status as determined by an IHC assay from other clinical studies
- Any approved anti-cancer therapy within 21 days prior to initiation of study treatment
- Treatment with any other investigational agent with therapeutic intent within 28 days prior to randomization
- Administration of a live, attenuated vaccine within 4 weeks before randomization or anticipation that such a live attenuated vaccine would be required during the study
- Prior treatment with CD137 agonists or immune checkpoint blockade therapies, anti-PD-1, and anti-PD-L1 therapeutic antibodies
- Prior anti-cytotoxic T-lymphocyte-associated protein (CTLA)-4 treatment was permitted, provided the last dose was at least 6 weeks prior to randomization and the patient had no history of severe immune-mediated adverse effects from anti-CTLA-4.
- Treatment with systemic immunostimulatory agents within 4 weeks or five half-lives of the drug, whichever was longer, prior to randomization
- Treatment with systemic immunosuppressive medications within 2 weeks prior to randomization

Treatments

Table 7: Treatment regimens

		Atezolizumab	Nab-Paclitaxel	Carboplatin
Dose and administration route		1200 mg IV	100 mg/m ² IV	AUC 6 mg/mL/min IV
Infusion rate		60 (±15) minutes for the first infusion, shortening to 30 (±10) minutes for subsequent infusions	Over 30 minutes	Over approximately 15–30 minutes
Frequency		Day 1 of every 21-day cycle ^a		
Treatment to be Administered				
Induction phase (four or six cycles)	Arm A	X	X	X
	Arm B	—	X	X
Maintenance phase	Arm A	X	—	—
	Arm B ^b	—	—	—

Arm A=atezolizumab+carboplatin+nab-paclitaxel; Arm B=carboplatin+nab-paclitaxel; AUC=area under the concentration-time curve; IV=intravenous.

^a Nab-paclitaxel was also administered on Days 8 and 15 using the same dose and infusion time. Guidelines for dose modification and treatment interruption or discontinuation for carboplatin and nab-paclitaxel are provided in [Protocol v7 Section 5.1.7](#) and [Section 5.1.9](#).

^b Switch maintenance to erlotinib (for patients treated under v1–v4 of the protocol) or pemetrexed was permitted for patients randomized to Arm B. For patients remaining on Arm B, best supportive care was administered.

Switch maintenance to pemetrexed was permitted for patients randomized to the CnP arm. Switch maintenance to erlotinib was no longer permitted in Protocol v5 and after. However, patients randomized to the CnP arm who had already started switch maintenance treatment with erlotinib under v1–v4 of the protocol were allowed to continue treatment with erlotinib upon discussion of the risks, potential benefits, and alternative treatment options with the investigator. Because erlotinib was considered an

investigational medicinal product (IMP), erlotinib was provided by the Sponsor if required by local health authority regulations. Pemetrexed was used in a commercially available formulation at each participating site.

Following PD, patients consented under Protocol v1-v4 and randomized to the CnP arm were given the option to receive atezolizumab monotherapy provided they met crossover eligibility criteria (with the exception of no prior treatment for Stage IV non-squamous NSCLC and no prior treatment with immune checkpoint blockade therapy).

Objectives

Efficacy objectives

In regard to the objectives below, the intent-to-treat (ITT)-wild type (WT) population refers to randomized patients who did not have an activating *EGFR* mutation or *ALK* translocation. The ITT population refers to all randomized patients. Some efficacy endpoints were analyzed in a population of randomized patients with a defined level of PD-L1 expression. Tumour specimens were evaluated for PD-L1 expression on tumour cells (TC) and tumour infiltrating immune cells (IC) as assayed by a centrally performed immunohistochemistry (IHC) test. The results were used to define the PD-L1 expression subgroups for the analyses described under statistical methods.

Co-Primary Efficacy Objectives

The co-primary objectives for this study were as follows:

- To evaluate the efficacy of Atezo-CnP compared with CnP as measured by investigator-assessed PFS according to Response Evaluation Criteria in Solid Tumours (RECIST) Version 1.1 in the ITT-WT population
- To evaluate the efficacy of Atezo-CnP compared with CnP as measured by OS in the ITT-WT population

Secondary Efficacy Objectives

The secondary efficacy objectives for this study were as follows:

- To evaluate the efficacy of atezolizumab as measured by investigator-assessed PFS according to RECIST v1.1 and OS in the ITT population
- To evaluate the efficacy of atezolizumab as measured by investigator-assessed PFS according to RECIST v1.1 and OS in the PD-L1 expression population and PD-L1 expression WT population
- To evaluate the efficacy of atezolizumab as measured by investigator-assessed objective response rate (ORR) according to RECIST v1.1 in the ITT-WT population
- To evaluate the efficacy of atezolizumab as measured by investigator-assessed duration of response (DOR) according to RECIST v1.1 in the ITT-WT population
- To evaluate the OS rates at 1 and 2 years after randomization in each treatment arm in the ITT-WT population
- To determine the impact of atezolizumab as measured by time to deterioration (TTD) in patient-reported lung cancer symptoms, using the European Organisation for Research and Treatment of Cancer (EORTC) Quality of Life Questionnaire-Core 30 (QLQ-C30) and the supplemental 13-Item Lung Cancer Module (LC13) in the ITT-WT population
- To determine the impact of atezolizumab as measured by change from baseline (i.e., improvement or deterioration based on presenting symptomatology) in patient-reported lung

cancer symptoms (chest pain, dyspnoea, and cough) using the Symptoms in Lung Cancer (SILC) scale symptom severity score in the patient-reported outcome (PRO)-evaluable WT population.

Safety objectives

The safety objectives for this study were as follows:

- To evaluate the incidence, nature, and severity of adverse events (AEs) graded according to the National Cancer Institute Common Terminology Criteria for Adverse Events (NCI CTCAE) v4.0
- To evaluate the changes in vital signs and clinical laboratory results during and following atezolizumab administration
- To evaluate the incidence and titers of anti-drug antibody (ADA; also known as anti-therapeutic antibody) response to atezolizumab and potential correlation with pharmacokinetic (PK), pharmacodynamic, safety, and efficacy parameters

Pharmacokinetic objectives

The PK objectives for this study were as follows:

- To evaluate the maximum observed serum concentration (C_{max}) of atezolizumab after infusion on Cycle 1 Day 1 (C1D1) and minimum observed serum concentration (C_{min}) of atezolizumab prior to infusion at selected cycles, at treatment discontinuation, and at 120 days (±30 days) after the last dose of atezolizumab for patients randomized to Atezo-CnP
- To evaluate the plasma concentrations of carboplatin (Atezo-CnP and CnP arms)
- To evaluate the plasma concentrations of nab-paclitaxel reported as total paclitaxel (Atezo-CnP and CnP arms)

Exploratory objectives

The exploratory objectives for this study were as follows:

- To evaluate the efficacy of atezolizumab as measured by investigator-assessed PFS, ORR, and DOR according to RECIST v1.1 and OS in the T-effector (Teff)-high population (with Teff gene signature expression in tumor tissue as determined by an RNA-based assay) and the Teff-high population excluding patients with an activating *EGFR* mutation or *ALK* translocation (Teff-high WT)
- To evaluate the PFS rates at 6 months and 1 year after randomization, as assessed by the investigator according to RECIST v1.1
- To evaluate the OS rate at 3 years after randomization
- To determine the impact of atezolizumab as measured by the utility scores of the EuroQol 5-Dimension 3-Level (EQ-5D-3L) questionnaire
- To evaluate the impact of atezolizumab on the change from baseline and the number and proportion of patients with a clinically meaningful change in PROs of health-related quality of life (HRQoL), lung cancer-related symptoms, and health status as assessed by the EORTC QLQ-C30 and EORTC QLQ-LC13.

Outcomes/endpoints

The endpoints and analyses described in this summary are based on the IMpower130 Statistical Analysis Plan (SAP) v4, dated 4 May 2018. The analyses described in the SAP supersede those specified in the study protocol, as applicable.

Definitions of key efficacy endpoints are given below table.

The co-primary efficacy outcome measures for IMpower130 were as follows:

- Investigator-assessed PFS in the ITT-WT population
- OS in the ITT-WT population

As per the pre-specified analysis hierarchy and α -spending algorithm, PFS in the ITT-WT population was tested first and, if this test was significant, the α allocated to PFS was to be recycled for the test of OS in the ITT-WT population.

The secondary efficacy outcome measures included:

- PFS and OS in the ITT population
- PFS and OS in the PD-L1 expression and PD-L1 expression WT populations
- Objective response rate (ORR; confirmation not required)
- Duration of response (DOR; confirmation not required)
- OS rates at the 1- and 2-year landmark timepoints
- Time to deterioration (TTD) in patient-reported lung cancer symptoms per European Organisation for Research and Treatment of Cancer (EORTC) questionnaires
- Change from baseline in patient-reported lung cancer symptoms using the Symptoms in Lung Cancer (SILC) scale
- Proportion of patients improved, remained stable, worsened in patients-reported lung cancer symptoms, HRQOL and functioning

Table 8: Key Efficacy Endpoint Definitions in IMpower130

Endpoint	Definition
PFS	Time from randomization to first documented PD, as determined by the investigator using RECIST v1.1, or death from any cause, whichever occurred first
OS	Time from randomization to death from any cause
ORR	Proportion of patients with an objective response (either CR or PR), as determined by the investigator using RECIST v1.1; confirmation not required
DOR	Time from first documented objective response (CR or PR) to documented PD, as determined by the investigator using RECIST v1.1, or death from any cause, whichever occurred first; confirmation not required

CR=complete response; DOR=duration of response; ORR=objective response rate; OS=overall survival; PD=progressive disease; PFS=progression-free survival; PR=partial response; RECIST=Response Evaluation Criteria in Solid Tumours.

Confirmation of response according to RECIST v1.1 was not required, but for exploratory purposes, ORR with confirmation may be reported. For exploratory purposes, DOR for patients with confirmed response may be reported.

Tumour assessments were conducted every 6 weeks for the first 48 weeks following Cycle 1, then every 9 weeks thereafter.

Sample size

This study planned to randomize approximately 715 patients. Patients with an activating *EGFR* mutation or *ALK* translocation were excluded from the primary analysis population (i.e., ITT-WT). There were expected to be approximately 650 patients in the ITT-WT population given the approximate 10% prevalence of these genetic variations.

The sample size of this study was based on the number of events required to demonstrate efficacy with regard to both PFS and OS (co-primary endpoints) for the comparison of the Atezo-CnP arm versus the CnP arm. The estimate of the number of events required to demonstrate efficacy with regard to PFS in the primary comparison of the Atezo-CnP arm versus the CnP arm was based on the following assumptions:

- One-sided significance level of 0.003 in the ITT-WT population

- 95% power to detect a hazard ratio (HR) of 0.65, corresponding to an improvement in median PFS from 6 months to 9.2 months in the ITT-WT population
- No interim PFS analysis
- Dropout rate of 5% per 12 months

The estimate of the number of events required to demonstrate efficacy with regard to OS in the comparison of the Atezo-CnP arm versus the CnP arm was based on the following assumptions:

- One-sided significance level of 0.022 in the ITT-WT population
- 80% power to detect an HR of 0.75, corresponding to an improvement in median OS from 12 months to 16 months in the ITT-WT population
- One interim OS analysis performed when approximately 77% of the total OS events required for the final analysis had occurred using the Lan-DeMets approximation to the O'Brien-Fleming boundary
- Dropout rate of 5% per 24 months

Randomisation

For patients who were eligible for enrollment, the study site obtained the patient's randomization number and treatment assignment from the IxRS. The number of cycles of induction treatment (four or six) was determined by the investigator and documented prior to randomization. Randomization to one of the two treatment arms occurred in a 2:1 ratio. Permuted-block randomization was applied to ensure a balanced assignment to each treatment arm.

Randomization was stratified by the following criteria:

- Sex (male vs. female)
- Presence of liver metastases at baseline (yes vs. no)
- PD-L1 tumour expression by IHC (TC3 and any IC vs. TC0/1/2 and IC2/3 vs. TC0/1/2 and IC0/1)

Following PD, patients who were consented under v1-v4 of the protocol and randomized to the CnP arm were given the option to receive atezolizumab monotherapy, provided they continued to meet eligibility criteria to receive atezolizumab (with the exception of no prior treatment for Stage IV non-squamous NSCLC and no prior treatment with immune checkpoint blockade therapy).

Blinding (masking)

IMpower130 is an open-label study. The Sponsor was blinded to treatment arm allocation and patient-level data on PD-L1 expression until the database had been locked for analysis.

Statistical methods

Analysis Population

Populations analysed are defined as follows, in alphabetical order:

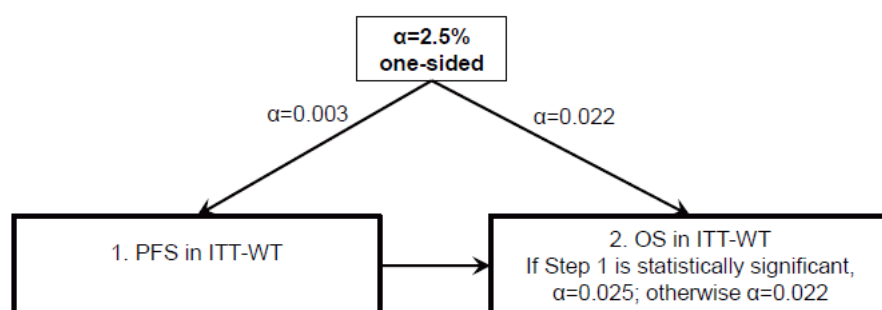
- **ITT:** All randomized patients, regardless of receipt of the assigned treatment.
- **ITT-WT:** All randomized patients regardless of assigned treatment, excluding patients with an activating EGFR mutation or ALK translocation.

- **PD-L1 expression population:** ITT patients with PD-L1 IHC TC1/2/3 (defined as PD-L1 staining in <1%, ≥1% to <5%, ≥5% to <50%, and ≥50% of TCs) or IC1/2/3 (defined as PD-L1-stained ICs covering <1%, ≥1% to <5%, ≥5% to <10%, and ≥10% of the tumour area) expression in baseline tumour tissue (TC1/2/3 or IC1/2/3 subgroup); ITT patients with PD-L1 IHC TC3 or IC3 expression in baseline tumour tissue (TC3 or IC3 subgroup); and ITT patients with PD-L1 IHC TC2/3 or IC2/3 expression in baseline tumour tissue (TC2/3 or IC2/3 subgroup).
- **PD-L1 expression WT population:** As defined above for the PD-L1 expression population, excluding patients with an activating EGFR mutation or ALK translocation.
- **Patient-reported outcome (PRO)-evaluable WT population:** Patients randomized into the ITT-WT population who completed a given PRO questionnaire at the baseline visit and at least one post-baseline visit.

Primary analysis

The co-primary efficacy endpoints are PFS and OS in the ITT-WT population.

To control the overall Type I error rate for the one-sided test at 0.025, a one-sided alpha of 0.003 and a one-sided alpha of 0.022 were allocated to PFS and OS, respectively. The primary comparison of PFS was tested at a one-sided alpha level of 0.003 in the ITT-WT population. The primary comparison of OS was to be tested in the ITT-WT population at the allocated alpha together with the alpha recycled from the PFS analysis if the PFS comparison was statistically significant. The PFS and OS analysis hierarchy and alpha-allocation, including possible alpha-recycling.



ITT=intent-to-treat; OS=overall survival; PFS=progression-free survival; WT=wild type.

^a To control the overall Type I error rate for the one-sided test at 0.025, a one-sided Type I error (α) of 0.003 and 0.022 (ratio of 3:22) was allocated to PFS and OS, respectively, for comparison of the Atezo+CnP arm versus the CnP arm.

Figure 7: PFS and OS analysis hierarchy, alpha allocation, and alpha recycling

OS was planned to be tested using the group sequential method at the interim and final OS analyses. Statistical significance at the interim and final OS analyses was planned to be tested.

The HRs, λ_A/λ_B , where λ_A and λ_B represent the hazard of having a PFS event in the Atezo-CnP and CnP arms, respectively, comparing the treatment effect between the two treatment arms, were estimated using a stratified Cox regression model with the same stratification variables used for the stratified log-rank test, and the 95% CIs were provided. For each comparison, analyses were conducted according to an analysis hierarchy and an alpha-spending algorithm to control for the Type I error rate and to account for an interim OS analysis.

The stratification factors were, in principle, planned to be those used during randomization as recorded in the IxRS. Due to the potential risk of over-stratification (Akazawa et al. 1997), if at least one stratum had fewer than 15 events (PFS or OS events), the stratification factor that contained the level with the smallest number of patients was to be removed from the stratified analyses. The removal of the stratification factor continued until there were no strata with fewer than 15 events (PFS or OS events).

The final set of stratification factors used in stratified analyses following the pre-specified rule in the SAP v4 were applied to all endpoints where stratified analyses were planned. Results from unstratified analysis were also provided.

Kaplan-Meier (KM) methodology was used to estimate median PFS and OS and to construct survival curves for each treatment arm for a visual description of the difference among arms. The Brookmeyer-Crowley methodology was used to construct the 95% CI for median PFS and OS (Brookmeyer and Crowley 1982).

Follow-up for OS was summarized for all patients included in the analysis (whether the patient was alive or had died). Follow-up was summarized using the KM method with data for patients who died, censored at the date of death.

Note that the OS boundaries were updated to use the Lan-DeMets approximation to the Pocock boundaries; therefore, the overall power for the OS analysis in the ITT-WT population was reduced from 80% to 77%.

Primary Analysis Timing for PFS

No interim analyses were planned for the co-primary endpoint of PFS in this study. The primary PFS analysis was conducted at the time of the interim OS analysis. At this timepoint, it was expected that approximately 475 PFS events would have occurred in the ITT-WT population based on the assumptions specified in the SAP v4. This number of events would allow for a minimum detectable difference in HR of approximately 0.76 in the ITT-WT population.

Table 9: Primary analysis for PFS

Population	Analysis Timing, Months from FPI	No. of Events, Event Ratio (%)	MDD in Hazard Ratio	Power
ITT-WT	32	475 (73%)	0.76	95%

FPI=first patient in; ITT=intent-to-treat; MDD=minimum detectable difference; WT=wild type.

Interim and Final Analysis Timing for Overall Survival

One interim analysis was planned for the co-primary endpoint of OS to be conducted when approximately 352 OS events had occurred in the ITT-WT population. The final OS analysis was planned to be conducted when there were approximately 457 OS events in the ITT-WT population.

The stopping boundaries for the interim and final OS analyses considering different scenarios with alpha-recycling were calculated using the Lan-DeMets approximation to the Pocock boundary, assuming the specified observed number of events (352 and 457, respectively), and are shown in Table 10. The p-value was used to claim crossing of the boundaries. For example, if approximately 352 events had occurred in the ITT-WT population at the time of the interim OS analysis, then statistical significance of the OS endpoint in ITT-WT population was declared if $p \leq 0.0211$ when PFS had claimed significance in the ITT-WT population.

Table 10: Analysis timing and stopping boundary for OS (ITT-WT population)

Analysis	Stopping Boundary in HR (p-value)	
	If $\alpha=0.022$	If $\alpha=0.025$
Interim OS analysis	HR ≤ 0.790 (p-value ≤ 0.0185)	HR ≤ 0.795 (p-value ≤ 0.0211)
Final OS analysis	HR ≤ 0.795 (p-value ≤ 0.0103)	HR ≤ 0.799 (p-value ≤ 0.0119)

HR=hazard ratio; ITT=intent-to-treat; OS=overall survival; WT=wild type.

Note: α and p-value are one-sided.

Sensitivity analyses

Impact of non-protocol-specified anti-cancer therapy

The impact of non-protocol-specified anti-cancer therapy (NPT) on PFS as determined by the investigator will be assessed, depending on the number of patients who received NPT before a PFS event. If > 5% of patients received NPT before a PFS event in the control arm, a sensitivity analysis may be performed for the comparisons between two treatment arms in which data from patients who receive NPT before a PFS event will be censored at the last tumor assessment date before receipt of NPT.

The impact of subsequent NPT on OS will be assessed depending on the number of patients who received NPT. If > 10% of patients received a NPT in the control arm, the following analyses may be performed to compare treatment arms:

- The discount method uses a 'discounted' survival time after switching for patients who switch treatments based on a user-specified assumption for the effect on OS. OS may be discounted according to a range of possible effects on OS of the subsequent NPT after treatment switching occurred (e.g., 10%, 20%, 30%, etc.).
- Rank preserving structural failure time provides an estimate of the OS time for the control group had NPT not occurred (Robins and Tsiatis 1991). It estimates OS measured from the time of NPT by applying an estimate of the benefit of the NPT. The total OS time (sum of time to NPT and the estimated survival time after NPT started) may then be analyzed using the same methodology as for the primary analysis of OS.
- The inverse probability of censoring weighting method censors patients at start of NPT and uses the control arm patients to create weights that represent how NPT treated patient is like a non-NPT-treated patient (Robins and Finkelstein 2000). These time-varying weights are included into the OS analysis to correct the effect of NPT by giving increased weight to non-censored patients with similar characteristics to censored patients.

Secondary analyses

PFS and OS in the ITT Population will be analyzed using the same methods as described for PFS and OS. If the difference in OS between Arm A and Arm B in the ITT-WT population was statistically significant, a comparison of Arm A versus Arm B was to be conducted in the ITT population. The alpha allocation will follow the same alpha-spending algorithm and allocation ratio (3:22) described for analysis of the co-primary efficacy endpoints.

PFS and OS in the TC1/2/3 or IC1/2/3-WT, and the TC1/2/3 or IC1/2/3 populations will be analyzed using similar methods as described for OS and PFS. Analyses of PFS and OS may be conducted in other PD-L1 expression populations and PD-L1 expression WT populations (e.g., TC2/3 or TC2/3-WT, TC3 or IC3-WT, TC2/3 or IC2/3, TC3 or IC3, and the corresponding complementary groups, as well as TC0 and IC0-WT, TC0 and IC0).

ORR will be compared between treatment arms using the stratified Cochran-Mantel-Haenszel test. The 95% CI for the difference in ORRs between the two treatment arms will be computed using the normal approximation to the binomial distribution. An estimate of ORR and its 95% CI will be calculated for each treatment arm using the Clopper-Pearson method.

Because the determination of DOR is based on a non-randomized subset of patients, formal hypothesis testing was not performed. DOR was estimated using KM methodology. Comparisons between treatment arms were made using the stratified and unstratified log-rank test for descriptive purposes only.

The OS rate at the 1- and 2-year landmark time points after randomization was estimated for each treatment arm using KM methodology, along with 95% CI calculated with the standard error derived from

the Greenwood formula. The 95% CI for the difference in OS rates between the two treatment arms will be estimated using the normal approximation method, with standard errors computed using the Greenwood method.

Patient-Reported Outcomes: Through the use of the SILC scale, the EORTC QLQ-C30, and EORTC QLQ-LC13, lung cancer symptom data will be collected about symptoms commonly associated with cancer treatments, and disease and treatment impact on patients' functioning.

The EORTC QLQ-C30 and EORTC QLQ-LC13 will be scored according to the EORTC scoring manual. For multi-item subscales, if $\leq 50\%$ of items within the multi-item subscale are missing at a given timepoint, the multi-item score will be calculated on the basis of the non-missing items. If $> 50\%$ of items are missing or if a single-item measure is missing, the subscale is missing.

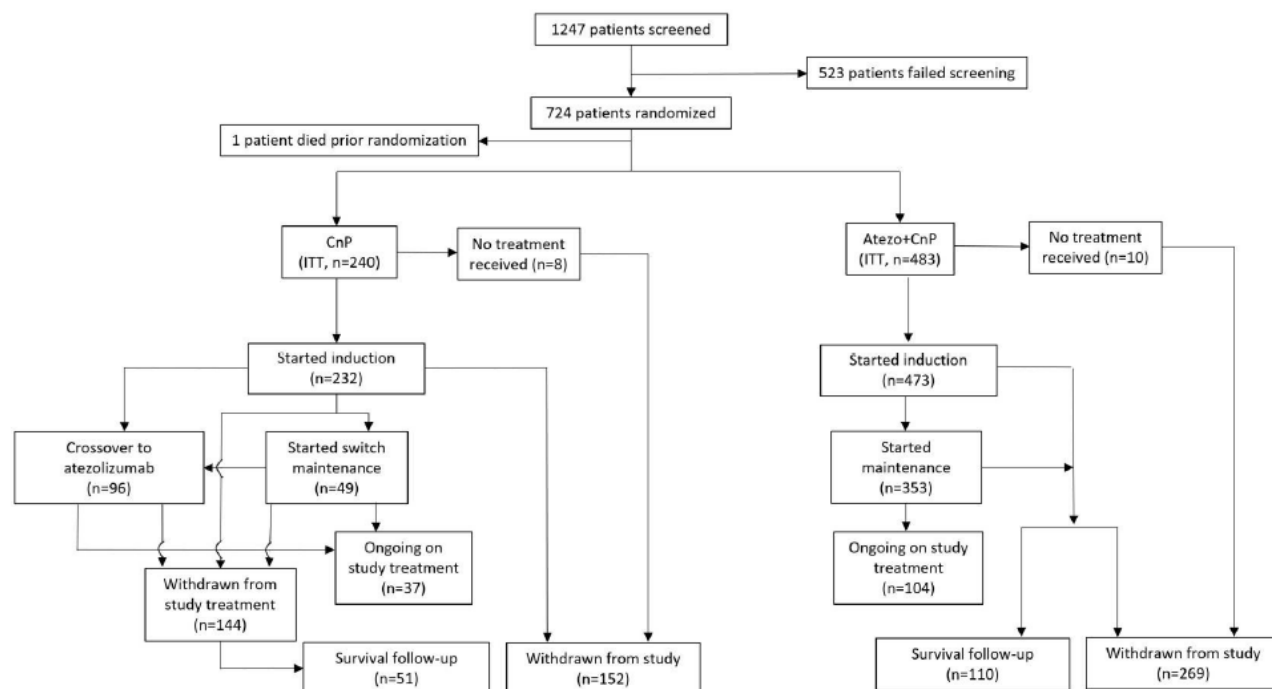
The SILC questionnaire will be scored at the individual symptom level, thus will have a dyspnea score, chest pain score, and cough score. Each individual symptom score will be calculated as the average of responses for the symptom items (e.g., Chest Pain Score = mean [item 1; item 2]). An increase in score is suggestive of a worsening in symptomology (i.e., frequency or severity). A score change of ≥ 0.3 points for the dyspnea and cough symptom scores is considered to be clinically significant; whereas a score change of ≥ 0.5 points for the chest pain score is considered to be clinically significant.

Time to deterioration in PRO: TTD with use of the EORTC is defined as the time from baseline to the first confirmed clinically meaningful deterioration in the EORTC symptom score. TTD will be documented for a 3-symptom composite endpoint using the following EORTC QLQ-LC13 symptom scores: cough, chest pain, and dyspnea multi-item scale. Confirmed clinically meaningful deterioration in lung cancer symptoms is defined as a ≥ 10 -point increase above baseline in a symptom score that must be held for at least two consecutive assessments or an initial ≥ 10 -point increase above baseline followed by death within 3 weeks from the last assessment. A ≥ 10 -point change in the EORTC scale score is perceived by patients as clinically significant.

Summary statistics (mean, SD, median, 25th and 75th percentiles, and range) of the change from baseline per SILC scale may be provided.

Results

Participant flow



Atezo=atezolizumab; CnP=carboplatin+nab-paclitaxel; ITT=intent-to-treat.

Figure 8: Patient disposition (ITT population)

Table 11: Patient Disposition from Study (ITT Population)

Patient Disposition from Study
Intent-to-Treat Patients
Protocol: G029537 (Data Cutoff: 15MAR2018)

	CnP (Randomized) (N=240)	Atezo+CnP (Randomized) (N=483)
Received Treatment	232 (96.7%)	473 (97.9%)
Received Treatment During The Induction Phase	232 (96.7%)	473 (97.9%)
Received Treatment During The Maintenance Phase	49 (20.4%)	353 (73.1%)
On-study Status	88 (36.7%)	214 (44.3%)
Alive: On Treatment	37 (15.4%)	104 (21.5%)
Alive: In Follow-Up	51 (21.3%)	110 (22.8%)
Discontinued Study	152 (63.3%)	269 (55.7%)
Death	136 (56.7%)	240 (49.7%)
Lost To Follow-Up	1 (0.4%)	0
Non-Compliance	0	1 (0.2%)
Other	4 (1.7%)	8 (1.7%)
Physician Decision	0	1 (0.2%)
Protocol Violation	1 (0.4%)	0
Withdrawal By Subject	10 (4.2%)	19 (3.9%)

Atezo=Atezolizumab, CnP=Carboplatin+Nab-Paclitaxel.
Data Cutoff: 15MAR2018; RAVE Data Extracted: 18 May 2018.

Recruitment

First patient randomized: 16 April 2015. Last patient randomized: 13 February 2017. Data cutoff date: 15 March 2018

131 centers in 8 countries: Belgium 21 (4), Canada 52 (5), France 45 (11), Germany 132 (25), Italy 52 (6), Israel 35 (10), Spain 71 (13), United States 315 (57).

Conduct of the study

Protocol Amendments

The first version (v1) of the protocol was issued on 21 November 2014 and amended (v2) on 29 March 2015, prior to the first patient randomized (16 April 2015). The protocol was subsequently amended five times (v3-v7). The key changes to the protocol are summarized below.

Table 12: Comparison of key features in IMpower130 Protocol versions 1-7

	Protocol Versions 1-4 11 November 2015 (v4)	Protocol Version 5 15 June 2016	Protocol Version 6 1 March 2017 (global) 15 July 2017 (Germany)	Protocol Version 7 28 February 2018 (VHP) 25 April 2018 (Germany)
Total number of patients to be randomized	550	650	715	No changes
Primary endpoints	Investigator-assessed PFS in PD-L1-selected and ITT population	Investigator-assessed PFS and OS in PD-L1-selected and ITT population	Investigator-assessed PFS in tGE-WT and ITT-WT population and OS in ITT-WT population	No changes
Secondary efficacy endpoints	ORR ^a , OS ^a , DOR, TIR, IRF-assessed PFS, 1- and 2-year OS, TTD, patient-reported outcomes	ORR, DOR, TTR, TIR, IRF-assessed PFS, 1- and 2-year OS, TTD, patient-reported outcomes	PFS and OS in secondary populations ^b , ORR, DOR, 1- and 2-year OS, TTD, patient-reported outcomes	No changes
Type I error rate (α) and α allocation	One-sided $\alpha=0.025$ PFS: 0.02 (PD-L1-selected) and 0.005 (ITT) ORR: 0.005 (PD-L1-selected) and 0.001 (ITT) OS: 0.02 or 0.015 ^c (PD-L1-selected) and 0.005 or 0.004 ^c (ITT)	One-sided $\alpha=0.025$ PFS: 0.003 (PD-L1-selected) then 0.003 (ITT) OS: 0.025 or 0.022 ^c (PD-L1-selected) then 0.025 or 0.022 ^c (ITT)	One-sided $\alpha=0.025$ PFS: 0.003 (tGE-WT) then 0.003 (ITT-WT) ^d OS: 0.025 or 0.022 ^c (ITT-WT) ^d	No changes
Crossover to receive atezolizumab monotherapy	Allowed	Not allowed	No change	No change
Biomarker subgroup (PD-L1-selected population)	Not specified	TC1/2/3 or IC1/2/3	TC2/3 or IC2/3 and TC1/2/3 or IC1/2/3	No changes
Stratification factors for randomization	Sex (male vs. female), presence of liver metastases at baseline (yes vs. no), PD-L1 tumor expression status (TC3 and any IC vs. TC0/1/2 and IC2/3 vs. TC0/1/2 and IC0/1)	No changes	No changes	No changes

DOR=duration of response; ECOG=Eastern Cooperative Oncology Group; IC=tumor-infiltrating immune cell; IRF=Independent Review Facility; ITT=intent-to-treat; ORR=objective response rate; OS=overall survival; PD-L1=programmed death-ligand 1; PFS=progression-free survival; TC=tumor cell; tGE=tumor gene expression; TIR=time in response; TTD=time to deterioration; TTR=time to response; WT=wild type.

^a ORR and OS will be formally tested if PFS is positive.

^b Secondary populations refer to tGE-WT (for OS) and TC2/3 or IC2/3-WT, TC1/2/3 or IC1/2/3-WT, tGE, and ITT populations.

^c First α was planned to be used if the preceding analysis in the hierarchy was statistically significant; otherwise, the second α would be used.

^d If OS in ITT-WT is positive, formal tests will be conducted in the tGE and ITT populations with same testing strategy and α allocation as PFS and OS in the tGE-WT and ITT-WT populations.

Changes to Planned Analyses

The first version (v1) of the SAP was issued on 19 May 2017 and was amended three times (v2-v4). The key changes to the SAP are summarized below.

Table 13: Comparison of key features in IMpower130 statistical analysis plan versions 1-4

	SAP Version 1 19 May 2017	SAP Version 2 17 October 2017	SAP Version 3 13 March 2018	SAP Version 4 4 May 2018
Primary endpoints	Investigator-assessed PFS in tGE-WT and ITT-WT population and OS in ITT-WT population	No changes	Investigator-assessed PFS and OS in ITT-WT population	No changes
Type I error rate (α) and α allocation	One-sided $\alpha=0.025$ PFS: 0.003 (tGE-WT) then 0.003 (ITT-WT) ^a OS: 0.025 or 0.022 ^b (ITT-WT) ^a	No changes	One-sided $\alpha=0.025$ PFS: 0.003 (ITT-WT) ^c OS: 0.025 or 0.022 ^b (ITT-WT) ^c	No changes
Primary analysis timing	Approximately 225 PFS events in the tGE-WT, approximately 475 PFS events in the ITT-WT, and last patient randomized, whichever occurs last	Approximately 352 OS events in the ITT-WT	No changes	No changes
Method to calculate stopping boundaries for interim and final OS analyses	Lan-DeMets approximation to the O'Brien-Fleming boundary	No changes	Lan-DeMets approximation to the Pocock boundary	No changes
Stratification factors for stratified analysis ^d	Sex (male vs. female), presence of liver metastases at baseline (yes vs. no), and PD-L1 tumor expression status (TC3 and any IC, TC0/1/2 and IC2/3 combined) vs. TC0/1/2 and IC0/1)	No changes	No changes	No changes to factors; however, if at least one stratum has <15 PFS or OS events, the factor that contains the level with the smallest number of patients would be removed from stratified analysis

HR=hazard ratio; IC=tumor-infiltrating immune cell; ITT=intent-to-treat; OR=objective response; OS=overall survival; PD-L1=programmed death-ligand 1; PFS=progression-free survival; SAP=Statistical Analysis Plan; TC=tumor cell; tGE=tumor gene expression; WT=wild type.

^a If OS in ITT-WT is positive, formal tests will be conducted in the tGE and ITT populations with same testing strategy and α allocation as PFS and OS in the tGE-WT and ITT-WT populations.

^b First α was planned to be used if the preceding analysis in the hierarchy was statistically significant; otherwise, the second α would be used.

^c If OS in ITT-WT is positive, formal tests will be conducted in PFS and OS in the ITT population with same testing strategy and α allocation as in the ITT-WT population.

^d Note that the factors used for the stratified analysis (i.e., two levels of PD-L1 stratification) differ from the factors used for randomization (three levels). In addition, stratification by PD-L1 expression is not used for the tGE-WT population, where applicable.

Protocol deviations

Table 14: Summary of major protocol deviations (ITT population) – data cut-off 15 March 2018

Protocol Deviation Category Protocol Deviation Description	CnP (Randomized) (N=240)	Atezo+CnP (Randomized) (N=483)	All Patients (N=723)
Total number of patients with at least one deviation	86 (35.8%)	179 (37.1%)	265 (36.7%)
Overall total number of deviations	150	346	496
PROCEDURAL			
Total number of patients with at least one deviation	67 (27.9%)	135 (28.0%)	202 (27.9%)
ON STUDY ASSESSMENT NOT DONE/OUTSIDE OF WINDOW	36 (15.0%)	68 (14.1%)	104 (14.4%)
OMISSION OF LAB TEST REQ. BY PROTOCOL	20 (8.3%)	33 (6.8%)	53 (7.3%)
ICF- OTHER (PROCEDURAL ISSUES)	14 (5.8%)	34 (7.0%)	48 (6.6%)
OMISSION OF TUMOR ASSESSMENT	7 (2.9%)	8 (1.7%)	15 (2.1%)
EX.CRITERIA PROCEDURAL (E.G. OUT OF WINDOW)	7 (2.9%)	6 (1.2%)	13 (1.8%)
FAILURE TO REPORT SAEs OR PREGNANCY PER PROTOCOL	2 (0.8%)	8 (1.7%)	10 (1.4%)
INC.CRITERIA PROCEDURAL E.G. OUT OF WINDOW	2 (0.8%)	7 (1.4%)	9 (1.2%)
INCLUSION CRITERIA			
Total number of patients with at least one deviation	14 (5.8%)	35 (7.2%)	49 (6.8%)
INCLUSION-RELATED TEST NOT DONE	7 (2.9%)	10 (2.1%)	17 (2.4%)
ICF - OTHER (E.G. PROCEDURAL ISSUES)	1 (0.4%)	11 (2.3%)	12 (1.7%)
OTHER INCLUSION CRITERIA	2 (0.8%)	7 (1.4%)	9 (1.2%)
MEDICATION			
Total number of patients with at least one deviation	16 (6.7%)	28 (5.8%)	44 (6.1%)
SIGNIFICANT DEVIATION FROM PLANNED DOSE	10 (4.2%)	11 (2.3%)	21 (2.9%)
RECEIVED INCORRECT STUDY TREATMENT	3 (1.3%)	7 (1.4%)	10 (1.4%)
DOSE MISSED OR SIGNIFICANT OUT OF WINDOW	4 (1.7%)	4 (0.8%)	8 (1.1%)
RECEIVED PROHIBITED CONCOMITANT THERAPY/MEDICATION	0	6 (1.2%)	6 (0.8%)
EXCLUSION CRITERIA			
Total number of patients with at least one deviation	6 (2.5%)	23 (4.8%)	29 (4.0%)
EXCLUSION RELATED TEST NOT DONE	4 (1.7%)	10 (2.1%)	14 (1.9%)
RECEIVED PRIOR PROHIBITED THERAPY OR MEDICATION	0	6 (1.2%)	6 (0.8%)

Atezo=Atezolizumab, CnP=Carboplatin+Nab-Paclitaxel.

A patient was counted once if received more than one deviation of the same type.

A patient was counted more than once if received more than one type of the deviation.

Data Cutoff: 15MAR2018; RAVE Data Extracted: 18 May 2018.

Baseline data

Table 15: Demographic and baseline disease characteristics (ITT-WT population)

Protocol: G029537 (Data Cutoff: 15MAR2018)

	CnP (Randomized) (N=228)	Atezo+CnP (Randomized) (N=451)	All Patients (N=679)
Sex			
n	228	451	679
Female	94 (41.2%)	185 (41.0%)	279 (41.1%)
Male	134 (58.8%)	266 (59.0%)	400 (58.9%)
Liver Metastasis at Enrollment			
n	228	451	679
Yes	31 (13.6%)	69 (15.3%)	100 (14.7%)
No	197 (86.4%)	382 (84.7%)	579 (85.3%)
TC/IC Strat Factor from IxRS			
n	228	451	679
TC3 or IC2/3	61 (26.8%)	121 (26.8%)	182 (26.8%)
TC0/1/2 and IC0/1	167 (73.2%)	330 (73.2%)	497 (73.2%)
Age (years)			
n	228	451	679
Mean (SD)	64.2 (8.9)	63.9 (9.5)	64.0 (9.3)
Median	64.5	64.0	64.0
Min - Max	38 - 85	18 - 86	18 - 86
Age Group (years)			
n	228	451	679
<65	114 (50.0%)	227 (50.3%)	341 (50.2%)
65 to 74	84 (36.8%)	174 (38.6%)	258 (38.0%)
75 to 84	29 (12.7%)	48 (10.6%)	77 (11.3%)
>=85	1 (0.4%)	2 (0.4%)	3 (0.4%)
Ethnicity			
n	228	451	679
Hispanic or Latino	10 (4.4%)	22 (4.9%)	32 (4.7%)
Not Hispanic or Latino	204 (89.5%)	397 (88.0%)	601 (88.5%)
Not reported	9 (3.9%)	18 (4.0%)	27 (4.0%)
Unknown	5 (2.2%)	14 (3.1%)	19 (2.8%)
Race			
n	228	451	679
Asian	3 (1.3%)	12 (2.7%)	15 (2.2%)
Black or African American	8 (3.5%)	17 (3.8%)	25 (3.7%)
White	210 (92.1%)	402 (89.1%)	612 (90.1%)
Multiple	0	1 (0.2%)	1 (0.1%)
Unknown	7 (3.1%)	19 (4.2%)	26 (3.8%)
Weight (kg) at baseline			
n	227	451	678
Mean (SD)	76.18 (18.36)	75.32 (16.67)	75.61 (17.24)
Median	74.00	72.00	72.55
Min - Max	40.9 - 145.2	42.0 - 144.0	40.9 - 145.2
Tobacco Use History			
n	228	451	679
Never	17 (7.5%)	48 (10.6%)	65 (9.6%)
Current	51 (22.4%)	92 (20.4%)	143 (21.1%)
Previous	160 (70.2%)	311 (69.0%)	471 (69.4%)
Pathology/Histology			
n	228	451	679
Non-Squamous - Adenocarcinoma	218 (95.6%)	432 (95.8%)	650 (95.7%)
Non-Squamous - Adenocarcinoma with neuroendocrine features	4 (1.8%)	4 (0.9%)	8 (1.2%)
Non-Squamous - Adenosquamous	0	4 (0.9%)	4 (0.6%)
Non-Squamous - Bronchioloalveolar carcinoma	1 (0.4%)	1 (0.2%)	2 (0.3%)
Non-Squamous - Large cell	2 (0.9%)	6 (1.3%)	8 (1.2%)
Non-Squamous - Sarcomatoid	0	2 (0.4%)	2 (0.3%)
Non-Squamous - Undifferentiated	2 (0.9%)	1 (0.2%)	3 (0.4%)
Non-Squamous - NA	0	1 (0.2%)	1 (0.1%)
Non-Squamous - Unknown	1 (0.4%)	0	1 (0.1%)
KRAS Mutation Status			
n	228	451	679
Positive	23 (10.1%)	56 (12.4%)	79 (11.6%)
Negative	36 (15.8%)	73 (16.2%)	109 (16.1%)
Unknown	169 (74.1%)	322 (71.4%)	491 (72.3%)
Baseline ECOG			
n	228	450	678
0	91 (39.9%)	189 (42.0%)	280 (41.3%)
1	136 (59.6%)	261 (58.0%)	397 (58.6%)
2	1 (0.4%)	0	1 (0.1%)
Intended Cycles at Induction Tx Period			
n	228	451	679
4 Cycles	119 (52.2%)	227 (50.3%)	346 (51.0%)
6 Cycles	109 (47.8%)	224 (49.7%)	333 (49.0%)
Sum of Longest Diameters (mm) at baseline			
n	228	449	677
Mean (SD)	78.90 (51.08)	84.01 (51.46)	82.29 (51.35)
Median	67.00	74.00	71.00
Min - Max	10.0 - 287.0	10.0 - 392.0	10.0 - 392.0

Atezo=Atezolizumab, CnP=Carboplatin+Nab-Paclitaxel.

EGFR mutation = Sensitizing EGFR mutations include all EGFR activating mutations in exons 18-21. ALK positive = ALK translocation.

Data Cutoff: 15MAR2018; RAVE Data Extracted: 18 May 2018.

Table 16: Baseline PD-L1 expression status (ITT-WT population)

	CnP (Randomized) (N=228)	Ateso+CnP (Randomized) (N=451)	All Patients (N=679)
TC3IC23 vs TC012IC01 from Initial Read			
n	228	451	679
TC3 or IC2/3	61 (26.8%)	121 (26.8%)	182 (26.8%)
TC0/1/2 and IC0/1	167 (73.2%)	330 (73.2%)	497 (73.2%)
TC3IC3 vs TC012IC012 per TC3IC3 Reread			
n	228	451	679
TC3 or IC3	42 (18.4%)	88 (19.5%)	130 (19.1%)
TC0/1/2 and IC0/1/2	186 (81.6%)	363 (80.5%)	549 (80.9%)
TC23IC23 vs TC01IC01 per TC2IC2 Reread			
n	228	451	679
TC2/3 or IC2/3	81 (35.5%)	158 (35.0%)	239 (35.2%)
TC0/1 and IC0/1	147 (64.5%)	293 (65.0%)	440 (64.8%)
TC123IC123 vs TC0IC0 per TC1IC1 Reread			
n	228	451	679
TC1/2/3 or IC1/2/3	107 (46.9%)	216 (47.9%)	323 (47.6%)
TC0 and IC0	121 (53.1%)	235 (52.1%)	356 (52.4%)
TC/IC 4 Incremental Subgroups			
n	228	451	679
TC3 or IC3	42 (18.4%)	88 (19.5%)	130 (19.1%)
TC2/3 or IC2/3 exclude TC3 or IC3	39 (17.1%)	70 (15.5%)	109 (16.1%)
TC1/2/3 or IC1/2/3 exclude TC2/3 or IC2/3	27 (11.8%)	63 (14.0%)	90 (13.3%)
TC0 and IC0	120 (52.6%)	230 (51.0%)	350 (51.5%)

Ateso=Atezolizumab, CnP=Carboplatin+Nab-Paclitaxel.

Wild Type patients are EGFR/ALK wild type patients who did not have an activating EGFR mutation or ALK-positive disease.

Data Cutoff: 15MAR2018; RAVE Data Extracted: 18 May 2018.

Prior Cancer Therapy

At the time of study enrollment, 10.0% of ITT-WT patients had received at least one prior anti-cancer therapy, with 9.0% in the adjuvant/neoadjuvant setting, 0.7% in other settings, and 0.3% in the metastatic setting. In the adjuvant/neoadjuvant setting, the majority of ITT-WT patients who received prior anti-cancer therapy had chemotherapy, with the most common chemotherapy treatments being cisplatin (6.3%), carboplatin (3.1%), vinorelbine (2.1%), pemetrexed disodium (1.6%), and paclitaxel (1.5%).

Prior Surgery and Radiotherapy

Overall, 49.3% of patients in the ITT-WT population had prior surgery for cancer, with other surgeries (33.4%), lobectomy (11.5%), and fine needle aspiration (8.7%) being the most commonly reported procedures. Similar findings were observed in the ITT population.

A total of 38.3% of ITT-WT patients had received prior radiotherapy, including 29.0% of patients in the metastatic setting, 6.2% in the adjuvant/neoadjuvant setting, and 5.0% in other settings. Similar findings were observed in the ITT population.

Numbers analysed

Table 17: Overview of analysis populations

	CnP (Randomized) (N=240)	Ateso+CnP (Randomized) (N=484)	All Patients (N=724)
Randomized Patients	240	484	724
Randomized Intent-to-Treat Patients	240	483	723
Safety Evaluable Patients	232	473	705
WT in Intent-to-Treat Patients	228	451	679
WT in Safety Evaluable Patients	222	442	664
PRO-Evaluable Patients for EORTC QLQ-C30 Among the Randomized Patients	184	386	570
PRO-Evaluable Patients for EORTC QLQ-C13 Among the Randomized Patients	178	369	547
PRO-Evaluable Patients for SILC Among the Randomized Patients	133	253	386
Measurable Disease at Baseline-INV Flag	227	480	717
Measurable Disease at Baseline-IRF Flag	227	467	694
ADA Evaluable	100	446	546

Ateso=Atezolizumab, CnP=Carboplatin+Nab-Paclitaxel.

ADA = anti-therapeutic antibodies; PRO = patient reported outcome; INV = Investigator; IRF = Independent Review Facility

Safety and ADA evaluable populations are summarized by actual treatment received.

All other populations are summarized by randomized treatment.

Data Cutoff: 15MAR2018; RAVE Data Extracted: 18 May 2018.

Outcomes and estimation

Co-primary endpoint: OS

Table 18: Time-to-event summary for OS (primary analysis – ITT-WT population)

	CnP (Randomized) (N=228)	Atezo+CnP (Randomized) (N=451)
Patients with event (%)	131 (57.5%)	226 (50.1%)
Earliest contributing event		
Death	131	226
Patients without event (%)	97 (42.5%)	225 (49.9%)
Time to Event (Months)		
Median	13.9	18.6
95% CI	(12.0, 18.7)	(16.0, 21.2)
25% and 75%-ile	6.1, NE	7.5, NE
Range	0.0* to 28.5*	0.0* to 32.0*
Unstratified Analysis		
p-value (log-rank)		0.0276
Hazard Ratio		0.785
95% CI		(0.633, 0.974)
Stratified Analysis		
p-value (log-rank)		0.0331
Hazard Ratio		0.791
95% CI		(0.637, 0.982)
Time Point Analysis		
1 year		
Patients remaining at risk	119	268
Event Free Rate (%)	55.53	63.13
95% CI	(48.89, 62.17)	(58.59, 67.66)
Difference in Event Free Rate		7.60
95% CI		(-0.44, 15.63)
p-value (Z-test)		0.0640
2 years		
Patients remaining at risk	13	40
Event Free Rate (%)	29.97	39.64
95% CI	(21.72, 38.23)	(33.57, 45.71)
Difference in Event Free Rate		9.67
95% CI		(-0.58, 19.91)
p-value (Z-test)		0.0644
3 years		
Patients remaining at risk	NE	NE
Event Free Rate (%)	NE	NE
95% CI	NE	NE
Difference in Event Free Rate		NE
95% CI		NE
p-value (Z-test)		NE

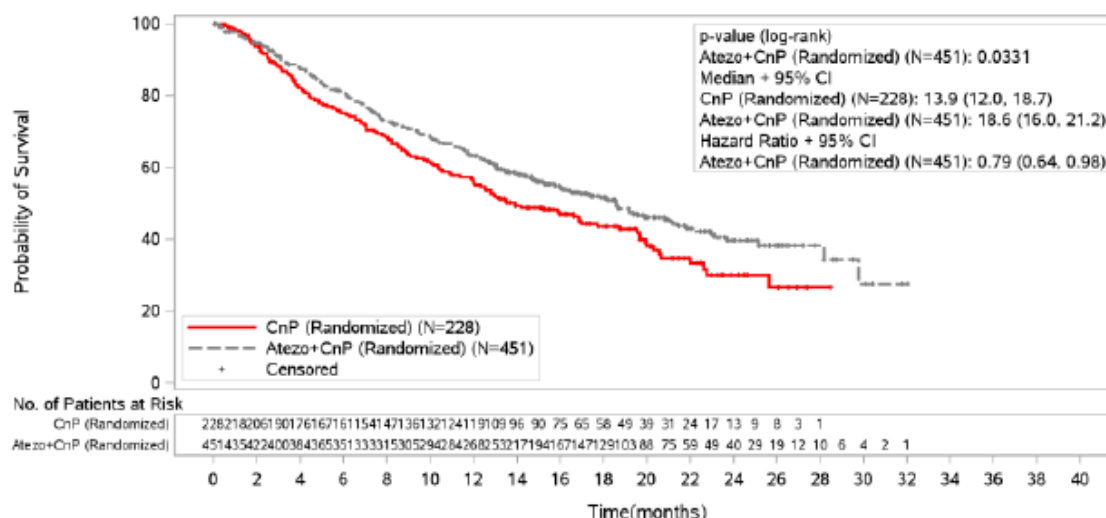
Atezo=Atezolizumab, CnP=Carboplatin+Nab-Paclitaxel.

Wild Type patients are EGFR/ALK wild type patients who did not have an activating EGFR mutation or ALK-positive disease.

* Censored, ^ Censored and event, NE = Not estimable.

Summaries of Time-to-Event (median, percentiles) were Kaplan-Meier estimates. 95% CI for median was computed using the method of Brookmeyer and Crowley. Hazard ratios were estimated by Cox regression. Stratification factors are: sex and PD-L1 tumor expression by IHC per IxRS.

Data Cutoff: 15MAR2018; RAVE Data Extracted: 18 May 2018.



Atezo=Atezolizumab, CnP=Carboplatin+Nab-Paclitaxel.

Wild Type patients are EGFR/ALK wild type patients who did not have an activating EGFR mutation or ALK-positive disease.

Randomized treatments are displayed.

Stratification factors are: sex and PD-L1 tumor expression by IHC per IxRS.

Data Cutoff: 15MAR2018; RAVE Data Extracted: 18 May 2018.

Figure 9: Kaplan-Meier plot for OS with stratified analysis (primary analysis – ITT-WT population)

Updated OS

Table 19: Time to event summary for overall survival (cut-off: 04 September 2018)

	CnP (Randomized) (N=229)	Atezo+CnP (Randomized) (N=456)
Patients with event (%)	151 (65.9%)	271 (59.4%)
Earliest contributing event		
Death	151	271
Patients without event (%)	78 (34.1%)	185 (40.6%)
Time to Event (Months)		
Median	13.5	18.1
95% CI	(11.9, 17.7)	(15.1, 20.8)
25% and 75%-ile	5.8, NE	7.3, NE
Range	0.5 to 34.4*	0.0* to 37.6*
Unstratified Analysis		
p-value (log-rank)	0.0516	
Hazard Ratio	0.821	
95% CI	(0.672, 1.002)	
Stratified Analysis		
p-value (log-rank)	0.0568	
Hazard Ratio	0.824	
95% CI	(0.675, 1.006)	

Atezo=Atezolizumab, CnP=Carboplatin+Nab-Paclitaxel.

* Censored, ^ Censored and event, NE = Not estimable.

Summaries of Time-to-Event (median, percentiles) were Kaplan-Meier estimates. 95% CI for median was computed using the method of Brookmeyer and Crowley. Hazard ratios were estimated by Cox regression. Stratification factors are: sex and PD-L1 tumor expression by IHC per IxRS.

Data Cutoff: 04SEP2018; RAVE Data Extracted: 19 Oct 2018.

Time to Event Summary for Overall Survival
 Intent-to-Treat Patients, Wild Type
 Protocol: GO29537 (Data Cutoff: 04SEP2018)

Overall Survival

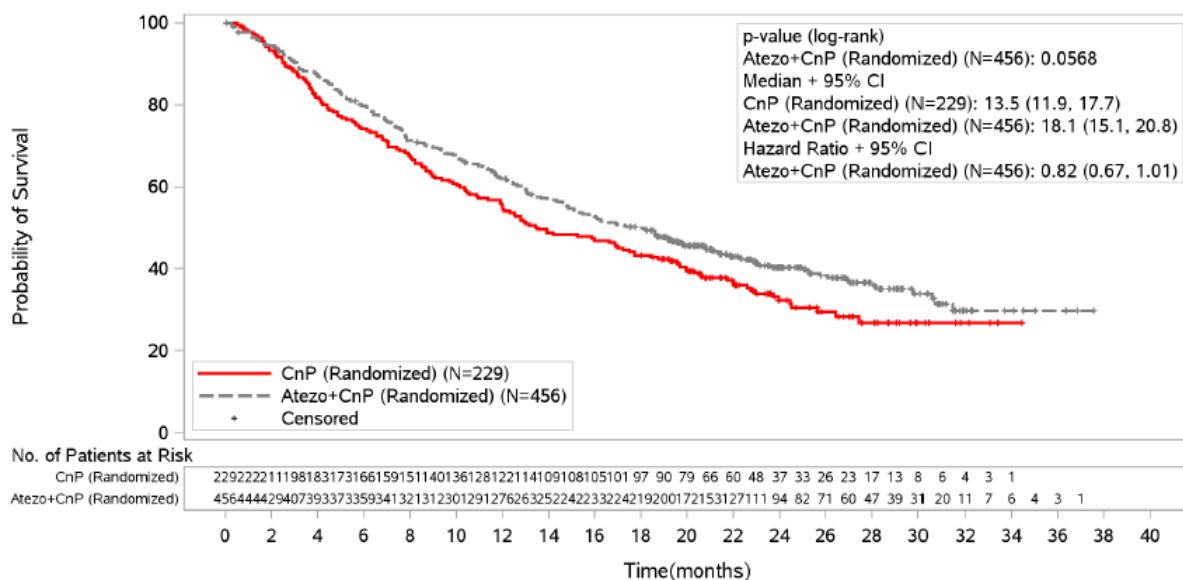
	CnP (Randomized) (N=229)	Atezo+CnP (Randomized) (N=456)
Time Point Analysis		
1 year		
Patients remaining at risk	122	276
Event Free Rate (%)	54.56	62.02
95% CI	(48.04, 61.08)	(57.53, 66.51)
Difference in Event Free Rate		7.46
95% CI		(-0.45, 15.37)
p-value (Z-test)		0.0647
2 years		
Patients remaining at risk	37	94
Event Free Rate (%)	32.36	40.43
95% CI	(25.80, 38.92)	(35.64, 45.22)
Difference in Event Free Rate		8.07
95% CI		(-0.06, 16.19)
p-value (Z-test)		0.0516
3 years		
Patients remaining at risk	NE	3
Event Free Rate (%)	NE	29.85
95% CI	NE	(23.22, 36.47)
Difference in Event Free Rate		NE
95% CI		NE
p-value (Z-test)		NE

Atezo=Atezolizumab, CnP=Carboplatin+Nab-Paclitaxel.

* Censored, * Censored and event, NE = Not estimable.

Summaries of Time-to-Event (median, percentiles) were Kaplan-Meier estimates. 95% CI for median was computed using the method of Brookmeyer and Crowley. Hazard ratios were estimated by Cox regression. Stratification factors are: sex and PD-L1 tumor expression by IHC per IxRS.

Data Cutoff: 04SEP2018; RAVE Data Extracted: 19 Oct 2018.



Atezo=Atezolizumab, CnP=Carboplatin+Nab-Paclitaxel.

Randomized treatments are displayed.

Stratification factors are: sex and PD-L1 tumor expression by IHC per IxRS.

Data Cutoff: 04SEP2018; RAVE Data Extracted: 19 Oct 2018.

Figure 10: Kaplan-Meier plot of OS with stratified analysis (ITT-WT patients)

Co-primary endpoint: PFS by INV

Table 20: Time-to-event summary for investigator-assessed PFS (primary analysis – ITT-WT population)

	CnP (Randomized) (N=228)	Atezo+CnP (Randomized) (N=451)
Patients with event (%)	198 (86.8%)	347 (76.9%)
Earliest contributing event		
Death	30	64
Disease Progression	168	283
Patients without event (%)	30 (13.2%)	104 (23.1%)
Time to Event (Months)		
Median	5.5	7.0
95% CI	(4.4, 5.9)	(6.2, 7.3)
25% and 75%-ile	2.8, 8.6	4.0, 14.2
Range	0.0 [^] to 26.3 [*]	0.0 [*] to 25.8 [*]
Unstratified Analysis		
p-value (log-rank)	<.0001	
Hazard Ratio	0.637	
95% CI	(0.534, 0.760)	
Stratified Analysis		
p-value (log-rank)	<.0001	
Hazard Ratio	0.643	
95% CI	(0.539, 0.768)	
Time Point Analysis		
6 Months		
Patients remaining at risk	90	242
Event Free Rate (%)	42.50	56.14
95% CI	(35.88, 49.12)	(51.48, 60.79)
Difference in Event Free Rate	13.64	
95% CI	(5.55, 21.73)	
p-value (Z-test)	0.0010	
1 year		
Patients remaining at risk	29	119
Event Free Rate (%)	14.06	29.14
95% CI	(9.37, 18.76)	(24.83, 33.44)
Difference in Event Free Rate	15.07	
95% CI	(8.70, 21.44)	
p-value (Z-test)	<.0001	

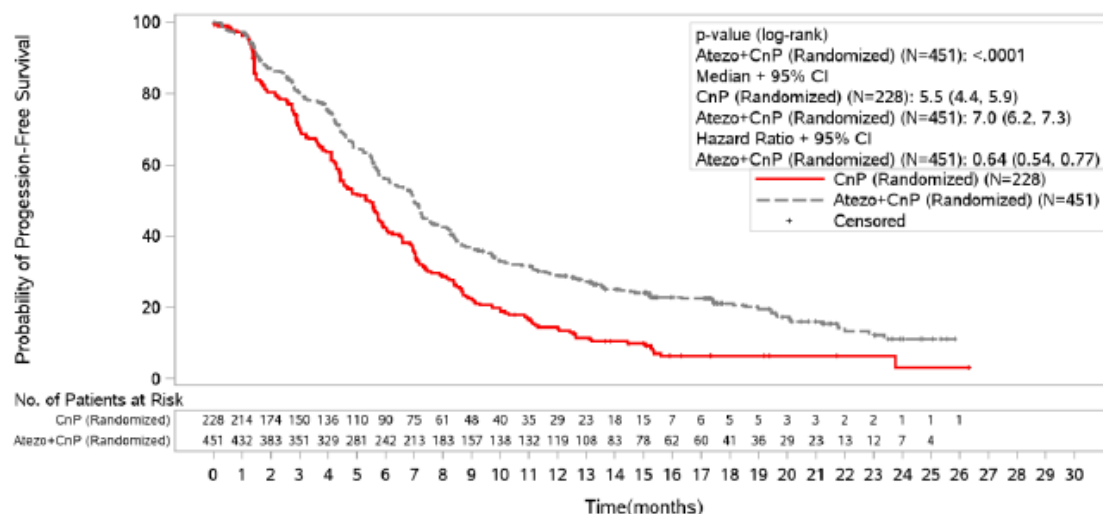
Atezo=Atezolizumab, CnP=Carboplatin+Nab-Paclitaxel.

Wild Type patients are EGFR/ALK wild type patients who did not have an activating EGFR mutation or ALK-positive disease.

^{*} Censored, [^] Censored and event, NE = Not estimable.

Summaries of Time-to-Event (median, percentiles) were Kaplan-Meier estimates. 95% CI for median was computed using the method of Brookmeyer and Crowley. Hazard ratios were estimated by Cox regression. Stratification factors are: sex and PD-L1 tumor expression by IHC per IxRS.

Data Cutoff: 15MAR2018; RAVE Data Extracted: 18 May 2018.



Atezo=Atezolizumab, CnP=Carboplatin+Nab-Paclitaxel.

Wild Type patients are EGFR/ALK wild type patients who did not have an activating EGFR mutation or ALK-positive disease.

Randomized treatments are displayed.

Stratification factors are: sex and PD-L1 tumor expression by IHC per IxRS.

Data Cutoff: 15MAR2018; RAVE Data Extracted: 18 May 2018.

Figure 11: Kaplan-meier plot for investigator-assessed PFS (primary analysis – ITT-WT population)

Censoring

Table 21: Reasons for Censoring from Primary PFS analysis for All Censored Patients (ITT-WT)

		CnP (N=228)	Atezo+CnP (N=451)
Reason for Censoring			
Absence of events at data cutoff date		21 (9.2%)	89 (19.7%)
Study discontinuation	Withdrawal by subject	6 (2.6%)	10 (2.2%)
	Other	2 (0.9%)	4 (0.9%)
	Protocol violation	1 (0.4%)	0
	Non-compliance	0	1 (0.2%)
Total number of patients censored		30 (13.2%)	104 (23.1%)

PFS=Progression free survival; ITT-WT = intent to treat wild-type.

The MAH conducted a placebo multiple imputation analysis to address potential informative censoring for those 24 patients censored due to early study discontinuation.

Table 22: Placebo multiple imputation analysis in patients censored due to early study discontinuation

PFS in ITT-WT	HR (95% CI)	p-value
Primary	0.643 (0.539, 0.768)	<0.0001
Placebo Multiple Imputation	0.650 (0.545, 0.776)	<0.0001*

HR = hazard ratio; ITT-WT = intent to treat wild-type; PFS = progression-free survival.

*p-value of sensitivity analysis is for descriptive purpose only.

Secondary endpoints

Secondary PFS analyses

Table 23: Time-to-event summary for investigator-assessed PFS (ITT, PD-L1 expression, and PD-L1 expression WT populations)

	CnP	Atezo+CnP
ITT	n=240	n=483
Patients with event (%)	209 (87.1%)	375 (77.6%)
Median PFS (95% CI), months	5.6 (4.5, 5.9)	7.0 (6.3, 7.3)
Stratified hazard ratio (95% CI)	0.65 (0.54, 0.77)	
p-value (log-rank)	<0.0001	
TC1/2/3 or IC1/2/3	n=111	n=230
Patients with event (%)	97 (87.4%)	168 (73.0%)
Median PFS (95% CI), months	5.7 (4.5, 6.6)	7.5 (7.0, 9.1)
Unstratified hazard ratio (95% CI)	0.55 (0.43, 0.71)	
TC1/2/3 or IC1/2/3-WT	n=107	n=216
Patients with event (%)	93 (86.9%)	157 (72.7%)
Median PFS (95% CI), months	5.9 (4.5, 6.6)	7.5 (7.0, 9.0)
Unstratified hazard ratio (95% CI)	0.56 (0.43, 0.73)	
Exploratory Subgroups		
TC3 or IC3-WT	n=42	n=88
Patients with event (%)	41 (97.6%)	61 (69.3%)
Median PFS, months	4.6	6.4
Unstratified hazard ratio (95% CI)	0.51 (0.34, 0.77)	
TC2/3 or IC2/3-WT	n=81	n=158
Patients with event (%)	72 (88.9%)	109 (69.0%)
Median PFS, months	5.9	7.1
Unstratified hazard ratio (95% CI)	0.54 (0.40, 0.73)	
TC1/2 or IC1/2-WT	n=65	n=128
Patients with event (%)	52 (80.0%)	96 (75.0%)
Median PFS, months	6.0	8.3
Unstratified hazard ratio (95% CI)	0.61 (0.43, 0.85)	
TC0/1/2 and IC0/1/2-WT	n=186	n=363
Patients with event (%)	157 (84.4%)	286 (78.8%)
Median PFS, months	5.5	7.0
Unstratified hazard ratio (95% CI)	0.67 (0.55, 0.82)	
TC0 and IC0-WT	n=121	n=235
Patients with event (%)	105 (86.8%)	190 (80.9%)
Median PFS, months	4.7	6.2
Unstratified hazard ratio (95% CI)	0.72 (0.56, 0.91)	

Atezo=atezolizumab; CnP=carboplatin+nab-paclitaxel; IC=tumor-infiltrating immune cell; ITT=intent-to-treat; PD-L1=programmed death–ligand 1; PFS=progression-free survival; TC=tumor cell; WT=wild type.

Secondary OS analyses

Table 24: Time-to-event summary for OS (ITT, PD-L1 expression, and PD-L1 expression WT populations)

	CnP	Atezo+CnP
ITT	n=240	n=483
Patients with event, %	138 (57.5%)	246 (50.9%)
Median OS (95% CI), months	13.9 (12.0, 18.2)	18.1 (15.3, 20.8)
Stratified hazard ratio (95% CI)	0.80 (0.65, 0.99)	
p-value (log-rank)	0.0393	
TC1/2/3 or IC1/2/3	n=111	n=230
Patients with event, %	58 (52.3%)	104 (45.2%)
Median OS (95% CI), months	16.9 (12.5, 22.0)	21.2 (17.3, 28.2)
Unstratified hazard ratio (95% CI)	0.75 (0.55, 1.04)	
TC1/2/3 or IC1/2/3-WT	n=107	n=216
Patients with event, %	56 (52.3%)	97 (44.9%)
Median OS (95% CI), months	16.9 (12.5, 22.0)	21.2 (18.1, 28.2)
Unstratified hazard ratio (95% CI)	0.75 (0.54, 1.04)	
Exploratory Subgroups		
TC3 or IC3-WT	n=42	n=88
Patients with event, %	23 (54.8%)	43 (48.9%)
Median OS, months	16.9	17.3
Unstratified hazard ratio (95% CI)	0.84 (0.51, 1.39)	
TC2/3 or IC2/3-WT	n=81	n=158
Patients with event, %	42 (51.9%)	75 (47.5%)
Median OS, months	16.9	18.6
Unstratified hazard ratio (95% CI)	0.81 (0.55, 1.18)	
TC1/2 or IC1/2-WT	n=65	n=128
Patients with event, %	33 (50.8%)	54 (42.2%)
Median OS, months	15.9	23.7
Unstratified hazard ratio (95% CI)	0.70 (0.45, 1.08)	
TC0/1/2 and IC0/1/2-WT	n=186	n=363
Patients with event, %	108 (58.1%)	183 (50.4%)
Median OS, months	13.4	18.6
Unstratified hazard ratio (95% CI)	0.78 (0.61, 0.99)	
TC0 and IC0-WT	n=121	n=235
Patients with event (%)	75 (62.0%)	129 (54.9%)
Median OS, months	12.0	15.2
Unstratified hazard ratio (95% CI)	0.81 (0.61, 1.08)	

Atezo=atezolizumab; CnP=carboplatin+nab-paclitaxel; IC=tumor-infiltrating immune cell; ITT=intent to treat; OS=overall survival; PD-L1=programmed death–ligand 1; TC=tumor cell; WT=wild type.

Secondary endpoint: Landmark Overall Survival Rates at the 1- and 2-Year Time points

OS rates in the ITT-WT population at landmark timepoints were higher in the Atezo-CnP arm compared with the CnP arm at 1 year (63.1% vs. 55.5%) and 2 years (39.6% vs. 30.0%) after randomization. The results of this analysis were consistent with those observed in the ITT population.

Secondary endpoint: Objective response rate (ORR; confirmation not required)

Table 25: Summary of unconfirmed objective response rate by investigator (ITT-WT population)

	CnP (Randomized) (N=226)		Atezo+CnP (Randomized) (N=447)
Responders	93 (41.2%)		267 (59.7%)
Non-Responders	133 (58.8%)		180 (40.3%)
95% CI for Response Rate (Clopper-Pearson)	(34.67, 47.87)		(55.02, 64.31)
Difference in Response Rates		18.58	
95% CI for Difference in Response Rates (Wald with Continuity Correction)		(10.38, 26.78)	
p-Value* (Cochran-Mantel-Haenszel)		<.0001	
Odds Ratio*		2.13	
95% CI for Odds Ratio*		(1.53, 2.94)	
Complete Response (CR)	4 (1.8%)		13 (2.9%)
95% CI	(0.48, 4.47)		(1.56, 4.92)
Partial Response (PR)	89 (39.4%)		254 (56.8%)
95% CI	(32.97, 46.08)		(52.09, 61.47)
Stable Disease (SD)	67 (29.6%)		91 (20.4%)
95% CI	(23.77, 36.06)		(16.72, 24.39)
Progressive Disease (PD)	40 (17.7%)		48 (10.7%)
95% CI	(12.96, 23.31)		(8.02, 13.98)
Non-CR / Non-PD	0		0
95% CI	(0.00, 1.62)		(0.00, 0.82)
Missing or unevaluable	26 (11.5%)		41 (9.2%)

Atezo=Atezolizumab, CnP=Carboplatin+Nab-Paclitaxel.

Wild Type patients are EGFR/ALK wild type patients who did not have an activating EGFR mutation or ALK-positive disease.

BL = Baseline. * Stratified by: sex and PD-L1 tumor expression by IHC per IxRS.

Wald is the normal approximation.

Patients were classified as missing or unevaluable when no post-baseline response assessments were available or all post-baseline response assessments were unevaluable.

Data Cutoff: 15MAR2018; RAVE Data Extracted: 18 May 2018.

Table 26: Summary of unconfirmed objective response rate by investigator (ITT population)

	CnP (Randomized) (N=237)		Atezo+CnP (Randomized) (N=479)
Responders	100 (42.2%)		283 (59.1%)
Non-Responders	137 (57.8%)		196 (40.9%)
95% CI for Response Rate (Clopper-Pearson)	(35.83, 48.76)		(54.53, 63.52)
Difference in Response Rates		16.89	
95% CI for Difference in Response Rates (Wald with Continuity Correction)		(8.90, 24.88)	
p-Value* (Cochran-Mantel-Haenszel)		<.0001	
Odds Ratio*		1.98	
95% CI for Odds Ratio*		(1.44, 2.71)	
Complete Response (CR)	4 (1.7%)		13 (2.7%)
95% CI	(0.46, 4.26)		(1.45, 4.60)
Partial Response (PR)	96 (40.5%)		270 (56.4%)
95% CI	(34.20, 47.05)		(51.79, 60.86)
Stable Disease (SD)	69 (29.1%)		102 (21.3%)
95% CI	(23.41, 35.35)		(17.71, 25.24)
Progressive Disease (PD)	41 (17.3%)		52 (10.9%)
95% CI	(12.71, 22.73)		(8.21, 13.99)
Non-CR / Non-PD	0		0
95% CI	(0.00, 1.54)		(0.00, 0.77)
Missing or unevaluable	27 (11.4%)		42 (8.8%)

Atezo=Atezolizumab, CnP=Carboplatin+Nab-Paclitaxel.

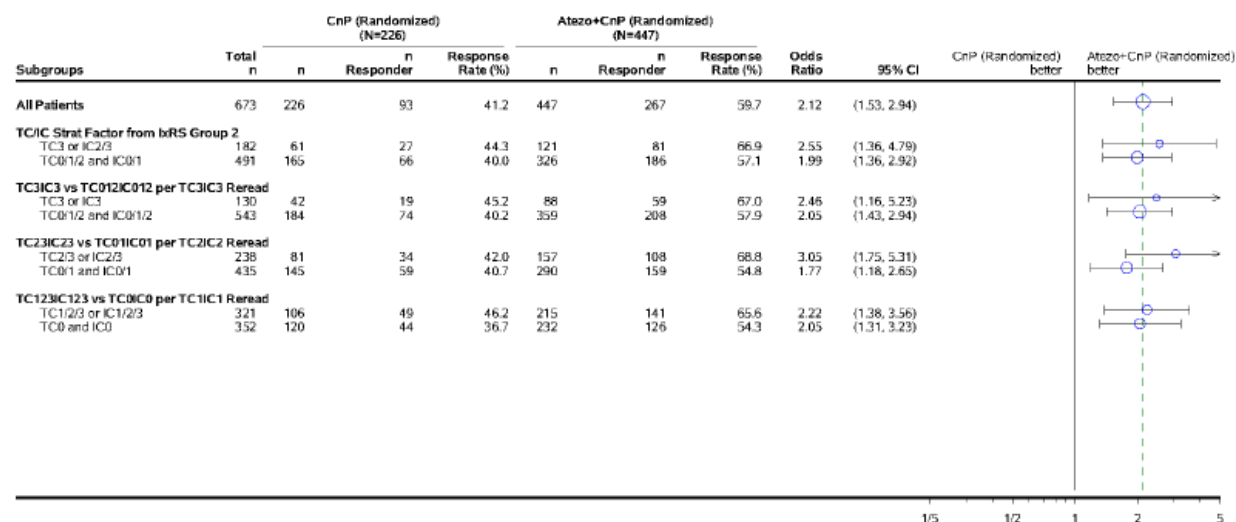
BL = Baseline. * Stratified by: sex and PD-L1 tumor expression by IHC per IxRS.

Wald is the normal approximation.

Patients were classified as missing or unevaluable when no post-baseline response assessments were available or all post-baseline response assessments were unevaluable.

Data Cutoff: 15MAR2018; RAVE Data Extracted: 18 May 2018.

Forest plot - Subgroup Analysis of Unconfirmed Best Overall Response per RECIST v1.1 - Investigator Measurable Disease at BL-Investigator in Intent-to-Treat Patients, Wild Type
Protocol: GO29537 (Data Cutoff: 15MAR2018)



Atezo=Atezolizumab, CnP=Carboplatin+Nab-Paclitaxel.

Wild Type patients are EGFR/ALK wild type patients who did not have an activating EGFR mutation or ALK-positive disease.

NE = Not estimable. BL = Baseline. Odds ratios relative to CnP and Wald confidence interval were estimated using unstratified logistic regression. The vertical dashed line indicates the odds ratio for all patients. The diameter of the circle is proportional to the square root of the total number of events. Randomized treatments are displayed.

Data Cutoff: 15MAR2018; RAVE Data Extracted: 18 May 2018.

Figure 12: Subgroup analysis of ORR by PD-L1 expression level (ITT-WT population)

Secondary endpoint: Duration of response (DOR; confirmation not required)

Table 27: Summary of unconfirmed duration of response (ITT-WT population, patients with unconfirmed response assessed by investigator per RECIST v1.1)*

	CnP (Randomized) (N=93)	Atezo+CnP (Randomized) (N=267)
Patients with event (%)	79 (84.9%)	183 (68.5%)
Earliest contributing event		
Death	6	17
Disease Progression	73	166
Patients without event (%)	14 (15.1%)	84 (31.5%)
Time to Event (Months)		
Median	5.4	6.7
95% CI	(3.9, 5.8)	(5.6, 8.0)
25% and 75%-ile	2.8, 9.1	3.4, 17.9
Range	0.3 to 17.8*	0.0* to 24.3*
Unstratified Analysis		
p-value (log-rank)	0.0002	
Hazard Ratio	0.600	
95% CI	(0.458, 0.785)	
Stratified Analysis		
p-value (log-rank)	0.0002	
Hazard Ratio	0.599	
95% CI	(0.457, 0.786)	

Atezo=Atezolizumab, CnP=Carboplatin+Nab-Paclitaxel.

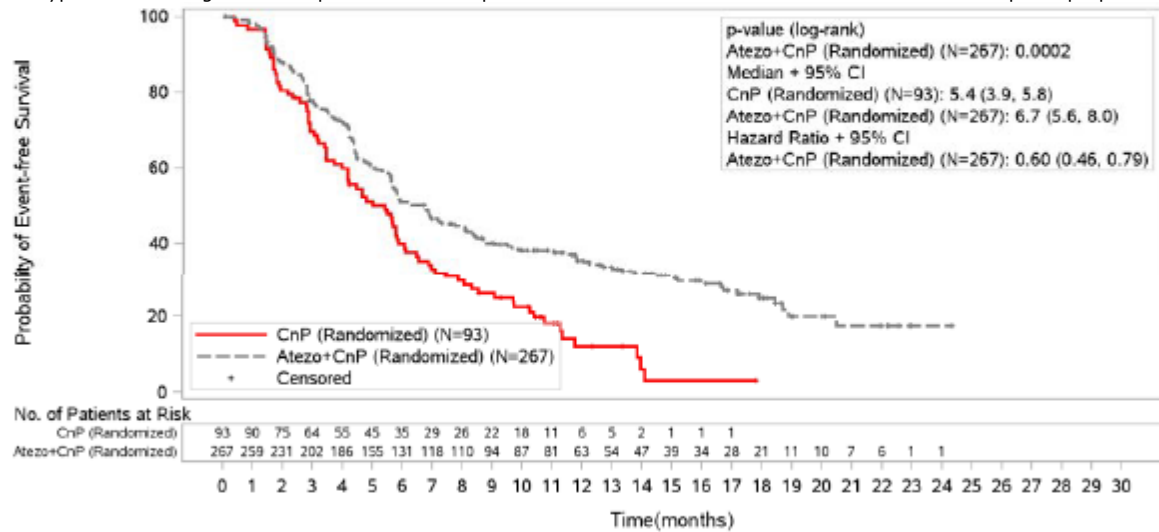
Wild Type patients are EGFR/ALK wild type patients who did not have an activating EGFR mutation or ALK-positive disease.

* Censored, ^ Censored and event, NE = Not estimable.

Summaries of Time-to-Event (median, percentiles) were Kaplan-Meier estimates. 95% CI for median was computed using the method of Brookmeyer and Crowley. Hazard ratios were estimated by Cox regression. Stratification factors are: sex and PD-L1 tumor expression by IHC per IxRS.

Data Cutoff: 15MAR2018; RAVE Data Extracted: 18 May 2018.

*Formal hypothesis testing was not performed. The p values and hazard ration estimates are for descriptive purpose only.



Atezo=Atezolizumab, CnP=Carboplatin+Nab-Paclitaxel.

Wild Type patients are EGFR/ALK wild type patients who did not have an activating EGFR mutation or ALK-positive disease.

Randomized treatments are displayed.

Data Cutoff: 15MAR2018; RAVE Data Extracted: 18 May 2018.

Figure 13: Kaplan-meier plot for unconfirmed duration of response (ITT-WT population, patients with unconfirmed response assessed by investigator per RECIST v1.1)

Table 28: Summary of unconfirmed duration of response (ITT population, patients with unconfirmed response assessed by investigator per RECIST v1.1)*

	CnP (Randomized) (N=100)	Atezo+CnP (Randomized) (N=283)
Patients with event (%)	85 (85.0%)	196 (69.3%)
Earliest contributing event		
Death	7	19
Disease Progression	78	177
Patients without event (%)	15 (15.0%)	87 (30.7%)
Time to Event (Months)		
Median	5.4	6.2
95% CI	(4.1, 5.8)	(5.6, 7.9)
25% and 75%-ile	2.8, 9.7	3.4, 17.9
Range	0.3 to 17.8*	0.0* to 26.0*
Unstratified Analysis		
p-value (log-rank)		0.0002
Hazard Ratio		0.614
95% CI		(0.473, 0.797)
Stratified Analysis		
p-value (log-rank)		0.0002
Hazard Ratio		0.614
95% CI		(0.473, 0.797)
Time Point Analysis		
6 Months		
Patients remaining at risk	38	138
Event Free Rate (%)	40.03	50.52
95% CI	(30.30, 49.75)	(44.62, 56.42)
Difference in Event Free Rate		10.49
95% CI		(-0.89, 21.87)
p-value (Z-test)		0.0707
1 year		
Patients remaining at risk	7	68
Event Free Rate (%)	12.13	35.25
95% CI	(4.65, 19.61)	(29.48, 41.03)
Difference in Event Free Rate		23.12
95% CI		(13.67, 32.57)
p-value (Z-test)		<.0001

Atezo=Atezolizumab, CnP=Carboplatin+Nab-Paclitaxel.

* Censored, ^ Censored and event, NE = Not estimable.

Summaries of Time-to-Event (median, percentiles) were Kaplan-Meier estimates. 95% CI for median was computed using the method of Brookmeyer and Crowley. Hazard ratios were estimated by Cox regression. Stratification factors are: sex and PD-L1 tumor expression by IHC per IxRS. Data Cutoff: 15MAR2018; RAVE Data Extracted: 18 May 2018.

*Formal hypothesis testing was not performed. The p values and hazard ration estimates are for descriptive purpose only.

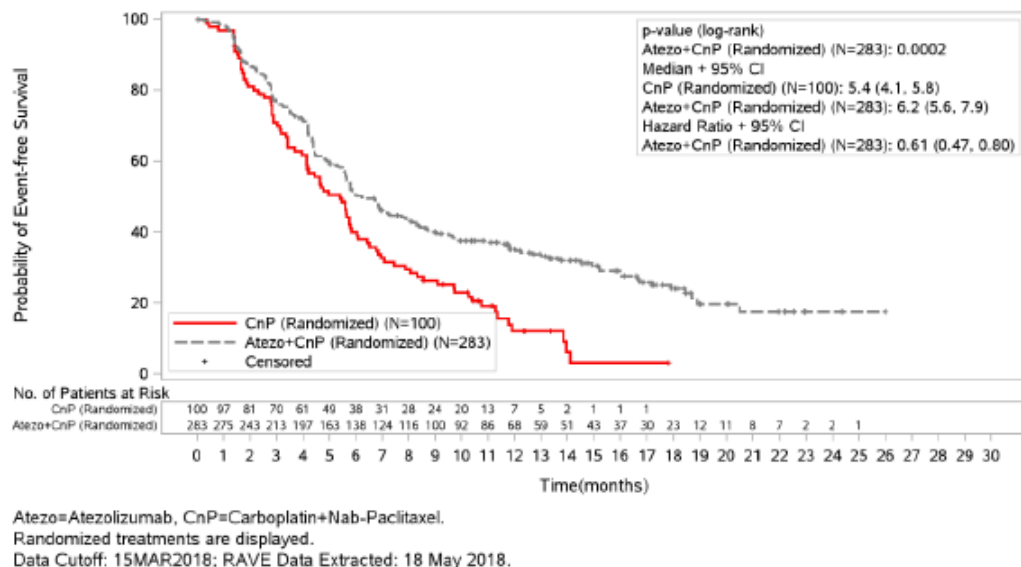


Figure 14: Kaplan-meier plot for unconfirmed duration of response among patients with unconfirmed response assessed by investigator per RECIST v1.1 (ITT population)

Secondary endpoint: Time to deterioration (TTD) in patient-reported lung cancer symptoms per EORTC questionnaires

At baseline (i.e., randomization/C1D1), before initiating study treatment, patients in both arms in the PRO-evaluable WT population had similar mean (SD) PRO scores in lung-related symptoms of cough (Atezo-CnP 40.76 [29.54] vs. CnP 41.76 [26.91]), chest pain (20.14 [27.86] vs. 18.82 [26.37]), and dyspnoea (27.66 [24.78] vs. 29.67 [23.37]) on the EORTC QLQ-LC13, based on a 0-100 scale, with 100 being the most severe, indicating that most patients reported low symptom burden at baseline with the exception of cough.

Delayed deterioration of lung-related symptoms (cough, chest pain, or dyspnoea) was observed in the ITT-WT population favoring Atezo-CnP (median 2.2 months) over CnP (median 1.9 months; HR=0.89; range: 0.71-1.12). A confirmed deterioration in symptoms or an initial deterioration in symptoms followed by death was documented in 71.6% versus 66.5% patients receiving Atezo-CnP versus CnP, respectively. The event-free rate at 6 months was higher in the Atezo-CnP arm (34.2%) versus the CnP arm (30.1%) and remained higher in the Atezo-CnP arm (26.1%) versus the CnP arm (17.6%) at 1 year.

Secondary endpoint: Change from baseline in patient-reported lung cancer symptoms using the Symptoms in Lung Cancer (SILC) scale

At baseline (i.e., randomization/C1D1), before initiating study treatment, patients in both arms in the PRO-evaluable WT population had similar mean (SD) PRO scores in lung-related symptoms of chest pain (Atezo-CnP 1.05 [1.08] vs. CnP 0.93 [1.03]), cough (1.48 [1.03] vs. 1.55 [0.99]), and dyspnoea (1.24 [1.02] vs. 1.16 [0.99]) on the SILC, based on a 0-4 scale, with 4 being the most severe, confirming minimal symptom burden at baseline.

Cough, chest pain, and dyspnoea scores were less severe throughout the study for patients receiving Atezo-CnP, with larger differences seen for dyspnoea followed by chest pain and minimal differences between arms for cough. Cough and chest pain were the first symptoms to show improvement as early as Week 3 following a first report of deterioration after treatment initiation. Dyspnoea scores showed clinically meaningful deterioration (mean change in score >0.3) between Weeks 8 and 14 in patients receiving Atezo-CnP and between Weeks 2 and 24 in patients receiving CnP; however, patients in the Atezo-CnP arm reported lower mean severity scores compared with patients receiving CnP when dyspnoea was at its worst (1.61 [1.08] at Week 11 vs. 1.75 [1.13] at Week 13, respectively).

Secondary endpoint: Proportion of patients improved, remained stable, worsened in patients-reported lung cancer symptoms, HRQoL and functioning

At baseline (i.e., randomization/C1D1), before initiating study treatment, patients in both arms in the PRO-evaluable WT population had similar mean (SD) PRO scores in HRQoL and functioning as measured by global health status (60.34 [23.92] vs. 57.43 [22.45]), physical functioning (72.81 [23.23] vs. 71.20 [22.01]), and role functioning (64.85 [32.30] vs. 62.19 [31.57]) on the EORTC QLQ-C30, based on a 0-100 scale, with 100 being the best HRQoL/functioning, indicating moderate-to-high HRQoL and functioning at baseline.

Patients in both arms reported no clinically meaningful changes (defined as a ≥ 10 point difference in 0-100 score) for HRQoL, physical functioning, and role functioning. Patients in both arms reported a worsening during the weeks of induction followed by a rebound towards baseline scores during the maintenance phase, following induction. The proportion of patients who improved, remained stable, or worsened on these domains were relatively similar (within approximately 10%) for patients in both arms.

At baseline (i.e., randomization/C1D1), before initiating study treatment, patients in both arms had similar mean (SD) PRO scores in lung-related symptoms as measured by fatigue (39.53 [27.77] vs. 42.03 [24.78]), diarrhoea (5.79 [14.27] vs. 7.81 [18.81]), constipation (20.07 [28.90] vs. 19.05 [27.08]), nausea/vomiting (8.82 [17.69] vs. 10.38 [20.42]), and sore mouth (3.71 [13.98] vs. 3.73 [13.78]) on the EORTC QLQ-C30 and EORTC QLQ-LC13, based on a 0-100 scale, with 100 being the most severe, indicating minimal symptom burden at baseline, with the exception of fatigue. There were no clinically meaningful changes detected from baseline in any of the common treatment-related symptoms assessed by the EORTC QLQ-C30 and EORTC QLQ-LC13 (i.e., fatigue, diarrhoea, constipation, nausea/vomiting, and sore mouth) in either treatment arm. Although patients in both arms reported worsening in symptoms during the induction phase, changes from baseline in symptoms were nominal in both arms during the maintenance phase. All treatment-related symptoms returned to baseline following the completion of platinum-based chemotherapy.

Exploratory endpoints: Confirmed ORR

Table 29: Confirmed Best Overall Response per RECIST v1.1 – ITT-WT

	CnP (Randomized) (N=226)	Atezo+CnP (Randomized) (N=447)
Responders	72 (31.9%)	220 (49.2%)
Non-Responders	154 (68.1%)	227 (50.8%)
95% CI for Response Rate (Clopper-Pearson)	(25.84, 38.36)	(44.49, 53.96)
Difference in Response Rates	17.36	
95% CI for Difference in Response Rates (Wald with Continuity Correction)	(9.38, 25.33)	
p-Value* (Cochran-Mantel-Haenszel)	<.0001	
Odds Ratio*	2.07	
95% CI for Odds Ratio*	(1.48, 2.89)	
Complete Response (CR)	3 (1.3%)	11 (2.5%)
95% CI	(0.27, 3.83)	(1.23, 4.36)
Partial Response (PR)	69 (30.5%)	209 (46.8%)
95% CI	(24.60, 36.98)	(42.05, 51.50)
Stable Disease (SD)	86 (38.1%)	136 (30.4%)
95% CI	(31.70, 44.73)	(26.19, 34.92)
Progressive Disease (PD)	41 (18.1%)	49 (11.0%)
95% CI	(13.34, 23.80)	(8.22, 14.23)
Non-CR / Non-PD	0	0
95% CI	(0.00, 1.62)	(0.00, 0.82)
Missing or unevaluable	27 (11.9%)	42 (9.4%)

Atezo=Atezolizumab, CnP=Carboplatin+Nab-Paclitaxel.

Wild Type patients are EGFR/ALK wild type patients who did not have an activating EGFR mutation or ALK-positive disease.

BL = Baseline. * Stratified by: sex and PD-L1 tumor expression by IHC per IxRS.

Wald is the normal approximation.

Patients were classified as missing or unevaluable when no post-baseline response assessments were available or all post-baseline response assessments were unevaluable.

Data Cutoff: 15MAR2018; RAVE Data Extracted: 18 May 2018.

Exploratory endpoints: Confirmed DOR

Table 30: Time to Event Summary for Duration of confirmed Response per RECIST v1.1 – ITT-WT*

	CnP (Randomized) (N=72)	Atezo+CnP (Randomized) (N=220)
Patients with event (%)	58 (80.6%)	139 (63.2%)
Earliest contributing event		
Death	3	8
Disease Progression	55	131
Patients without event (%)	14 (19.4%)	81 (36.8%)
Time to Event (Months)		
Median	6.1	8.4
95% CI	(5.5, 7.9)	(6.9, 11.8)
25% and 75%-ile	4.1, 10.7	4.4, 18.9
Range	2.4* to 17.8*	1.4* to 24.3*
Unstratified Analysis		
p-value (log-rank)		0.0004
Hazard Ratio		0.573
95% CI		(0.418, 0.784)
Stratified Analysis		
p-value (log-rank)		0.0004
Hazard Ratio		0.569
95% CI		(0.415, 0.781)

Atezo=Atezolizumab, CnP=Carboplatin+Nab-Paclitaxel.

Wild Type patients are EGFR/ALK wild type patients who did not have an activating EGFR mutation or ALK-positive disease.

* Censored, ^ Censored and event, NE = Not estimable.

Summaries of Time-to-Event (median, percentiles) were Kaplan-Meier estimates. 95% CI for median was computed using the method of Brookmeyer and Crowley. Hazard ratios were estimated by Cox regression. Stratification factors are: sex and PD-L1 tumor expression by IHC per IxRS.

Data Cutoff: 15MAR2018; RAVE Data Extracted: 18 May 2018.

*Formal hypothesis testing was not performed. The p values and hazard ration estimates are for descriptive purpose only.

Ancillary analyses

Time to event summary for patients who received pemetrexed in the CnP arm in the WT population

Overall, 49 patients (21%) in the CnP arm received at least 1 cycle of pemetrexed; in the WT population, 45 patients (20%) received at least 1 cycle of pemetrexed

Table 31: Time to Event Summary for Investigator Assessed PFS (ITT-WT Population, Assigned CnP and Pemetrexed Treated)

Earliest Contributing Event to Investigator Assessed PFS

	CnP (Randomized) (N=45)
Patients with event (%)	38 (84.4%)
Earliest contributing event	
Death	2
Disease Progression	36
Patients without event (%)	7 (15.6%)
Time to Event (Months)	
Median	8.4
95% CI	(7.2, 11.1)
25% and 75%-ile	7.0, 14.4
Range	4.1 to 26.3*
Time Point Analysis	
6 Months	
Patients remaining at risk	37
Event Free Rate (%)	84.33
95% CI	(73.67, 95.00)
1 year	
Patients remaining at risk	15
Event Free Rate (%)	34.19
95% CI	(20.16, 48.21)

Wild Type patients are EGFR/ALK wild type patients who did not have an activating EGFR mutation or ALK-positive disease.

Atezo=Atezolizumab, CnP=Carboplatin+Nab-Paclitaxel.

* Censored, ^ Censored and event, NE = Not estimable.

Summaries of Time-to-Event (median, percentiles) were Kaplan-Meier estimates. 95% CI for median was computed using the method of Brookmeyer and Crowley.

Data Cutoff: 15MAR2018; RAVE Data Extracted: 18 May 2018.

Table 32: Time to Event Summary for Overall Survival (ITT-WT Population, Assigned CnP and Pemetrexed Treated)

	CnP (Randomized) (N=45)
Patients with event (%)	20 (44.4%)
Earliest contributing event	
Death	20
Patients without event (%)	25 (55.6%)
Time to Event (Months)	
Median	19.9
95% CI	(16.9, NE)
25% and 75%-ile	13.9, NE
Range	5.7 to 27.4*
Time Point Analysis	
1 year	
Patients remaining at risk	36
Event Free Rate (%)	80.00
95% CI	(68.31, 91.69)
2 years	
Patients remaining at risk	5
Event Free Rate (%)	48.09
95% CI	(31.35, 64.82)

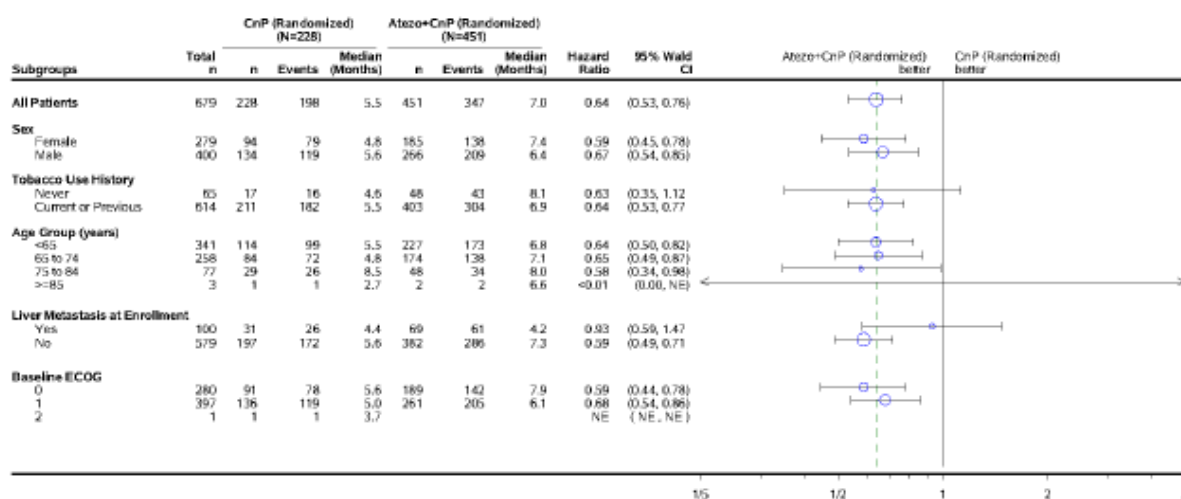
Wild Type patients are EGFR/ALK wild type patients who did not have an activating EGFR mutation or ALK-positive disease. Atezo=Atezolizumab, CnP=Carboplatin+Nab-Paclitaxel.

* Censored, ^ Censored and event, NE = Not estimable.

Summaries of Time-to-Event (median, percentiles) were Kaplan-Meier estimates. 95% CI for median was computed using the method of Brookmeyer and Crowley.

Data Cutoff: 15MAR2018; RAVE Data Extracted: 18 May 2018.

Subgroup analysis for PFS:



Atezo=Atezolizumab, CnP=Carboplatin+Nab-Paclitaxel.

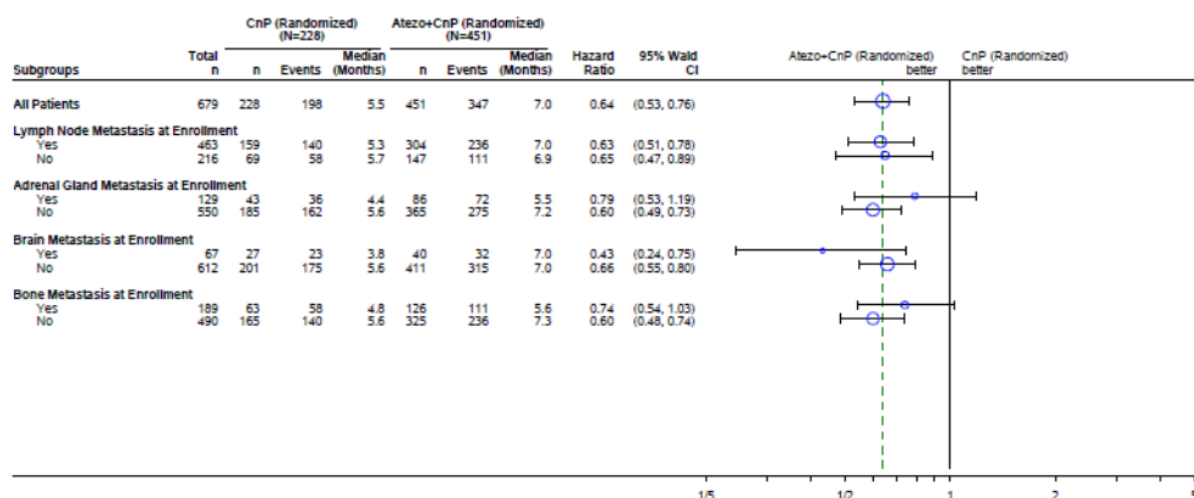
Wild Type patients are EGFR/ALK wild type patients who did not have an activating EGFR mutation or ALK-positive disease.

NE = Not estimable; Median progression-free survival was estimated from Kaplan-Meier method. Unstratified hazard ratio relative to CnP and 95% CI for the hazard ratio were estimated using Cox regression. The vertical dashed line indicates the hazard ratio for all patients. The diameter of the circle is proportional to the square root of the total number of events.

Randomized treatments are displayed.

Data Cutoff: 15MAR2018; RAVE Data Extracted: 18 May 2018.

Figure 15: Subgroup analysis of investigator-assessed PFS by selected demographics and baseline disease characteristics (ITT-WT population)



Atezo=Atezolizumab, CnP=Carboplatin+Nab-Paclitaxel.

Wild Type patients are EGFR/ALK wild type patients who did not have an activating EGFR mutation or ALK-positive disease.

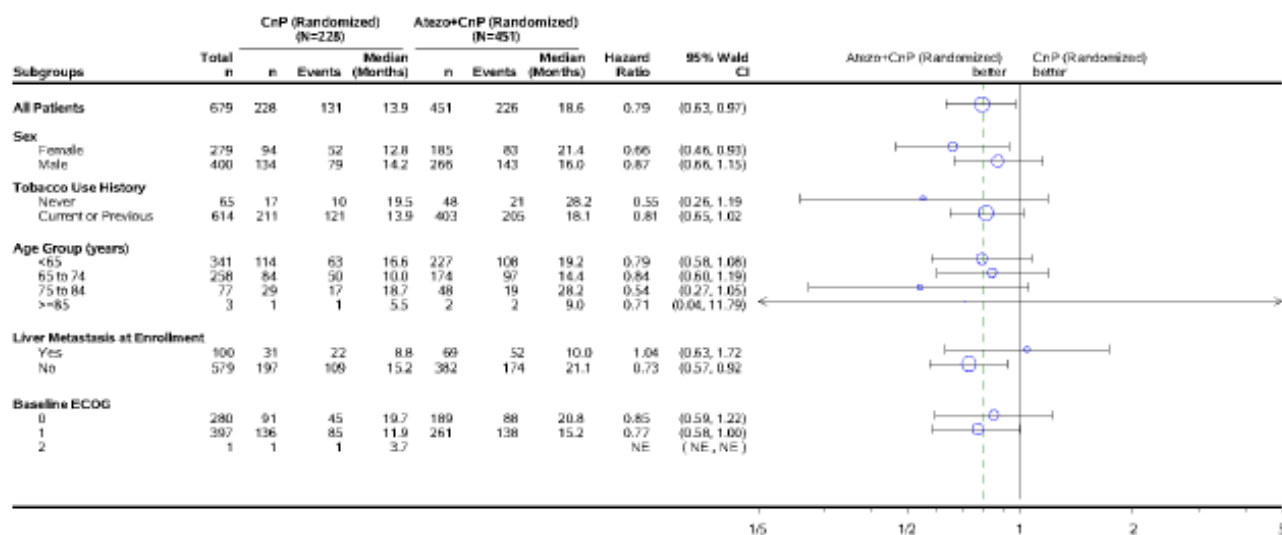
NE = Not estimable; Median progression-free survival was estimated from Kaplan-Meier method. Unstratified hazard ratio relative to CnP and 95% CI for the hazard ratio were estimated using Cox regression. The vertical dashed line indicates the hazard ratio for all patients. The diameter of the circle is proportional to the square root of the total number of events.

Randomized treatments are displayed.

Data Cutoff: 15MAR2018; RAVE Data Extracted: 18 May 2018.

Figure 16: Forest Plot – Subgroup Analysis of Investigator-Assessed PFS (ITT-WT patients)

Subgroup analysis for OS:



Atezo=Atezolizumab, CnP=Carboplatin+Nab-Paclitaxel.

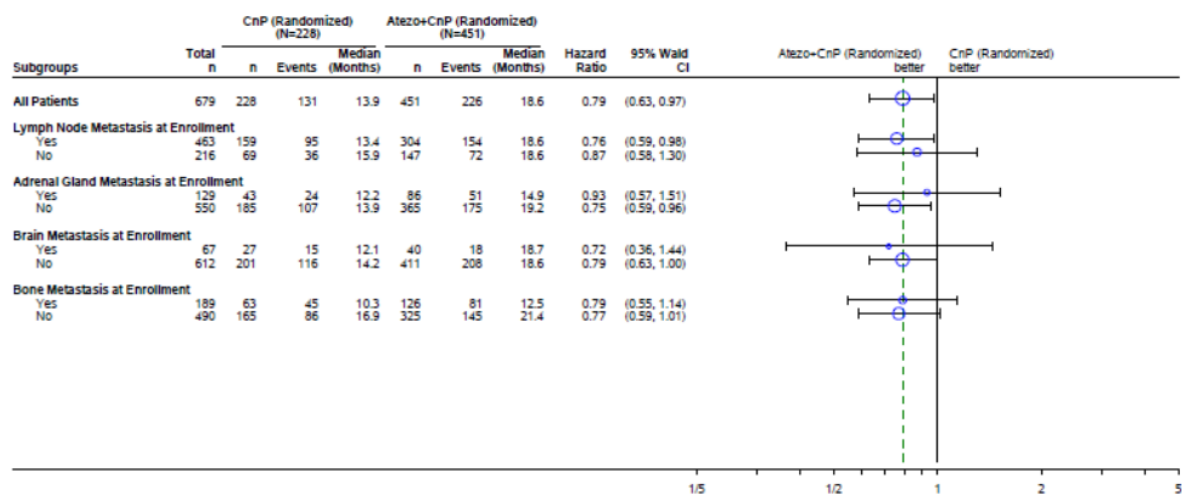
Wild Type patients are EGFR/ALK wild type patients who did not have an activating EGFR mutation or ALK-positive disease.

NE = Not estimable; Median survival was estimated from Kaplan-Meier method. Unstratified hazard ratio relative to CnP and 95% CI for the hazard ratio were estimated using Cox regression. The vertical dashed line indicates the hazard ratio for all patients. The diameter of the circle is proportional to the square root of the total number of events.

Randomized treatments are displayed.

Data Cutoff: 15MAR2018; RAVE Data Extracted: 18 May 2018.

Figure 17: Subgroup analysis of overall survival by selected demographics and baseline characteristics (ITT-WT population)



Atezo=Atezolizumab, CnP=Carboplatin+Nab-Paclitaxel.

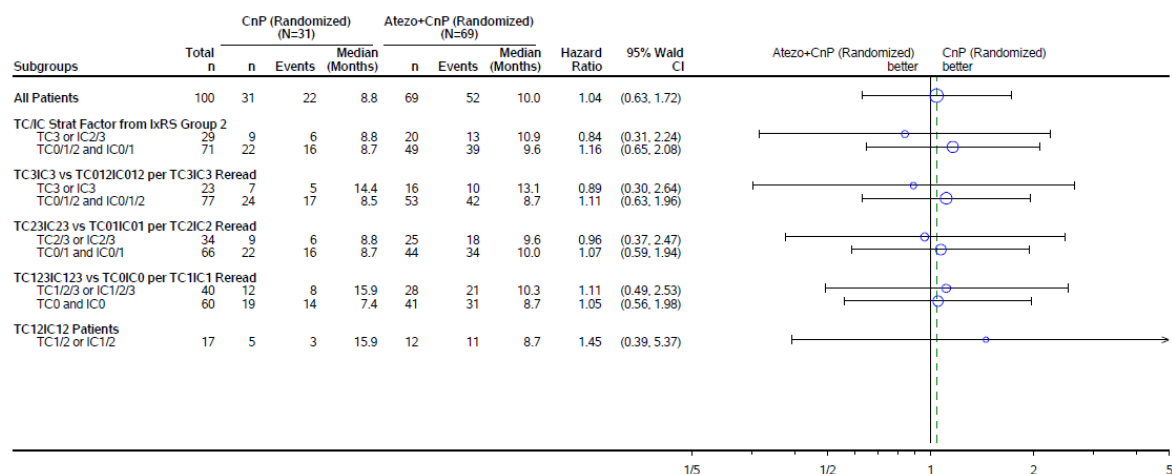
Wild Type patients are EGFR/ALK wild type patients who did not have an activating EGFR mutation or ALK-positive disease.

NE = Not estimable; Median survival was estimated from Kaplan-Meier method. Unstratified hazard ratio relative to CnP and 95% CI for the hazard ratio were estimated using Cox regression. The vertical dashed line indicates the hazard ratio for all patients. The diameter of the circle is proportional to the square root of the total number of events.

Randomized treatments are displayed.

Data Cutoff: 15MAR2018; RAVE Data Extracted: 18 May 2018.

Figure 18: Forest Plot – Subgroup Analysis of Overall Survival (ITT-WT patients)



Wild Type patients are EGFR/ALK wild type patients who did not have an activating EGFR mutation or ALK-positive disease.

Atezo=Atezolizumab, CnP=Carboplatin+Nab-Paclitaxel.

NE = Not estimable; Median survival was estimated from Kaplan-Meier method. Unstratified hazard ratio relative to CnP and 95% CI for the hazard ratio were estimated using Cox regression. The vertical dashed line indicates the hazard ratio for all patients. The diameter of the circle is proportional to the square root of the total number of events.

Randomized treatments are displayed.

Data Cutoff: 15MAR2018; RAVE Data Extracted: 18 May 2018.

Figure 19: Forest plot – subgroup analysis of OS (ITT-WT population) liver metastasis at baseline

Sensitivity Analysis for PFS

Missing Tumour Assessment

At the time of clinical cutoff date (CCOD), 6 patients in Atezo+CnP arm and 3 patients in CnP arm missed two or more tumor assessments prior to the date of disease progression or death in ITT-WT population. The potential impact of missing tumor assessment on PFS was assessed by the following sensitivity analyses:

Patients who missed two or more scheduled assessments immediately prior to the date of disease progression per RECIST v1.1 or death were censored at the last tumor assessment prior to the missed visits.

Patients who missed two or more scheduled assessments immediately prior to the date of disease progression per RECIST v1.1 or death were counted as having progressed on the date of the first of these missing assessments.

PFS in ITT-WT	HR (95% CI)	p-value
Primary Analysis	0.643 (0.539, 0.768)	<0.0001
Sensitivity Analysis a)	0.641 (0.536, 0.766)	<0.0001*
Sensitivity Analysis b)	0.643 (0.539, 0.768)	<0.0001*

HR = hazard ratio; ITT-WT = intent to treat- wild type; PFS = progression-free survival.

*p-value of sensitivity analysis is for descriptive purpose only.

Potential Impact of Non-Protocol-Specified Anti-Cancer Therapy (NPT)

At the time of CCOD, 9 patients in Atezo+CnP arm and 8 patients in CnP arm received NPT before a PFS event. The potential impact of NPT on PFS was assessed using a sensitivity analysis by which patients who received NPT before a PFS event were censored at the last tumor assessment date before receipt of NPT.

PFS in ITT-WT	HR (95% CI)	p-value
Primary Analysis	0.643 (0.539, 0.768)	<0.0001
Sensitivity Analysis	0.626 (0.522, 0.750)	<0.0001*

HR = hazard ratio; ITT-WT = intent to treat- wild type; PFS = progression-free survival.

*p-value of sensitivity analysis is for descriptive purpose only.

PFS by Independent Review Facility (IRF)

PFS as assessed by the IRF was defined as the time from randomization to the first documented PD, as determined by the IRF using RECIST v1.1, or death from any cause, whichever occurred first.

At the time of the CCOD, 506 deaths or PD events had occurred among patients in the ITT-WT population for the Atezo-CnP and CnP arms combined; the event/patient ratio was 72.9% in the Atezo-CnP arm, 77.6% in the CnP arm, and 74.5% overall.

The stratified HR was 0.75 (95% CI: 0.63, 0.91) in the ITT-WT population. PFS as assessed by the IRF in the ITT-WT population was consistent with the primary analysis of investigator-assessed PFS.

Table 33: Time-to-event summary for IRF-assessed PFS (sensitivity analysis – ITT-WT population)

	CnP (Randomized) (N=228)	Atezo+CnP (Randomized) (N=451)
Patients with event (%)	177 (77.6%)	329 (72.9%)
Earliest contributing event		
Death	71	91
Disease Progression	106	238
Patients without event (%)	51 (22.4%)	122 (27.1%)
Time to Event (Months)		
Median	6.4	7.2
95% CI	(5.6, 7.4)	(6.7, 8.3)
25% and 75%-ile	3.0, 11.2	4.0, 15.7
Range	0.0^ to 25.6	0.0^ to 29.8
Unstratified Analysis		
p-value (log-rank)	0.0017	
Hazard Ratio	0.746	
95% CI	(0.621, 0.897)	
Stratified Analysis		
p-value (log-rank)	0.0026	
Hazard Ratio	0.754	
95% CI	(0.627, 0.907)	
Time Point Analysis		
6 Months		
Patients remaining at risk	102	249
Event Free Rate (%)	51.82	57.97
95% CI	(45.03, 58.60)	(53.34, 62.60)
Difference in Event Free Rate	6.15	
95% CI	(-2.06, 14.37)	
p-value (Z-test)	0.1421	
1 year		
Patients remaining at risk	36	128
Event Free Rate (%)	23.16	32.32
95% CI	(17.07, 29.25)	(27.85, 36.79)
Difference in Event Free Rate	9.16	
95% CI	(1.61, 16.71)	
p-value (Z-test)	0.0174	

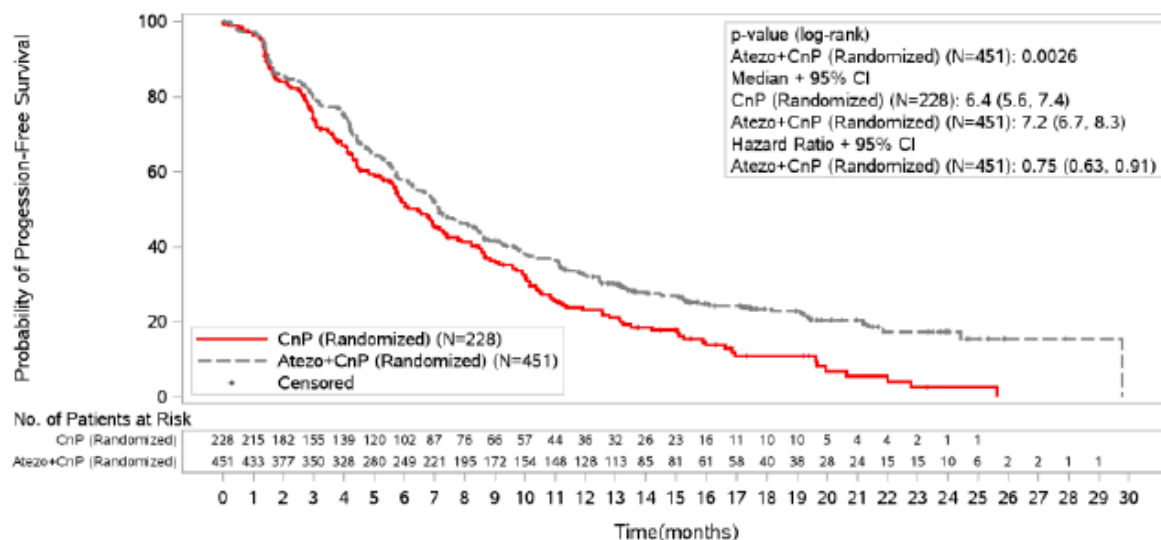
Atezo=Atezolizumab, CnP=Carboplatin+Nab-Paclitaxel.

Wild Type patients are EGFR/ALK wild type patients who did not have an activating EGFR mutation or ALK-positive disease.

* Censored, ^ Censored and event, NE = Not estimable.

Summaries of Time-to-Event (median, percentiles) were Kaplan-Meier estimates. 95% CI for median was computed using the method of Brookmeyer and Crowley. Hazard ratios were estimated by Cox regression. Stratification factors are: sex and PD-L1 tumor expression by IHC per IxRS.

Data Cutoff: 15MAR2018; RAVE Data Extracted: 18 May 2018.



Atezo=Atezolizumab, CnP=Carboplatin+Nab-Paclitaxel.

Wild Type patients are EGFR/ALK wild type patients who did not have an activating EGFR mutation or ALK-positive disease.

Randomized treatments are displayed.

Stratification factors are: sex and PD-L1 tumor expression by IHC per IxRS.

Data Cutoff: 15MAR2018; RAVE Data Extracted: 18 May 2018.

Figure 20: Kaplan-Meier plot for IRF-assessed PFS with stratified analysis (sensitivity analysis – ITT-WT population)

Patients eligible for maintenance therapy

Table 34: Subgroup Analyses of PFS for Patients Eligible for Maintenance Therapy According to Maintenance Received (ITT-WT)

	CnP Pem+ (N=40)	CnP Pem - (N=67)	Atezo+CnP Atezo main. (N=274)
Progression-Free Survival			
Event	33 (83%)	59 (88%)	190 (69%)
Median (mon, 95%CI)	8.6 (8.1, 12.6)	7.4 (6.5, 9.0)	10.0 (8.6, 11.4)
Unstratified HR (95%CI)*		1.43 (0.93, 2.20)	0.78 (0.53, 1.13)

*Reference group: CnP maintenance eligible, received pemetrexed maintenance

CnP = carboplatin+Nab-Paclitaxel; CI = confidence interval; HR=hazard ratio; mon=months; NE=not evaluable; OS=overall survival; PFS=progression free survival; ITT = intent-to-treat; WT =wild type.

Sensitivity Analysis for OS

Potential Impact of NPT

At the time of CCOD, the proportion of patients who received at least one NPT was similar between the two treatment arms in the ITT-WT population (Atezo+CnP 39.0% vs. CnP 39.5%)

Table 35: Non-Protocol Follow-up Cancer Therapy Regimens (ITT-WT)

	CnP (All Randomized) (n=228)	CnP (Randomized And Received Pem Maintenance) (n= 45)	Atezo+CnP (All Randomized) (n=451)
Patient with >=1 regimen	90 (39.5%)	15 (33.3%)	176 (39.0%)
Patient with >=2 regimen	20 (8.8%)	1 (2.2%)	55 (12.2%)
Patient with >=3 regimen	4 (1.8%)	0	16 (3.5%)

ITT-WT=Intent-to-treat, Wild Type. Data cut-off: 15 March 2018

In the Atezo+CnP arm, the most common treatments received were chemotherapy (34.1%) and targeted therapy (11.1%). In the CnP arm, the most common treatments received were chemotherapy (23.2%) and immunotherapies (19.3%). Among immunotherapies, anti-PD-1 or anti-PD-L1 agents were the most common.

The potential impact of NPT on OS was assessed by a discount method in the ITT-WT population. Survival time after patients received NPT was discounted at 30% for both treatment arms.

The results of this analysis are presented in the table below.

Table 36: Time-to-event summary for OS discounting for non-protocol-specified anti-cancer therapy (sensitivity analysis – ITT-WT population)

	CnP (Randomized) (N=228)	Atezo+CnP (Randomized) (N=451)
Patients with event (%)	131 (57.5%)	226 (50.1%)
Earliest contributing event		
Death	131	226
Patients without event (%)	97 (42.5%)	225 (49.9%)
Time to Event (Months)		
Median	13.1	17.2
95% CI	(10.9, 16.0)	(14.6, 19.6)
25% and 75%-ile	5.5, NE	7.0, NE
Range	0.0* to 27.4*	0.0* to 32.0*
Unstratified Analysis		
p-value (log-rank)	0.0279	
Hazard Ratio	0.786	
95% CI	(0.633, 0.975)	
Stratified Analysis		
p-value (log-rank)	0.0322	
Hazard Ratio	0.790	
95% CI	(0.637, 0.981)	
Time Point Analysis		
1 year		
Patients remaining at risk	110	243
Event Free Rate (%)	53.53	59.34
95% CI	(46.84, 60.21)	(54.72, 63.97)
Difference in Event Free Rate	5.82	
95% CI	(-2.31, 13.94)	
p-value (Z-test)	0.1607	
2 years		
Patients remaining at risk	7	29
Event Free Rate (%)	31.76	39.06
95% CI	(23.96, 39.56)	(32.71, 45.41)
Difference in Event Free Rate	7.30	
95% CI	(-2.76, 17.36)	
p-value (Z-test)	0.1547	

Atezo=Atezolizumab, CnP=Carboplatin+Nab-Paclitaxel.

Wild Type patients are EGFR/ALK wild type patients who did not have an activating EGFR mutation or ALK-positive disease.

NPT = Non-protocol therapy received after date of randomization.

* Censored, ^ Censored and event, NE = Not estimable.

Summaries of Time-to-Event (median, percentiles) were Kaplan-Meier estimates. 95% CI for median was computed using the method of Brookmeyer and Crowley. Hazard ratios were estimated by Cox regression. Stratification factors are: sex and PD-L1 tumor expression by IHC per IxRS.

Data Cutoff: 15MAR2018; RAVE Data Extracted: 18 May 2018.

OS in ITT-WT	HR (95% CI)	p-value
Primary Analysis	0.791 (0.637, 0.982)	0.0331
Sensitivity Analysis	0.790 (0.637, 0.981)	0.0322*

HR = hazard ratio; ITT-WT = intent to treat- wild type; OS = overall survival.

*p-value of sensitivity analysis is for descriptive purpose only.

Source: Primary CSR Section 5.2.3.1.

The potential impact of NPT on OS was also assessed by a rank preserving structural failure time method which provides an estimate of the OS time for the control group had NPT not occurred (Robins and Tsiatis 1991). The HR was estimated to be 0.778.

Delayed Clinical Effect

The OS K-M curves started to separate at approximately 2 months (minimal delay). The potential impact of delayed clinical effect on OS was assessed by a weighted log-rank analysis in the ITT-WT population, whereby a piecewise linear weight function was specified (Lin and Leon 2017) assuming no treatment effect during the first month or the first two months, respectively, and then a full treatment effect afterwards. The corresponding unstratified weighted log-rank test p-values of Atezo+CnP compared with CnP were 0.0197 and 0.0328, respectively.

Loss to Follow-up

At the time of CCOD, one patient (CnP arm) was lost to follow-up. The potential impact of loss to follow-up on OS was assessed using a sensitivity analysis by which the two treatment arms are compared for any patients lost to follow-up and considered as having died at the last date they were known to be alive.

OS in ITT-WT	HR (95% CI)	p-value
Primary Analysis	0.791 (0.637, 0.982)	0.0331
Sensitivity Analysis	0.785 (0.633, 0.974)	0.0276*

HR = hazard ratio; ITT-WT = intent to treat- wild type; OS = overall survival.

*p-value of sensitivity analysis is for descriptive purpose only.

Source: t_ef_lm_IT_WT_os_15Mar2018_29537; t_ef_lm_IT_WT_osltfu_15Mar2018_29537.

Patients eligible for maintenance therapy

Table 37: Subgroup Analyses of OS for Patients Eligible for Maintenance Therapy According to Maintenance Received (ITT-WT) (Cut-off 4. September 2018)

	CnP Pem+ (N=40)	CnP Pem - (N=68)	Atezo+CnP Atezo main. (N=279)
Overall Survival			
Event	19 (48%)	32 (47%)	124 (44%)
Median (mon, 95%CI)	24.4 (17.0, NE)	24.5 (20.6, NE)	28.2 (25.0, NE)
Unstratified HR (95%CI)*		1.05 (0.59, 1.86)	0.91 (0.56, 1.47)

*Reference group: CnP maintenance eligible, received pemetrexed maintenance

CnP = carboplatin+Nab-Paclitaxel; CI = confidence interval; HR=hazard ratio; mon=months; NE=not evaluable; OS=overall survival;

ITT = intent-to-treat; WT =wild type.

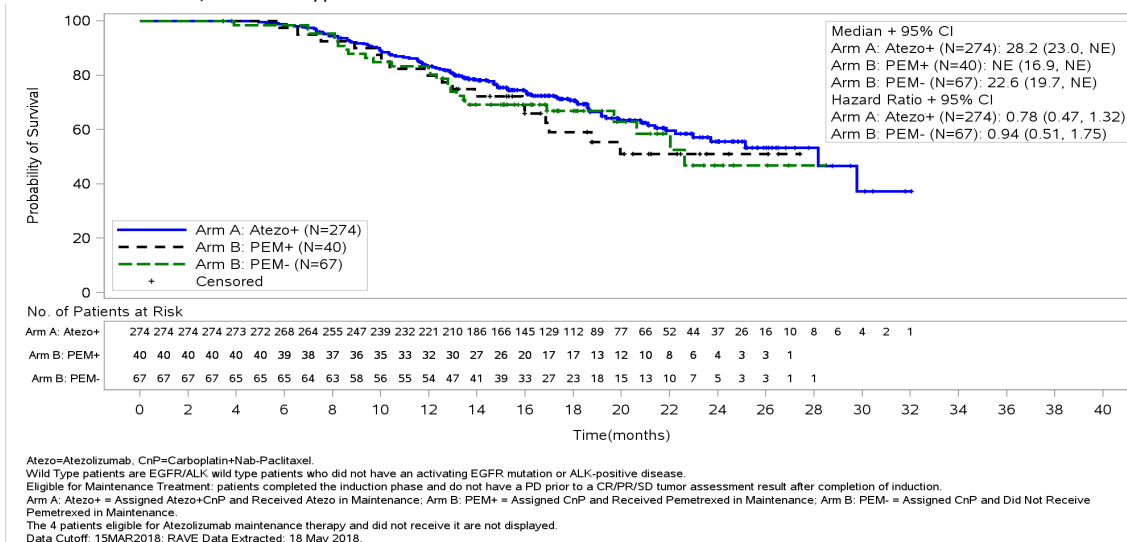
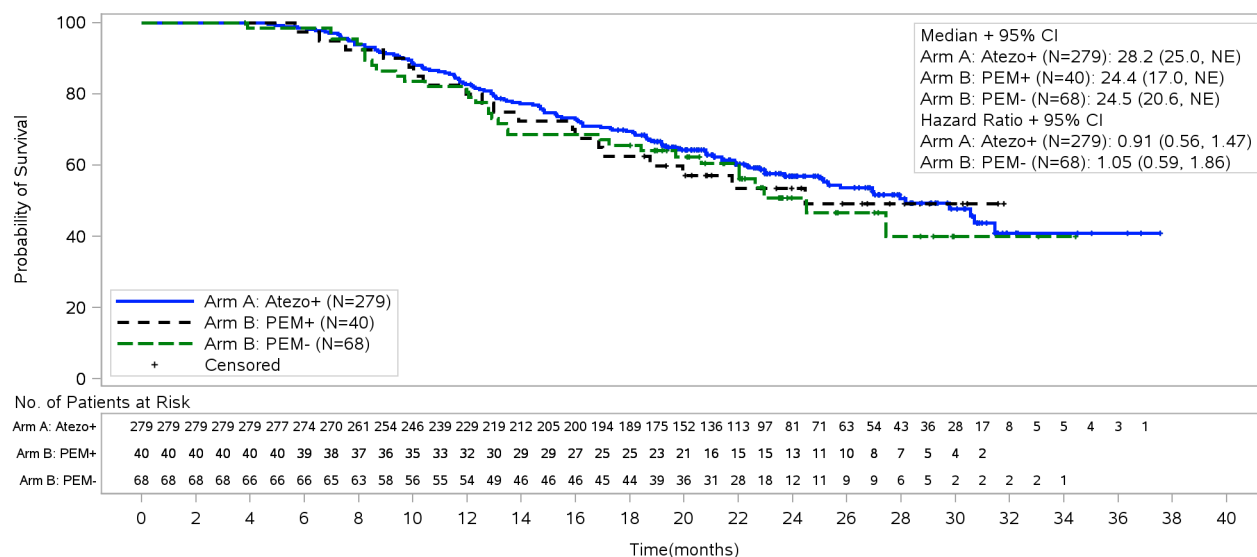


Figure 21: Kaplan-Meier plot of OS by assigned arm and by maintenance treatment received (ITT-WT)



Atezo=Atezolizumab, CnP=Carboplatin+Nab-Paclitaxel.
 Wild Type patients are EGFR/ALK wild type patients who did not have an activating EGFR mutation or ALK-positive disease.
 Arm A: Atezo+ = Assigned Atezo+CnP and Received Atezo in Maintenance; Arm B: PEM+ = Assigned CnP and Received Pemetrexed in Maintenance; Arm B: PEM- = Assigned CnP and Did Not Receive Pemetrexed in Maintenance.
 The 5 patients eligible for Atezolizumab maintenance therapy and did not receive it are not displayed.
 Data Cutoff: 04SEP2018; RAVE Data Extracted: 19 Oct 2018.

Figure 22: Kaplan-Meier plot of overall survival in patients received maintenance treatment (ITT-WT) (Cut-off 4. September 2018)

Table 38: Additional Sensitivity Analysis of updated OS (ITT-WT) (Cut-off 4 September 2018)

	HR (95% CI)
Supplemental Results Report	0.824 (0.675, 1.006)
Multiple Imputation	0.840 (0.677, 1.043)

HR= hazard ratio; OS=Overall Survival; ITT=Intent-to-Treat; Wild Type.

Sensitivity analyses to evaluate the impact of crossover on OS:

The MAH conducted the following sensitivity analyses to evaluate the impact of crossover on OS:

1. Discount Methods

Survival time after patients crossed over was discounted at 10%, 20%, and 30%, as follows:

- 10% discount: The stratified HR was 0.77 (95% CI: 0.62, 0.95), and the median OS discounting for crossover with benefit reduction of 10% was 13.5 months for the CnP arm in the ITT-WT population
 - 20% discount: The stratified HR was 0.74 (95% CI: 0.60, 0.92), and the median OS discounting for crossover with benefit reduction of 20% was 13.5 months for the CnP arm in the ITT-WT population
 - 30% discount: The stratified HR was 0.72 (95% CI: 0.58, 0.89), and the median OS discounting for crossover with benefit reduction of 30% was 12.8 months for the CnP arm in the ITT-WT population
2. Rank Preserving Structural Failure Time (RPSFT): The stratified HR was 0.71, and the median OS was estimated to be 12.9 months in the CnP arm in the ITT-WT population.

Immunogenicity

Table 39: Baseline prevalence and post-baseline incidence of ADA to atezolizumab among atezolizumab-treated patients (ADA-evaluable population)

	Atezo+CnP	CnP (Crossover Patients)
Baseline ADA-evaluable patients	n=451	n=84
No. of patients positive for ADA (%)	14 (3.1%)	4 (4.8%)
No. of patients negative for ADA	437	80
Post-baseline ADA-evaluable patients	n=446	n=85
No. of patients positive for ADA (%)	100 (22.4%)	20 (23.5%)
No. of patients with treatment-induced ADA ^a	98	20
No. of patients with treatment-enhanced ADA ^b	2	0
No. of patients negative for ADA	346	65
No. of patients with treatment-unaaffected ADA ^c	12	4

ADA=anti-drug antibody (also known as anti-therapeutic antibody); Atezo=atezolizumab; CnP=carboplatin+nab-paclitaxel.

- ^a Treatment-induced ADA refers to patients who had a baseline negative ADA result who developed ADA at any time after initial drug administration.
- ^b Treatment-enhanced ADA refers to patients who had a baseline positive ADA result in whom the assay signal was enhanced (greater than baseline titer by ≥ 0.60 titer units) at any time after initial drug administration.
- ^c Treatment-unaaffected ADA refers to patients who had a baseline positive ADA result in whom the assay signal was not enhanced (not greater than baseline titer by ≥ 0.60 titer units) at any time after initial drug administration. These patients are considered post-baseline negative for ADA.

Note: See Shankar et al. (2014) for further details on the definition of treatment-induced, treatment-enhanced, transient, and persistent ADA.

Table 40: PFS by investigator and OS by atezolizumab ADA status (ADA-evaluable WT population)

Parameter	Atezo+CnP	
	Atezo ADA-Negative	Atezo ADA-Positive
Investigator-Assessed Progression-Free Survival		
N	320	95
No. of patients with event (%)	242 (75.6%)	75 (78.9%)
Median time to event, months	7.3	7.2
95% CI	(6.9, 8.3)	(6.1, 8.3)
6 months		
No. of patients remaining at risk	186	54
Event-free rate, %	59.40	60.85
95% CI	(53.97, 64.82)	(50.85, 70.85)
1 year		
No. of patients remaining at risk	96	23
Event-free rate, %	32.52	26.78
95% CI	(27.30, 37.74)	(17.55, 36.01)
Overall Survival		
N	320	95
No. of patients with event (%)	150 (46.9%)	46 (48.4%)
Median time to event, months	20.8	19.2
95% CI	(18.2, 29.8)	(15.7, 25.1)
1 year		
No. of patients remaining at risk	205	62
Event-free rate, %	66.84	69.59
95% CI	(61.62, 72.06)	(60.18, 79.00)
2 years		
No. of patients remaining at risk	36	4
Event-free rate, %	44.30	33.02
95% CI	(37.30, 51.30)	(14.92, 51.12)

ADA=anti-drug antibody; Atezo=atezolizumab; CnP=carboplatin+nab-paclitaxel; WT=wild type.

Treatment beyond progression

A total of 308 patients randomized to the Atezo+CnP arm from the safety-evaluable population experienced PD and were included in these analyses. Of these 308 patients:

- 253 patients received anti-cancer therapy beyond the first PD.
 - 151 patients were treated on or after first PD with atezolizumab (group 1).
 - 102 patients received other NPT (Group 2).
- 55 patients did not receive any anti-cancer treatment after the first PD (Group 3).

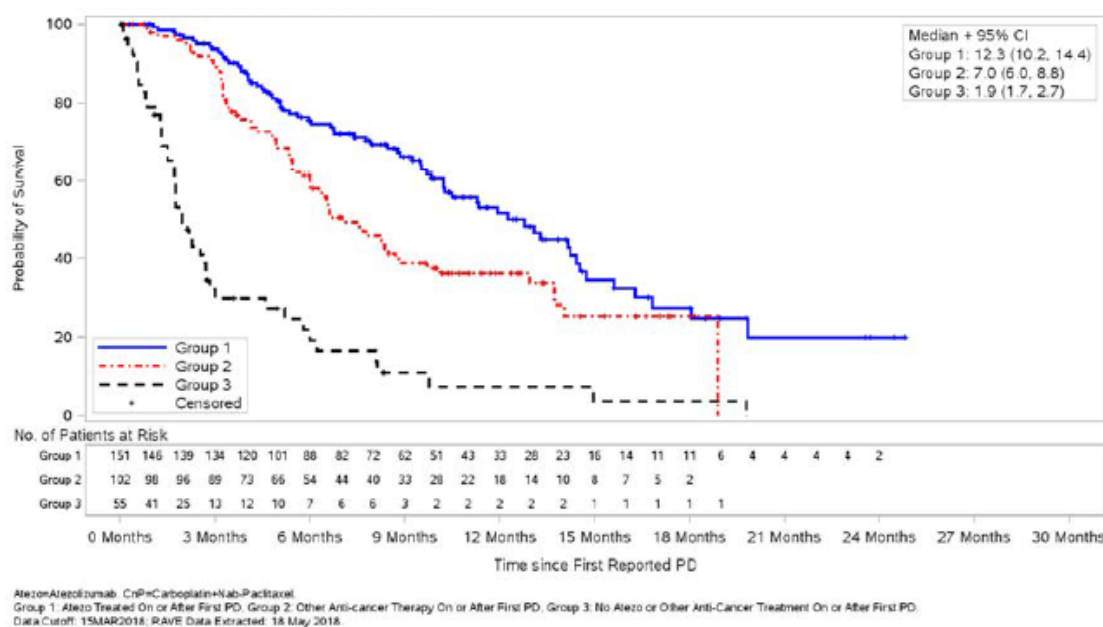


Figure 23: Kaplan-meier plot for post-progression OS in patients treated with atezolizumab beyond PD in the atezo+CnP arm (safety-evaluable population)

Table 41: Time-to-event summary for post-progression OS in patients treated with atezolizumab beyond PD in the atezo+CnP arm (safety-evaluable population)

Time to Event Summary for Overall Survival on or after First PD
Safety-Evaluable Patients, Reported Disease Progression, Atezo+CnP (Arm A)
Protocol: G029537 (Data Cutoff: 15MAR2018)

Overall Survival from First PD

	Atezo Treated On or After First PD (N=151)	Other Anti-cancer Therapy On or After First PD (N=102)	No Atezo or NPT Treatment On or After First PD (N=55)	Atezo Treated Patients Reported Disease Progression (N=308)
Patients with event (%)	69 (45.7%)	65 (63.7%)	45 (81.8%)	179 (58.1%)
Earliest contributing event				
Death	69	65	45	179
Patients without event (%)	82 (54.3%)	37 (36.3%)	10 (18.2%)	129 (41.9%)
Time Since First PD to Event (Months)				
Median	12.3	7.0	1.9	8.3
95% CI	(10.2, 14.4)	(6.0, 8.8)	(1.7, 2.7)	(6.7, 9.8)
25% and 75%-ile	6.0, 18.0	4.1, 18.9	1.3, 5.2	3.8, 15.6
Range	0.1* to 24.8*	0.4* to 18.9	0.0* to 19.8	0.0* to 24.8*
Time Point Analysis				
6 Months				
Patients remaining at risk	88	54	7	149
Event Free Rate (%)	75.57	58.03	19.16	59.98
95% CI	(68.27, 82.87)	(48.21, 67.85)	(7.31, 31.01)	(54.21, 65.75)
1 year				
Patients remaining at risk	33	18	2	53
Event Free Rate (%)	51.54	36.26	7.30	38.79
95% CI	(41.64, 61.44)	(26.33, 46.18)	(0.00, 16.04)	(32.44, 45.14)
1 1/2 years				
Patients remaining at risk	11	2	1	14
Event Free Rate (%)	27.57	25.38	3.65	22.50
95% CI	(16.05, 39.09)	(14.03, 36.73)	(0.00, 10.33)	(15.38, 29.62)

Results from the OS analysis on or after first PD in the CnP arm for each of the three groups, i.e., patients who received immunotherapy, patients who received other anti-cancer therapies, and patients who didn't receive any further treatment after documented PD, are shown in the figure below.

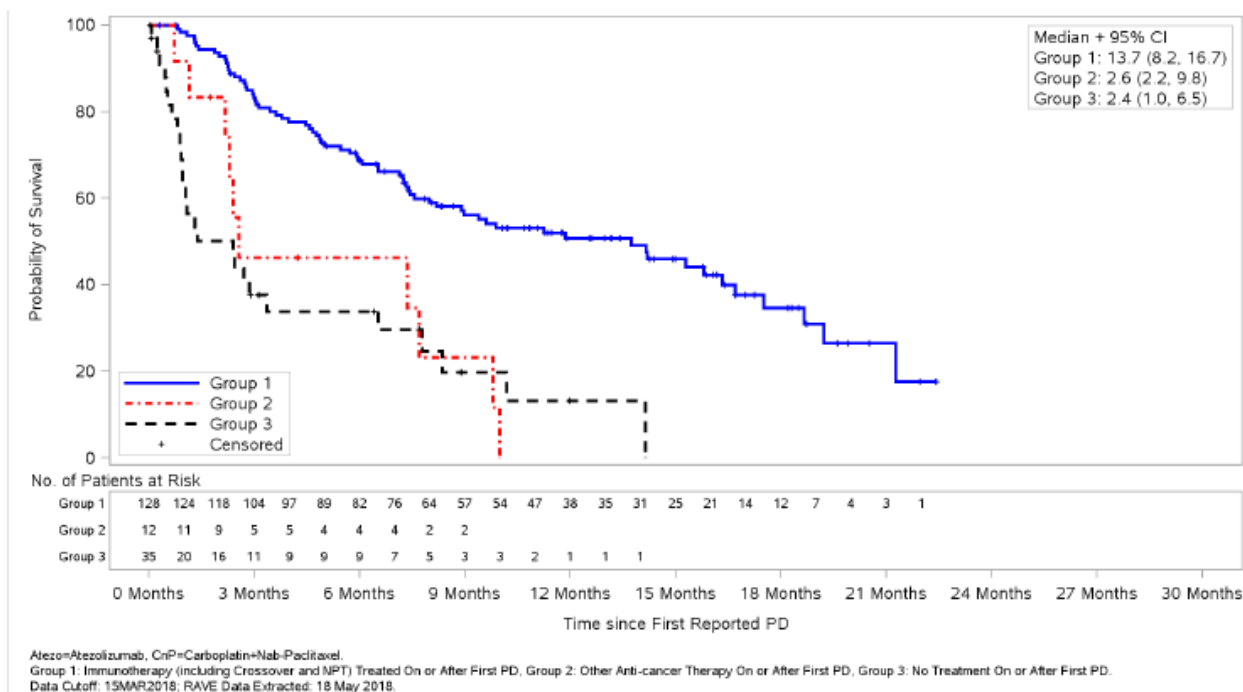


Figure 24: Kaplan-Meier Plot of Overall Survival on or after First PD in CnP arm (safety-evaluable patients)

Overall survival on or after the first instance of progressive disease (PD) in the atezo+CnP arm and pemetrexed-treated CnP arm was examined for each of the three groups, i.e., patients who received subsequent immunotherapy (including crossover to atezo, treatment beyond progression with atezo, and non-protocol therapies), patients who received other anti-cancer therapies, and patients who did not receive any further treatment after documented PD.

In the atezo+CnP arm (numbers updated for the ITT-WT population), 283 atezo treated patients who reported PD were included into this analysis. Out of the 283 patients, 162 had died (57.2%).

In the CnP arm, among the 45 pemetrexed-treated ITT-WT patients, 36 patients had PD and were included in this analysis (the other 9 patients had no PD as of the clinical cut-off date [CCOD]). Out of the 36 patients, 18 (50%) patients had died.

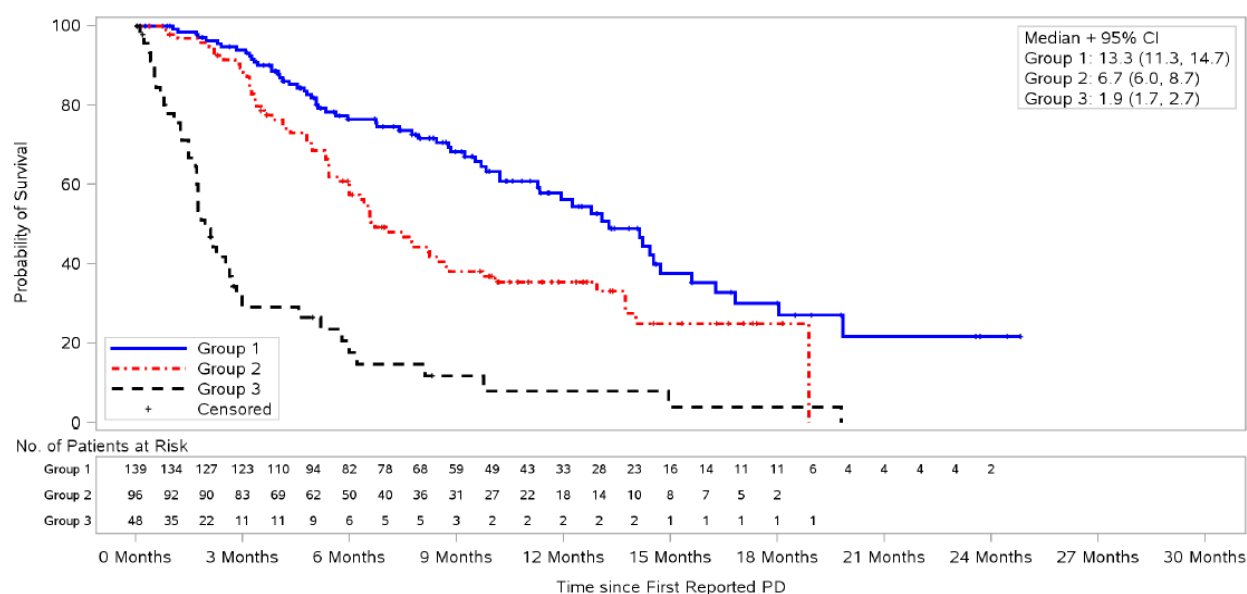
Distribution of the patients between three groups in both the arms (atezo+CnP and CnP arm) is provided in the table below.

Table 42: Number of patients who received subsequent therapies on or after first progressive disease- ITT –WT population

	Atezo+CnP n= 283	CnP+ pemetrexed-treated n=36
Subsequent Therapies Received		
Any immunotherapy as anti-cancer treatment in second line (2L) and subsequent lines of therapy (2L+)*	139 (49%)	29 (80%)
Other anti-cancer therapy	96 (34%)	1 (3%)
No treatment	48 (17%)	6 (17%)

ITT-WT = Intention-to treat, wild type.

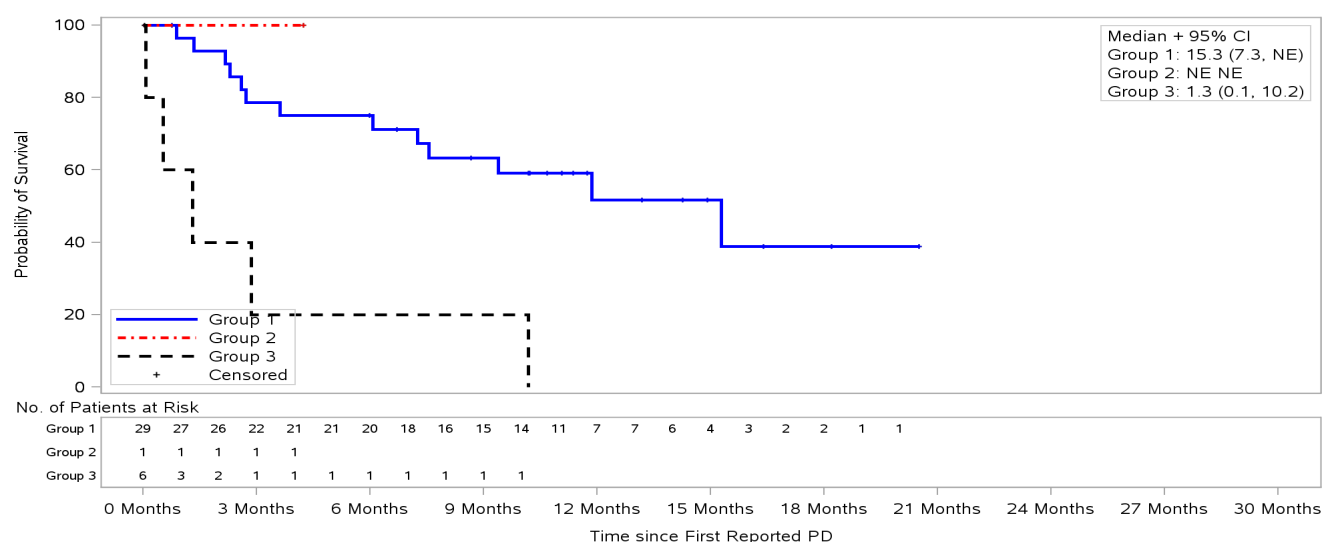
*Including crossover to atezo, treatment beyond progression with atezo, and non-protocol therapies.



Atezo=Atezolizumab, CnP=Carboplatin+Nab-Paclitaxel.
 Wild Type patients are EGFR/ALK wild type patients who did not have an activating EGFR mutation or ALK-positive disease.
 Group 1: Atezo Treated On or After First PD, Group 2: Other Anti-cancer Therapy On or After First PD, Group 3: No Atezo or Other Anti-Cancer Treatment On or After First PD.
 Data Cutoff: 15MAR2018; RAVE Data Extracted: 18 May 2018.

PD = Progressive disease.

Figure 25: Kaplan-Meier Plot of Overall Survival on or after First PD in atezo+CnP arm (safety-evaluable patients, wild type)



Atezo=Atezolizumab, CnP=Carboplatin+Nab-Paclitaxel.
 Wild Type patients are EGFR/ALK wild type patients who did not have an activating EGFR mutation or ALK-positive disease.
 Group 1: Immunotherapy (including Crossover and NPT) Treated On or After First PD, Group 2: Other Anti-cancer Therapy On or After First PD, Group 3: No Treatment On or After First PD.
 Data Cutoff: 15MAR2018; RAVE Data Extracted: 18 May 2018.

PD=Progressive Disease; NE=Not evaluable.

Figure 26: Kaplan-Meier Plot of Overall Survival on or after First PD in CnP arm pemetrexed-treated (safety evaluable patients, WT)

Table 43: Number of patients in CnP arm who received subsequent therapies on or after first progressive disease - ITT -WT population

	Immunotherapy (including Crossover and NPT) Treated On or After First PD	Other Anti-cancer On or After First PD	No Treatment On or After First PD	CnP Treated Patients Reported Disease Progression
	(n=29)	(n=1)	(n=6)	(n=36)
Patients with event (%)	13 (44.8%)	0	5 (83.3%)	18 (50.0%)

Patients without event (%)	16 (55.2%)	1 (100%)	1 (16.7%)	18 (50.0%)
Time Since First PD to Event (Months, median 95% CI)	15.3 (7.3, NE)	NE	1.3 (0.1, 10.2)	11.9 (6.1, NE)

NE=not evaluable; NPT=non protocol therapy; PD=progressive disease; ITT-WT=Intent-to-Treat, Wild Type.

Summary of main study

The following table summarises the efficacy results from the main study supporting the present application. This summary should be read in conjunction with the discussion on clinical efficacy as well as the benefit risk assessment (see later sections).

Summary of efficacy for trial GO29537

Title: Phase III, Multicenter, Randomized, Open-Label Study Evaluating the Efficacy and Safety of Atezolizumab (MPDL3280A, Anti-PD-L1 Antibody) in Combination with Carboplatin+Nab-Paclitaxel for Chemotherapy-Naïve Patients with Stage IV Non-Squamous Non-Small Cell Lung Cancer.			
Study identifier	GO29537 (IMpower130)		
Design	A multicenter, randomized, open-label, controlled trial		
	Duration of main phase:		16-April-2015 to 04-Sep 2018 (data cutoff)
	Duration of Run-in phase:		not applicable
	Duration of Extension phase:		not applicable
Hypothesis	Superiority		
Treatments groups	Atezo+CnP		Atezolizumab+carboplatin+nab-paclitaxel, n = 483
	CnP		carboplatin+nab-paclitaxel, n = 240
Endpoints and definitions	Primary (ITT-WT)	OS	Death due to any cause
		PFS	Progression free survival by RECIST 1.1 according investigator
	Secondary (ITT)	OS	Death due to any cause
		PFS	Progression free survival by RECIST 1.1 according investigator
Database lock	18 May 2018		
<u>Results and Analysis</u>			
Analysis description	Primary Analysis		
Analysis population and time point description	ITT-WT population (subset of ITT excluding patients with activating EGFR mutation or ALK translocation) time point: Primary analysis PFS, OS (CCOD 15 March 2018) <i>Updated OS (CCOD 4-Sep 2018)</i>		
Descriptive statistics and estimate variability (stratified)	Treatment group	Atezo+CnP	CnP
	ITT-WT - Number of subject	451 <i>456</i>	228 <i>229</i>
	OS (Median (months))	18.6 <i>18.1</i>	13.9 <i>13.5</i>
	95%-CI	(16.0, 21.2) <i>(15.1, 20.8)</i>	(12.0, 18.7) <i>(11.9, 17.7)</i>
	PFS (Median (months))	7.0	5.5
	95%-CI	(6.2, 7.3)	(4.4, 5.9)
Effect estimate per comparison	OS	Comparison groups	Atezo+CnP vs. CnP
		HR	0.791 <i>0.824</i>
		95%-CI	(0.637-0.982) <i>(0.675-1.006)</i>
		P-value	0.0331 <i>0.0568</i>

	PFS	HR	0.643
		95%-CI	(0.539 – 0.768)
		P-value	< 0.0001
Notes	The analysis indicates a statistical significant superiority with regard to PFS, but not in terms of OS of Atezo+CnP vs. CnP alone in the ITT-WT population		
Analysis description	Secondary Analyses		
Analysis population and time point description	Intent to treat time point: final analysis PFS, 1 st interim analysis OS		
Descriptive statistics and estimate variability	Treatment group	Atezo+CnP	CnP
	ITT- Number of subject	483	240
	OS (Median (months))	18.1	13.9
	95%-CI	(15.2, 20.8)	(12.0, 18.2)
	PFS (Median (months))	7.0	5.6
	95%-CI	(6.3, 7.3)	(4.5, 5.9)
Effect estimate per comparison	OS	Comparison groups	Atezo+CnP vs. CnP
		HR	0.80
		95%-CI	(0.65 – 0.99)
		P-value	0.0393
	PFS	HR	0.65
		95%-CI	(0.54 – 0.77)
		P-value	< 0.0001
Notes	The secondary analysis confirms the result of the primary analysis. The primary analysis results are also supported by pre-specified subgroup analyses on the PD-L1 WT population and further secondary and exploratory analyses.		

Clinical studies in special populations

Table 44: Elderly patients in study IMpower130

	Age 65-74 (Older subjects number /total number)	Age 75-84 (Older subjects number /total number)	Age 85+ (Older subjects number /total number)
Controlled Trials	276/723	82/723	3/723

2.4.3. Discussion on clinical efficacy

The purpose of this application is to extend the indication for atezolizumab to the use in combination with nab-paclitaxel and carboplatin for the first-line (1L) treatment of adult patients with metastatic, non-squamous NSCLC who do not have *EGFR*-mutant or *ALK*-positive NSCLC.

Design and conduct of clinical studies

The MAH provided data from the primary analysis of IMpower130 (Study GO29537; n=723), a Phase III trial designed to evaluate the safety and efficacy of atezolizumab-carboplatin-nab-paclitaxel (Atezo-CnP) versus treatment with carboplatin-nab-paclitaxel (CnP) in chemotherapy-naïve patients with Stage IV non-squamous NSCLC. Atezo is added to CnP in an open-label fashion. It is not clear, why a blinded design was not preferred, thus, PFS and QoL data should be interpreted with caution.

This study had two co-primary endpoints OS and PFS. The overall type I error was 5 %. The 1-sided alpha was distributed among PFS (0.003) and OS (0.022). The study was to be considered positive if at least one of the co-primary endpoints was statistically significant. The alpha allocated to the PFS was to be recycled for OS, if statistical significance was demonstrated for PFS. This is acceptable. An interim analysis of OS was pre-planned and appropriate stopping rules for OS were defined. Both OS and PFS

were tested in two different populations: ITT-WT (all randomized patients without activating EGFR mutation or ALK translocation) and ITT (all randomized patients). The MAH has appropriately adjusted for multiplicity caused by testing in two different populations by a hierarchical testing method.

The MAH amended the protocol seven times. The majority of the protocol amendments are not controversial, however with protocol amendment 4 (version 5) the co-primary endpoint OS was added. This was, according to the MAH, motivated by the fact that emerging and accumulating knowledge about immune checkpoint inhibitors showed that PFS may not be as sensitive endpoint as OS. Patients were allowed to cross over after the addition of OS as co-primary endpoint. Analyses showed that those patients that crossed over prior to the amendment had little effect on the final results.

As a consequence of the addition of OS as co-primary endpoint the SAP was accordingly amended. Also, the testing of PFS in ITT-WT Teff-high was removed from the primary testing and listed as an exploratory analysis. The final set of stratification factors were amended "due to the potential risk of over-stratification". According to the guideline on adjustment for baseline covariates in clinical trials (EMA/CHMP/295050/2013)", the factors that are the basis of stratification should normally be included as covariates or stratification variables in the primary outcome model, except where stratification was done purely for an administrative reason.

Overall, no major concerns have been identified with regards to design and conduct of the study.

Efficacy data and additional analyses

The demographics and baseline disease characteristics of the study population (n=723) were well balanced between the treatment arms. The median age was 64 years (range 18 to 86 years). The majority of the patients were male (57%) and white (90%). Fourteen point eight percent (14.8%) of patients had liver metastases at baseline, and most patients were current or previous smokers (88%). The majority of patients had a baseline ECOG performance status of 1 (58.7%) and PD-L1 expression <1% (approximately 53%). Among 107 Arm B patients who had a response status of stable disease, partial response, or complete response after induction therapy, 40 received pemetrexed switch maintenance therapy. Prior cancer therapies at baseline were similarly reported in the ITT population and in general were well balanced between treatment arms.

The study met both its co-primary endpoints at the data cut-off 15 March 2018. With regards to OS in ITT-WT, the HR was 0.79 (95%CI; 0.64, 0.98), $p = 0.033$. Median OS was 18.6 months vs 13.9 months in the atezo+CnP and CnP arms respectively, an improvement of median OS by 4.7 months, which is clinically encouraging in this patient population with a dismal prognosis. However, the p-value shows that the results are not highly statistically significant and barely mature since only approximately half of the population has died. Updated OS data in the ITT-WT population are in line with the primary analysis, the HR is now 0.82 (95% CI: 0.67, 1.01) and the p-value = 0.057. As OS data has matured, the difference between the two treatment arms is no longer statistically significant, but as discussed above the study will be considered positive if at least one of the co-primary endpoints is statistically significant. The updated results are also reflected in section 5.1 of the SmPC.

PFS data at the primary analysis are considered mature. PFS results in the ITT-WT patients showed a HR of 0.64 (95%CI: 0.54, 0.77; $p < 0.0001$). The median duration of PFS was 7.0 months in the atezo+CnP arm. The landmark analyses of PFS rates at 6 and 12 months are in line with the primary endpoint, however, these analyses were exploratory, not adjusted for multiplicity, and should therefore be interpreted with caution.

During the procedure, the MAH was requested to provide efficacy outcomes for those patients in the CnP arm, who received maintenance treatment with pemetrexed. The OS and PFS results are comparable to the atezolizumab arm and better than the ITT population in the CnP arm. The subgroup analyses of OS are immature (event rates are between 34% and 43%). Patients in the CnP arm eligible for maintenance pemetrexed and who received switch maintenance with pemetrexed had a better outcome compared to patients, who were eligible for maintenance, but did not receive switch maintenance with pemetrexed.

In order to support the claim of efficacy, the whole trial population was re-analysed assuming that all patients eligible for maintenance therapy received it. It is understood that in this analysis, all patients were included and only patients in the CnP arm who would have been eligible for maintenance but did not receive it were imputed by replacing their actual survival time with a randomly drawn time from the CnP patients who received maintenance. This does not take advantage of any baseline covariate information and might have led to even lower survival times than actually observed. No marked differences to the primary analyses were observed for PFS and OS.

However, the results presented need to be interpreted with caution as they rely on a post-randomization criterion, the eligibility for maintenance therapy, and thus do not compare randomized patients anymore. Furthermore, they compare patients who were eligible to maintenance under CnP with patients who were eligible to maintenance under Atezo. This might further bias the results, which can also be seen in the markedly lower number of patients under CnP who were eligible for maintenance.

Usage of pemetrexed maintenance in the CnP arm was consistent with clinical practice and the prevalence of its use did not appear to impact the outcomes of the CnP arm. Since most patients were adequately treated, the overall treatment comparison based on all patients in the ITT-WT population is the most relevant comparison.

The subgroup analysis showed consistent results in line with the primary endpoint. However, some of the subgroups were small and the CIs were wide. Most interestingly is the estimate in the subgroup of patients with liver metastasis at baseline. The hazard ratios for PFS and OS in the subgroup of patients with liver metastases at baseline, regardless of PD-L1 status, were 0.93 (0.59, 1.43) and 1.04 (0.63, 1.72) respectively. These patients do not seem to derive benefit by adding atezolizumab. Although it is difficult to draw conclusions from small subgroup and post-hoc analyses, this information is relevant for individual treatment decisions and is reflected in section 5.1 of the SmPC.

Results of other secondary endpoints were overall in line with the primary analyses in the ITT-WT population. A similar treatment effect is observed across all the different PD-L1 subgroups. These results are in line with the primary estimate. Secondary endpoints were not adjusted for multiplicity.

2.4.4. Conclusions on the clinical efficacy

Study IMpower 130 shows statistically significant and clinically relevant results in the proposed patient population across different PD-L1 subgroups in terms of PFS. Updated OS is no longer statistically significant. Patients in the CnP arm could according to the protocol at the discretion of the investigator continue with pemetrexed maintenance treatment if they were in CR, PR or SD. Data seem to suggest no relevant difference between Atezo+CnP + atezo maintenance and patients that received CnP + pemetrexed maintenance. However, one should be careful with such conclusions due to several limiting factors.

2.5. Clinical safety

Introduction

The evidence for safety is based on data from the primary analysis of IMpower130 (Study GO29537). In addition, pooled safety data from five atezolizumab combination studies in 1L lung cancer (IMpower130, IMpower131, IMpower132, IMpower133, and IMpower150) as well as eight atezolizumab monotherapy studies regardless of tumour type (OAK, POPLAR, BIRCH, FIR, IMvigor211, IMvigor210, IMmotion150, and PCD4989g) were provided for reference.

The Atezo-Chemo Combo population safety analyses are based on safety data from a total of 2421 atezolizumab-treated, safety-evaluable patients from IMpower130 (n=473 with NSCLC), IMpower131 (n=666 with NSCLC), IMpower132 (n=291 with NSCLC), IMpower133 (n=198 with small cell lung cancer [SCLC]), and IMpower150 (n=793 with NSCLC) in the following arms:

- IMpower130 Arm A (Atezo-CnP)
- IMpower131 Arm A (Atezo-CP) and Arm B (Atezo-CnP)
- IMpower132 Arm A (atezolizumab-cisplatin or carboplatin-pemetrexed)
- IMpower133 Arm A (atezolizumab-carboplatin-etoposide)
- IMpower150 Arm A (Atezo-CP) and Arm B (Atezo-CP-bevacizumab)

Safety data from this pooled population provide the largest dataset to evaluate the clinical safety of atezolizumab in combination with platinum-based chemotherapy in patients with NSCLC (n=2223) and SCLC (n=198).

Note: Some listings for the Atezo-Chemo Combo population may display GO30081 (IMpower133) as the protocol number at the header of the output. Nonetheless, these listings reflect the same studies included in all other Atezo-Chemo Combo pooled data presentations throughout this SCS, as can be identified from the footer of each output.

Patient exposure

Table 45: Exposure to atezolizumab (safety-evaluable population) – Study IMpower130

	CnP (Actual) (N=232)	Atezo+CnP (Actual) (N=473)	All Patients (N=705)
4 cycles planned, number of cycles received during induction			
n	0	240	240
1 - 3	0	64 (26.7%)	64 (26.7%)
4 - 5	0	175 (72.9%)	175 (72.9%)
6	0	1 (0.4%)	1 (0.4%)
6 cycles planned, number of cycles received during induction			
n	0	233	233
1 - 3	0	51 (21.9%)	51 (21.9%)
4 - 5	0	73 (31.3%)	73 (31.3%)
6	0	107 (45.9%)	107 (45.9%)
> 6	0	2 (0.9%)	2 (0.9%)
Treatment duration (M)			
n	96	473	569
Mean (SD)	5.3 (5.4)	8.9 (7.2)	8.3 (7.1)
Median	2.8	6.9	6.3
Min - Max	0 - 19	0 - 32	0 - 32
Treatment duration (M)			
n	96	473	569
0 to <= 3 months	49 (51.0%)	120 (25.4%)	169 (29.7%)
>3 months to <= 6 months	14 (14.6%)	94 (19.9%)	108 (19.0%)
>6 months to <= 12 months	20 (20.8%)	96 (20.3%)	116 (20.4%)
>12 months	13 (13.5%)	163 (34.5%)	176 (30.9%)
Dose Intensity (%)			
n	96	473	569
Mean (SD)	96.5 (7.1)	92.4 (8.2)	93.1 (8.2)
Median	100.0	94.9	95.6
Min - Max	64 - 105	60 - 102	60 - 105
Number of doses			
n	96	473	569
Mean (SD)	8.4 (7.7)	12.8 (9.8)	12.1 (9.6)
Median	5.0	10.0	9.0
Min - Max	1 - 29	1 - 45	1 - 45
Total cumulative dose (mg)			
n	96	473	569
Mean (SD)	10137.3 (9204.4)	15367.2 (11782.0)	14484.9 (11548.5)
Median	6000.0	12000.0	10800.0
Min - Max	1200 - 34800	1200 - 54000	1200 - 54000

Atezo=Atezolizumab, CnP=Carboplatin+Nab-Facitaxel.

M=Months. Treatment duration is the date of the last dose of study medication minus the date of the first dose plus one day.

Dose intensity is the number of doses actually received divided by the expected number of doses.

Data Cutoff: 15MAR2018; RAVE Data Extracted: 18 May 2018.

Table 46: Exposure to carboplatin (safety-evaluable population) - Study IMpower130

	CnP (Actual) (N=232)	Ateso+CnP (Actual) (N=473)	All Patients (N=705)
4 cycles planned, number of cycles received during induction			
n	121	240	361
1 - 3	35 (28.9%)	70 (29.2%)	105 (29.1%)
4 - 5	86 (71.1%)	170 (70.8%)	256 (70.9%)
6 cycles planned, number of cycles received during induction			
n	111	233	344
1 - 3	39 (35.1%)	52 (22.3%)	91 (26.5%)
4 - 5	33 (29.7%)	79 (33.9%)	112 (32.6%)
6	39 (35.1%)	102 (43.8%)	141 (41.0%)
Treatment duration (M)			
n	232	473	705
Mean (SD)	2.2 (1.2)	2.4 (1.2)	2.4 (1.2)
Median	2.3	2.3	2.3
Min - Max	0 - 5	0 - 6	0 - 6
Treatment duration (M)			
n	232	473	705
0 to <= 3 months	181 (78.0%)	332 (70.2%)	513 (72.8%)
>3 months to <= 6 months	51 (22.0%)	140 (29.6%)	191 (27.1%)
>6 months to <= 12 months	0	1 (0.2%)	1 (0.1%)
Dose Intensity (%)			
n	232	473	705
Mean (SD)	91.3 (9.4)	90.2 (10.2)	90.6 (9.9)
Median	92.3	92.3	92.3
Min - Max	55 - 102	50 - 105	50 - 105
Number of doses			
n	232	473	705
Mean (SD)	3.7 (1.5)	4.0 (1.4)	3.9 (1.4)
Median	4.0	4.0	4.0
Min - Max	1 - 6	1 - 6	1 - 6
Total cumulative dose (mg)			
n	232	473	705
Mean (SD)	2288.5 (1101.5)	2438.6 (1072.0)	2389.2 (1083.3)
Median	2177.7	2420.0	2340.0
Min - Max	298 - 6173	375 - 5400	298 - 6173

Ateso=Atesolisumab, CnP=Carboplatin+Nab-Paclitaxel.

M=Months. Treatment duration is the date of the last dose of study medication minus the date of the first dose plus one day. Dose intensity is the number of doses actually received divided by the expected number of doses.

Data Cutoff: 15MAR2018; RAVE Data Extracted: 18 May 2018.

Table 47: Exposure to Nab-paclitaxel (safety-evaluable population) - Study IMpower130

	CnP (Actual) (N=232)	Atezo+CnP (Actual) (N=473)	All Patients (N=705)
4 cycles planned, number of cycles received during induction			
n	121	239	360
1 - 3	35 (28.9%)	59 (24.7%)	94 (26.1%)
4 - 5	86 (71.1%)	180 (75.3%)	266 (73.9%)
6 cycles planned, number of cycles received during induction			
n	111	233	344
1 - 3	38 (34.2%)	51 (21.9%)	89 (25.9%)
4 - 5	32 (28.8%)	71 (30.5%)	103 (29.9%)
6	41 (36.9%)	111 (47.6%)	152 (44.2%)
Treatment duration (M)			
n	232	472	704
Mean (SD)	2.6 (1.3)	2.8 (1.3)	2.7 (1.3)
Median	2.7	2.8	2.8
Min - Max	0 - 5	0 - 7	0 - 7
Treatment duration (M)			
n	232	472	704
0 to <= 3 months	159 (68.5%)	299 (63.3%)	458 (65.1%)
>3 months to <= 6 months	73 (31.5%)	172 (36.4%)	245 (34.8%)
>6 months to <= 12 months	0	1 (0.2%)	1 (0.1%)
Dose Intensity (%)			
n	232	472	704
Mean (SD)	84.2 (13.8)	82.6 (13.5)	83.1 (13.6)
Median	85.5	83.3	84.1
Min - Max	33 - 108	36 - 108	33 - 108
Number of doses			
n	232	472	704
Mean (SD)	10.0 (4.4)	10.6 (4.4)	10.4 (4.4)
Median	11.0	11.0	11.0
Min - Max	1 - 18	1 - 18	1 - 18
Total cumulative dose (mg)			
n	232	472	704
Mean (SD)	1702.7 (830.4)	1799.2 (811.6)	1767.4 (818.5)
Median	1664.5	1784.5	1738.0
Min - Max	182 - 4410	125 - 4464	125 - 4464

Atezo=Atezolizumab, CnP=Carboplatin+Nab-Paclitaxel.

M=Months. Treatment duration is the date of the last dose of study medication minus the date of the first dose plus one day.

Dose intensity is the number of doses actually received divided by the expected number of doses. Data Cutoff: 15MAR2018; RAVE Data Extracted: 18 May 2018.

Adverse events

Overview of adverse events

Table 48: Safety Summary Comparing IMpower130 and IMpower150

Parameter	IMpower 130 Atezo+CnP (n = 473)	IMpower 150 Atezo+CP (n = 400)
Total number of adverse events	8766	4851
Total number of patients with at least one:		
Adverse event	471 (99.6%)	391 (97.8%)
Treatment-related AE	455 (96.2%)	377 (94.3%)
Grade 3-4 AE	381 (80.5%)	230 (57.5%)
Treatment-related Grade 3-4 AE	346 (73.2%)	172 (43.0%)
Grade 5 AE	25 (5.3%)	10 (2.5%)
Treatment-related Grade 5 AE	8 (1.7%)	4 (1.0%)
Serious AE	240 (50.7%)	157 (39.3%)
Treatment-related serious AE	112 (23.7%)	78 (19.5%)
AE leading to withdrawal from any treatment	125 (26.4%)	53 (13.3%)
AE leading to any dose modification/interruption	402 (85.0%)	207 (51.8%)
Total number of patients with at least one AESI	213 (45.0%)	192 (48.0%)

AE = adverse event; AESI = adverse event of special interest; Atezo = atezolizumab; CCOD = clinical cutoff date; CnP = carboplatin+nab-paclitaxel; CP = carboplatin+paclitaxel.

Only events reported in the Adverse Events Form are included. Investigator text for AEs encoded using MedDRA v20.1. Percentages are based on N in the column headings. Multiple occurrences of the same AE in one individual are counted only once except for "Total number of events" row in which multiple occurrences of the same AE are counted separately. AEs collected after first treatment dose are included.

Table 49: Overview of Adverse Events (Safety-Evaluable Population), Safety Summary Profile, Adverse Event with Onset Date before Cross Over - Study IMpower130

	CnP (Actual) (N=232)	Atezo+CnP (Actual) (N=473)
Total number of patients with at least one adverse event	230 (99.1%)	471 (99.6%)
Total number of events	2990	8766
Total number of patients with at least one		
Treatment-related AE	215 (92.7%)	455 (96.2%)
Grade 3-4 AE	164 (70.7%)	381 (80.5%)
Treatment-related Grade 3-4 AE	140 (60.3%)	346 (73.2%)
Grade 5 AE	13 (5.6%)	25 (5.3%)
Treatment-related Grade 5 AE	1 (0.4%)	8 (1.7%)
Serious Adverse Event	88 (37.9%)	240 (50.7%)
Treatment-Related Serious Adverse Event	30 (12.9%)	112 (23.7%)
AE leading to withdrawal from any treatment	51 (22.0%)	125 (26.4%)
-MPD3280A	0	59 (12.5%)
-Nab-Paclitaxel	44 (19.0%)	97 (20.5%)
-Carboplatin	36 (15.5%)	85 (18.0%)
-Pemetrexed	6 (2.6%)	1 (0.2%)
AE leading to any dose modification/interruption	186 (80.2%)	402 (85.0%)
-MPD3280A	6 (2.6%)	297 (62.8%)
-Nab-Paclitaxel	181 (78.0%)	377 (79.7%)
-Carboplatin	158 (68.1%)	328 (69.3%)
-Pemetrexed	23 (9.9%)	8 (1.7%)

Atezo=Atezolizumab, CnP=Carboplatin+Nab-Paclitaxel.

Only events reported in the Adverse Events Form are included.

Investigator text for AEs encoded using MedDRA v21.0. Percentages are based on N in the column headings. Multiple occurrences of the same AE in one individual are counted only once except for "Total number of events" row in which multiple occurrences of the same AE are counted separately. Counts in "Grade 3-4 AE" are number of patients whose highest grades of AE are 3 or 4.

Data Cutoff: 15MAR2018; RAVE Data Extracted: 18 May 2018.

Most common adverse events

Table 50: Adverse events with an incidence of at least 10% in any treatment arm by system organ class and preferred term (safety evaluable population) - Study IMpower130

MedDRA System Organ Class MedDRA Preferred Term	CnP (Actual) (N=232)	Ateso+CnP (Actual) (N=473)
Total number of patients with at least one adverse event	230 (99.1%)	471 (99.6%)
GASTROINTESTINAL DISORDERS		
Total number of patients with at least one adverse event	173 (74.6%)	381 (80.5%)
NAUSEA	107 (46.1%)	234 (49.5%)
DIARRHOEA	73 (31.5%)	201 (42.5%)
CONSTIPATION	72 (31.0%)	171 (36.2%)
VOMITING	45 (19.4%)	128 (27.1%)
ABDOMINAL PAIN	18 (7.8%)	53 (11.2%)
BLOOD AND LYMPHATIC SYSTEM DISORDERS		
Total number of patients with at least one adverse event	181 (78.0%)	370 (78.2%)
ANEMIA	124 (53.4%)	265 (56.0%)
NEUTROPENIA	105 (45.3%)	220 (46.5%)
THROMBOCYTOPENIA	60 (25.9%)	133 (28.1%)
LEUKOPENIA	18 (7.8%)	51 (10.8%)
GENERAL DISORDERS AND ADMINISTRATION SITE CONDITIONS		
Total number of patients with at least one adverse event	161 (69.4%)	372 (78.6%)
FATIGUE	109 (47.0%)	223 (47.1%)
ASTHENIA	38 (16.4%)	87 (18.4%)
PYREXIA	23 (9.9%)	82 (17.3%)
OEDEMA PERIPHERAL	26 (11.2%)	64 (13.5%)
RESPIRATORY, THORACIC AND MEDIASTINAL DISORDERS		
Total number of patients with at least one adverse event	123 (53.0%)	304 (64.3%)
DYSPNOEA	47 (20.3%)	133 (28.1%)
COUGH	39 (16.8%)	126 (26.6%)
EPISTAXIS	28 (12.1%)	68 (14.4%)
NERVOUS SYSTEM DISORDERS		
Total number of patients with at least one adverse event	106 (45.7%)	279 (59.0%)
HEADACHE	23 (9.9%)	77 (16.3%)
DIZZINESS	25 (10.8%)	76 (16.1%)
PERIPHERAL SENSORY NEUROPATHY	23 (9.9%)	60 (12.7%)
DYSGEUSIA	14 (6.0%)	57 (12.1%)
NEUROPATHY PERIPHERAL	21 (9.1%)	53 (11.2%)
METABOLISM AND NUTRITION DISORDERS		
Total number of patients with at least one adverse event	114 (49.1%)	260 (55.0%)
DECREASED APPETITE	60 (25.9%)	142 (30.0%)
HYPOMAGNESEMIA	39 (16.8%)	95 (20.1%)
HYPOGLAECIA	26 (11.2%)	74 (15.6%)
DEHYDRATION	25 (10.8%)	49 (10.4%)
INFECTIONS AND INFESTATIONS		
Total number of patients with at least one adverse event	99 (42.7%)	252 (53.3%)
URINARY TRACT INFECTION	20 (8.6%)	62 (13.1%)
PNEUMONIA	19 (8.2%)	54 (11.4%)
SKIN AND SUBCUTANEOUS TISSUE DISORDERS		
Total number of patients with at least one adverse event	93 (40.1%)	247 (52.2%)
ALOPECIA	63 (27.2%)	151 (31.9%)
RASH	16 (6.9%)	66 (14.0%)
PRURITUS	12 (5.2%)	53 (11.2%)
INVESTIGATIONS		
Total number of patients with at least one adverse event	101 (43.5%)	227 (48.0%)
PLATELET COUNT DECREASED	39 (16.8%)	108 (22.8%)
NEUTROPHIL COUNT DECREASED	35 (15.1%)	95 (20.1%)
WEIGHT DECREASED	28 (12.1%)	53 (11.2%)
WHITE BLOOD CELL COUNT DECREASED	18 (7.8%)	52 (11.0%)
MUSCULOSKELETAL AND CONNECTIVE TISSUE DISORDERS		
Total number of patients with at least one adverse event	85 (36.6%)	239 (50.5%)
BACK PAIN	16 (6.9%)	82 (17.3%)
ARTHRALGIA	24 (10.3%)	72 (15.2%)
PAIN IN EXTREMITY	14 (6.0%)	52 (11.0%)
PSYCHIATRIC DISORDERS		
Total number of patients with at least one adverse event	45 (19.4%)	129 (27.3%)
INSOMNIA	31 (13.4%)	68 (14.4%)
ENDOCRINE DISORDERS		
Total number of patients with at least one adverse event	2 (0.9%)	74 (15.6%)
HYPOTHYROIDISM	1 (0.4%)	53 (11.2%)

Ateso=Atesolizumab, CnP=Carboplatin+Nab-Paclitaxel.

Investigator text for AEs encoded using MedDRA v21.0. Percentages are based on N in the column headings. For frequency counts by preferred term, multiple occurrences of the same AE in an individual are counted only once. For frequency counts of "Total number of events" rows, the multiple occurrences of the same AE in an individual are counted separately. AEs collected after first treatment dose are included.

Data Cutoff: 15MAR2018; RAVE Data Extracted: 18 May 2018.

Table 51: Summary of adverse events with at least 5% difference between treatment arms by system organ class and preferred term (safety evaluable population) - Study IMpower130

MedDRA System Organ Class MedDRA Preferred Term	CnP (Actual) (N=232)	Ateso+CnP (Actual) (N=473)
Total number of patients with at least one adverse event	230 (99.1%)	471 (99.6%)
Overall total number of events	2990	8766
ENDOCRINE DISORDERS		
Total number of patients with at least one adverse event	2 (0.9%)	74 (15.6%)
Total number of events	2	95
HYPOTHYROIDISM	1 (0.4%)	53 (11.2%)
GASTROINTESTINAL DISORDERS		
Total number of patients with at least one adverse event	173 (74.6%)	381 (80.5%)
Total number of events	544	1460
DIARRHOEA	73 (31.5%)	201 (42.5%)
CONSTIPATION	72 (31.0%)	171 (36.2%)
VOMITING	45 (19.4%)	128 (27.1%)
GENERAL DISORDERS AND ADMINISTRATION SITE CONDITIONS		
Total number of patients with at least one adverse event	161 (69.4%)	372 (78.6%)
Total number of events	331	870
PYREXIA	23 (9.9%)	82 (17.3%)
INVESTIGATIONS		
Total number of patients with at least one adverse event	101 (43.5%)	227 (48.0%)
Total number of events	276	778
PLATELET COUNT DECREASED	39 (16.8%)	108 (22.8%)
MUSCULOSKELETAL AND CONNECTIVE TISSUE DISORDERS		
Total number of patients with at least one adverse event	85 (36.6%)	239 (50.5%)
Total number of events	133	541
BACK PAIN	16 (6.9%)	82 (17.3%)
MUSCULOSKELETAL PAIN	10 (4.3%)	45 (9.5%)
NERVOUS SYSTEM DISORDERS		
Total number of patients with at least one adverse event	106 (45.7%)	279 (59.0%)
Total number of events	194	591
DIZZINESS	25 (10.8%)	76 (16.1%)
HEADACHE	23 (9.9%)	77 (16.3%)
DYSGEUSIA	14 (6.0%)	57 (12.1%)
RESPIRATORY, THORACIC AND MEDIASTINAL DISORDERS		
Total number of patients with at least one adverse event	123 (53.0%)	304 (64.3%)
Total number of events	231	775
DYSPNOEA	47 (20.3%)	133 (28.1%)
COUGH	39 (16.8%)	126 (26.6%)
SKIN AND SUBCUTANEOUS TISSUE DISORDERS		
Total number of patients with at least one adverse event	93 (40.1%)	247 (52.2%)
Total number of events	146	483
RASH	16 (6.9%)	66 (14.0%)
PRURITUS	12 (5.2%)	53 (11.2%)

Ateso=Atesolizumab, CnP=Carboplatin+Nab-Paclitaxel.

Investigator text for AEs encoded using MedDRA v21.0. Percentages are based on N in the column headings. For frequency counts by preferred term, multiple occurrences of the same AE in an individual are counted only once. For frequency counts of "Total number of events" rows, the multiple occurrences of the same AE in an individual are counted separately. AEs collected after first treatment dose are included.

Data Cutoff: 15MAR2018; RAVE Data Extracted: 18 May 2018.

Table 52: Adverse events related to study treatment with an incidence of at least 10% in any treatment arm by system organ class and preferred term (safety evaluable population) - Study IMpower130

MedDRA System Organ Class MedDRA Preferred Term	CnP (Actual) (N=232)	Atezo+CnP (Actual) (N=473)
Total number of patients with at least one adverse event	215 (92.7%)	455 (96.2%)
Overall total number of events	1720	4665
BLOOD AND LYMPHATIC SYSTEM DISORDERS		
Total number of patients with at least one adverse event	169 (72.8%)	354 (74.8%)
ANAEMIA	109 (47.0%)	248 (52.4%)
NEUTROPENIA	104 (44.8%)	218 (46.1%)
THROMBOCYTOPENIA	58 (25.0%)	127 (26.8%)
LEUKOPENIA	17 (7.3%)	48 (10.1%)
GASTROINTESTINAL DISORDERS		
Total number of patients with at least one adverse event	137 (59.1%)	325 (68.7%)
NAUSEA	96 (41.4%)	207 (43.8%)
DIARRHOEA	55 (23.7%)	150 (31.7%)
VOMITING	33 (14.2%)	99 (20.9%)
CONSTIPATION	33 (14.2%)	73 (15.4%)
GENERAL DISORDERS AND ADMINISTRATION SITE CONDITIONS		
Total number of patients with at least one adverse event	123 (53.0%)	277 (58.6%)
FATIGUE	91 (39.2%)	185 (39.1%)
ASTHENIA	31 (13.4%)	61 (12.9%)
SKIN AND SUBCUTANEOUS TISSUE DISORDERS		
Total number of patients with at least one adverse event	79 (34.1%)	207 (43.8%)
ALOPECIA	61 (26.3%)	145 (30.7%)
INVESTIGATIONS		
Total number of patients with at least one adverse event	73 (31.5%)	196 (41.4%)
PLATELET COUNT DECREASED	38 (16.4%)	105 (22.2%)
NEUTROPHIL COUNT DECREASED	33 (14.2%)	93 (19.7%)
WHITE BLOOD CELL COUNT DECREASED	17 (7.3%)	51 (10.8%)
NERVOUS SYSTEM DISORDERS		
Total number of patients with at least one adverse event	71 (30.6%)	196 (41.4%)
DYSGEUSIA	11 (4.7%)	51 (10.8%)
PERIPHERAL SENSORY NEUROPATHY	22 (9.5%)	50 (10.6%)
NEUROPATHY PERIPHERAL	17 (7.3%)	49 (10.4%)
METABOLISM AND NUTRITION DISORDERS		
Total number of patients with at least one adverse event	77 (33.2%)	177 (37.4%)
DECREASED APPETITE	46 (19.8%)	107 (22.6%)
HYPOMAGNEAEMIA	24 (10.3%)	62 (13.1%)
ENDOCRINE DISORDERS		
Total number of patients with at least one adverse event	1 (0.4%)	61 (12.9%)
HYPOTHYROIDISM	0	49 (10.4%)

Atezo=Atezolizumab, CnP=Carboplatin+Nab-Paclitaxel.
Investigator text for AEs encoded using MedDRA v21.0. Percentages are based on N in the column headings. For frequency counts by preferred term, multiple occurrences of the same AE in an individual are counted only once. For frequency counts of "Total number of events" rows, the multiple occurrences of the same AE in an individual are counted separately. AEs collected after first treatment dose are included.
Data Cutoff: 15MAR2018; RAVE Data Extracted: 18 May 2018.

Grade 3/4 AEs

Table 53: Adverse Events with Onset Date before Cross Over by System Organ Class, Preferred Term and Highest NCI CTCAE Grade - Study IMpower130

MedDRA System Organ Class MedDRA Preferred Term	Grade	CnP (Actual) (N=232)	Atezo+CnP (Actual) (N=473)
- Any adverse events -	- Any Grade -	230 (99.1%)	471 (99.6%)
	1	5 (2.2%)	9 (1.9%)
	2	48 (20.7%)	56 (11.8%)
	3	127 (54.7%)	250 (52.9%)
	4	37 (15.9%)	131 (27.7%)
	5	13 (5.6%)	25 (5.3%)

Table 54: Grade ≥3 adverse events with at least 2% difference between treatment arms by preferred term (safety evaluable population) - Study IMpower130

MedDRA Preferred Term	CnP (Actual) (N=232)	Ateso+CnP (Actual) (N=473)
Total number of patients with at least one adverse event	173 (74.6%)	400 (84.6%)
Total number of events	398	1176
NEUTROPENIA	66 (28.4%)	153 (32.3%)
ANEMIA	56 (24.1%)	151 (31.9%)
NEUTROPHIL COUNT DECREASED	20 (8.6%)	59 (12.5%)
THROMBOCYTOPENIA	16 (6.9%)	47 (9.9%)
WHITE BLOOD CELL COUNT DECREASED	7 (3.0%)	32 (6.8%)
HYPOKALAEMIA	6 (2.6%)	22 (4.7%)
DYSPNOEA	2 (0.9%)	22 (4.7%)
LUNG INFECTION	1 (0.4%)	12 (2.5%)
SYNCOPE	0	13 (2.7%)

Ateso=Atezolizumab, CnP=Carboplatin+Nab-Paclitaxel.

Investigator text for AEs encoded using MedDRA v21.0. Percentages are based on N in the column headings. For frequency counts by preferred term, multiple occurrences of the same AE in an individual are counted only once. For frequency counts of "Total number of events" rows, the multiple occurrences of the same AE in an individual are counted separately. AEs collected after first treatment dose are included.

Data Cutoff: 15MAR2018; RAVE Data Extracted: 18 May 2018.

Table 55: Grade 3-4 adverse events with incidence of at least 2% in either arm/population (safety evaluable population) - Study IMpower130 and pooled safety datasets

MedDRA System Organ Class MedDRA Preferred Term	IMpower130				
	CnP (N=232)	Ateso+CnP (N=473)	Ateso+Chemo Combo (N=2421)	Ateso Mono NSCLC (N=1636)	Ateso Mono (N=3178)
Total number of patients with at least one adverse event	173 (74.6%)	402 (85.0%)	1702 (70.3%)	708 (43.3%)	1564 (49.2%)
Overall total number of events	472	1415	5073	1402	3366
BLOOD AND LYMPHATIC SYSTEM DISORDERS					
Total number of patients with at least one adverse event	112 (48.3%)	270 (57.1%)	880 (36.3%)	65 (4.0%)	218 (6.9%)
Total number of events	204	528	1573	83	257
NEUTROPENIA	66 (28.4%)	153 (32.3%)	414 (17.1%)	5 (0.3%)	11 (0.3%)
ANEMIA	56 (24.1%)	151 (31.9%)	415 (17.1%)	40 (2.4%)	160 (5.0%)
THROMBOCYTOPENIA	16 (6.9%)	47 (9.9%)	162 (6.7%)	7 (0.4%)	21 (0.7%)
LEUKOPENIA	7 (3.0%)	22 (4.7%)	70 (2.9%)	0	0
FEBRILE NEUTROPENIA	6 (2.6%)	12 (2.5%)	137 (5.7%)	1 (<0.1%)	8 (0.3%)
INVESTIGATIONS					
Total number of patients with at least one adverse event	37 (15.9%)	107 (22.6%)	445 (18.4%)	59 (3.6%)	189 (5.9%)
Total number of events	72	224	891	82	274
NEUTROPHIL COUNT DECREASED	20 (8.6%)	59 (12.5%)	229 (9.5%)	1 (<0.1%)	2 (<0.1%)
PLATELET COUNT DECREASED	14 (6.0%)	37 (7.8%)	110 (4.5%)	1 (<0.1%)	6 (0.2%)
WHITE BLOOD CELL COUNT DECREASED	7 (3.0%)	32 (6.8%)	90 (3.7%)	2 (0.1%)	5 (0.2%)
INFECTIONS AND INFESTATIONS					
Total number of patients with at least one adverse event	22 (9.5%)	83 (17.5%)	355 (14.7%)	154 (9.4%)	347 (10.9%)
Total number of events	27	104	437	180	419
PNEUMONIA	10 (4.3%)	31 (6.6%)	132 (5.5%)	67 (4.1%)	89 (2.8%)
LUNG INFECTION	1 (0.4%)	12 (2.5%)	27 (1.1%)	8 (0.5%)	13 (0.4%)
URINARY TRACT INFECTION	1 (0.4%)	6 (1.3%)	24 (1.0%)	5 (0.3%)	72 (2.3%)
GENERAL DISORDERS AND ADMINISTRATION SITE CONDITIONS					
Total number of patients with at least one adverse event	24 (10.3%)	78 (16.5%)	272 (11.2%)	118 (7.2%)	265 (8.3%)
Total number of events	27	88	308	133	315
FATIGUE	14 (6.0%)	36 (7.6%)	105 (4.3%)	47 (2.9%)	109 (3.4%)
ASTHENIA	5 (2.2%)	16 (3.4%)	78 (3.2%)	22 (1.3%)	63 (2.0%)
METABOLISM AND NUTRITION DISORDERS					
Total number of patients with at least one adverse event	24 (10.3%)	70 (14.8%)	273 (11.3%)	118 (7.2%)	308 (9.7%)
Total number of events	32	97	367	150	387
HYPOKALAEMIA	6 (2.6%)	22 (4.7%)	51 (2.1%)	17 (1.0%)	32 (1.0%)
HYPONATRAEMIA	3 (1.3%)	11 (2.3%)	60 (2.5%)	42 (2.6%)	98 (3.1%)
DECREASED APPETITE	5 (2.2%)	10 (2.1%)	49 (2.0%)	12 (0.7%)	35 (1.1%)
HYPMAGNEAEMIA	4 (1.7%)	10 (2.1%)	20 (0.8%)	1 (<0.1%)	2 (<0.1%)
RESPIRATORY, THORACIC AND MEDIASTINAL DISORDERS					
Total number of patients with at least one adverse event	22 (9.5%)	65 (13.7%)	242 (10.0%)	195 (11.9%)	323 (10.2%)
Total number of events	23	90	303	256	415
DYSPNOEA	2 (0.9%)	22 (4.7%)	59 (2.4%)	68 (4.2%)	117 (3.7%)
PULMONARY EMBOLISM	7 (3.0%)	16 (3.4%)	50 (2.1%)	32 (2.0%)	59 (1.9%)
CHRONIC OBSTRUCTIVE PULMONARY DISEASE	2 (0.9%)	11 (2.3%)	29 (1.2%)	7 (0.4%)	9 (0.3%)
GASTROINTESTINAL DISORDERS					
Total number of patients with at least one adverse event	29 (12.5%)	63 (13.3%)	257 (10.6%)	91 (5.6%)	242 (7.6%)
Total number of events	40	93	342	113	319
DIARRHOEA	14 (6.0%)	25 (5.3%)	85 (3.5%)	14 (0.9%)	36 (1.1%)
NAUSEA	5 (2.2%)	16 (3.4%)	53 (2.2%)	17 (1.0%)	35 (1.1%)
VOMITING	5 (2.2%)	13 (2.7%)	43 (1.8%)	14 (0.9%)	26 (0.8%)

NERVOUS SYSTEM DISORDERS					
Total number of patients with at least one adverse event	13 (5.6%)	44 (9.3%)	169 (7.0%)	66 (4.0%)	127 (4.0%)
Total number of events	17	48	185	74	143
SYNCOPE	0	13 (2.7%)	26 (1.1%)	16 (1.0%)	29 (0.9%)
VASCULAR DISORDERS					
Total number of patients with at least one adverse event	6 (2.6%)	18 (3.8%)	114 (4.7%)	47 (2.9%)	103 (3.2%)
Total number of events	7	20	130	49	107
HYPERTENSION	2 (0.9%)	6 (1.3%)	61 (2.5%)	17 (1.0%)	42 (1.3%)

Investigator text for AEs encoded using MedDRA v21.0. Percentages are based on N in the column headings. Patients with one AE of highest grade 5 and other AEs of highest grade 3-4 are included in the overall and SOC total number rows counting for grade 3-4 in this table. For frequency counts by preferred term, multiple occurrences of the same AE in an individual are counted only once. For frequency counts of "Total number of events" rows, the multiple occurrences of the same AE in an individual are counted separately. All treatment emergent AEs are included. Sort order is by decreasing frequency in Atezo+CnP treatment arm.
Atezo+Rtezo+atumab, CnP=Carboplatin+Nab-Paclitaxel. Atezo+Chemo Combo: G029537 (IMPOWER130) Arm A + G029437 (IMPOWER131) Arm A+B + G029436 (IMPOWER150) Arm A+B + G029438 (IMPOWER132) Arm A + G030081 (IMPOWER133) Arm A. Atezo Mono NSCLC: G028915 (OAK) + G028753 (FOPLAR) + G028754 (BIRCH) +G028625 (FIR) + G027831 (PCD4985g - NSCLC Cohort). Atezo Mono: G028915 (OAK) + G028753 (FOPLAR) + G028754 (BIRCH) + G028625 (FIR) + G027831 (PCD4985g - All Cohorts) + G029293 (IMVIGOR210) + G029294 (IMVIGOR211) + W029074 (DAMOTION150 Arm B). Clinical cut-off dates: G027831:31MAR2016, G028625:07JAN2015, G028753:01DEC2015, G028754:01DEC2015, G029293:04JUL2016, G028915:07JUL2016, G029294:13MAR2017, G029436:22JAN2018, G029437:20APR2018, G029438:18MAY2018, G029537:15MAR2018, G030081:24APR2018, W029074:17OCT2016.

Table 56: Grade 4 AEs by SOC - Study IMpower130

Grade 4 Adverse Events (by System Organ Class)	CnP (n = 232)	Atezo+CnP (n = 473)
Total Number of Grade 4 AEs	37 (15.9%)	131 (27.7%)
Blood and lymphatic system disorders	24 (10.3%)	75 (15.9%)
Cardiac disorders	0	7 (1.5%)
Congenital, familial and genetic disorders	0	0
Ear and labyrinth disorders	0	0
Endocrine disorders	0	0
Eye disorders	0	0
Gastrointestinal disorders	0	3 (0.6%)
General disorders and administration site conditions	0	2 (0.4%)
Hepatobiliary disorders	0	1 (0.2%)
Immune system disorders	0	0
Infections and infestations	1 (0.4%)	16 (3.4%)
Injury, poisoning and procedural complications	0	1 (0.2%)
Investigations	10 (4.3%)	33 (7.0%)
Metabolism and nutrition disorders	4 (1.7%)	11 (2.3%)
Musculoskeletal and connective tissue disorders	1 (0.4%)	1 (0.2%)
Neoplasms benign, malignant and unspecified (incl cysts and polyps)	0	0
Nervous system disorders	1 (0.4%)	3 (0.6%)
Pregnancy, puerperium and perinatal conditions	0	0
Product Issues	0	0
Psychiatric disorders	0	3 (0.6%)
Renal and urinary disorders	0	3 (0.6%)
Reproductive system and breast disorders	0	0
Respiratory, thoracic and mediastinal disorders	4 (1.7%)	14 (3.0%)
Skin and subcutaneous tissue disorders	0	0
Social circumstances	0	0
Surgical and medical procedures	0	0
Vascular disorders	0	0

Atezo = atezolizumab; CnP = carboplatin+nab-paclitaxel. Investigator text for AEs encoded using MedDRA v21.0. Multiple occurrences of the same AE in one individual are counted once at the highest grade for this patient. To the SOC (System Organ Class) Overall row counts, a patient contributes only with the AE occurring with the highest grade within the SOC. Percentages are based on N in the column headings. AEs collected after first treatment dose are included. Data Cutoff: 15MAR2018; RAVE Data.

Table 57: Time to Onset for First Grade 4 Event of Neutropenia and Thrombocytopenia

Parameter	CnP (N = 232)	Atezo+CnP (N = 473)	All patients (N = 705)
Time to onset for first Grade 4 event of neutropenia (months):			
n	19 (8.2%)	60 (12.7%)	79 (11.2%)
Mean (SD)	1.77 (2.71)	1.29 (0.83)	1.40 (1.50)
Median	1.08	1.18	1.15
Min-Max	0.4-12.5	0.2-3.6	0.2-12.5
Time to onset for first Grade 4 event of thrombocytopenia (months):			
n	3 (1.3%)	16 (3.4%)	19 (2.7%)
Mean (SD)	1.76 (1.09)	0.85 (0.59)	1.00 (0.73)
Median	1.87	0.64	0.69
Min-Max	0.6-2.8	0.0-2.3	0.0-2.8

Table 58: Duration of Event for First Grade 4 Event of Neutropenia and Thrombocytopenia

Parameter	CnP (N = 232)	Atezo+CnP (N = 473)	All Patients (N = 705)
Time from onset to time of resolution for first Grade 4 event of neutropenia (months):			
n	19 (8.2%)	60 (12.7%)	79 (11.2%)
Median	0.26	0.44	0.33
95% CI	(0.23, 0.36)	(0.26, 0.49)	(0.26, 0.49)
25% and 75% percentile	0.23, 0.49	0.26, 0.95	0.23, 0.72
Range	0.2 to 1.9	0.1 to 16.8*	0.1 to 16.8*
Time from onset to time of resolution for first Grade 4 event of thrombocytopenia (months)			
n	3 (1.3%)	16 (3.4%)	19 (2.7%)
Median	0.49	0.57	0.56
95% CI	(0.13, 3.06)	(0.49, 1.41)	(0.49, 1.41)
25% and 75% percentile	0.13, 3.06	0.46, 2.22	0.43, 3.02
Range	0.1 to 3.1	0.2 to 20.1*	0.1 to 20.1*

Atezo = atezolizumab; CnP = carboplatin+nab–paclitaxel. * Censored, ^ Censored and event, NE = Not estimable. Summaries of duration of adverse events (median, percentiles) are Kaplan–Meier estimates. 95% CI for median was computed using the method of Brookmeyer and Crowley. Investigator text for AEs encoded using MedDRA v21.0. Data Cutoff: 15MAR2018; RAVE Data Extracted: 18 May 2018.

Adverse events by relationship/Treatment-related AEs

The most common treatment-related AEs with an incidence of $\geq 10\%$ in any arm are presented in the table below AEs considered by the investigator as related to any study treatment

Table 59: Most common treatment-related AEs with an incidence of >10% in any arm with Onset Date before Cross Over - Study IMpower130

MedDRA System Organ Class MedDRA Preferred Term	CnP (Actual) (N=232)	Atezo+CnP (Actual) (N=473)
Total number of patients with at least one adverse event	215 (92.7%)	455 (96.2%)
Overall total number of events	1720	4665
BLOOD AND LYMPHATIC SYSTEM DISORDERS		
Total number of patients with at least one adverse event	169 (72.8%)	354 (74.8%)
ANAEMIA	109 (47.0%)	248 (52.4%)
NEUTROPENIA	104 (44.8%)	218 (46.1%)
THROMBOCYTOPENIA	58 (25.0%)	127 (26.8%)
LEUKOPENIA	17 (7.3%)	48 (10.1%)
GASTROINTESTINAL DISORDERS		
Total number of patients with at least one adverse event	137 (59.1%)	325 (68.7%)
NAUSEA	96 (41.4%)	207 (43.8%)
DIARRHOEA	55 (23.7%)	150 (31.7%)
VOMITING	33 (14.2%)	99 (20.9%)
CONSTIPATION	33 (14.2%)	73 (15.4%)
GENERAL DISORDERS AND ADMINISTRATION SITE CONDITIONS		
Total number of patients with at least one adverse event	123 (53.0%)	277 (58.6%)
FATIGUE	91 (39.2%)	185 (39.1%)
ASTHENIA	31 (13.4%)	61 (12.9%)
SKIN AND SUBCUTANEOUS TISSUE DISORDERS		
Total number of patients with at least one adverse event	79 (34.1%)	207 (43.8%)
ALOPECIA	61 (26.3%)	145 (30.7%)
INVESTIGATIONS		
Total number of patients with at least one adverse event	73 (31.5%)	196 (41.4%)
PLATELET COUNT DECREASED	38 (16.4%)	105 (22.2%)
NEUTROPHIL COUNT DECREASED	33 (14.2%)	93 (19.7%)
WHITE BLOOD CELL COUNT DECREASED	17 (7.3%)	51 (10.8%)
NERVOUS SYSTEM DISORDERS		
Total number of patients with at least one adverse event	71 (30.6%)	196 (41.4%)
DYSGEUSIA	11 (4.7%)	51 (10.8%)
PERIPHERAL SENSORY NEUROPATHY	22 (9.5%)	50 (10.6%)
NEUROPATHY PERIPHERAL	17 (7.3%)	49 (10.4%)
METABOLISM AND NUTRITION DISORDERS		
Total number of patients with at least one adverse event	77 (33.2%)	177 (37.4%)
DECREASED APPETITE	46 (19.8%)	107 (22.6%)
HYPOMAGNEAEMIA	24 (10.3%)	62 (13.1%)
ENDOCRINE DISORDERS		
Total number of patients with at least one adverse event	1 (0.4%)	61 (12.9%)
HYPOTHYROIDISM	0	49 (10.4%)

Atezo=Atezolizumab, CnP=Carboplatin+Nab-Paclitaxel.
Investigator text for AEs encoded using MedDRA v21.0. Percentages are based on N in the column headings. For frequency counts by preferred term, multiple occurrences of the same AE in an individual are counted only once. For frequency counts of "Total number of events" rows, the multiple occurrences of the same AE in an individual are counted separately. AEs collected after first treatment dose are included.
Data Cutoff: 15MAR2018; RAVE Data Extracted: 18 May 2018.

Table 60: Summary of All Study Treatment-Related AEs and Atezolizumab Related AEs in Study IMpower130

Summary of AEs by System Organ Class	All Study Treatment Related IMpower 130 Atezo+CnP (n = 473)	Atezolizumab Only Related IMpower 130 Atezo+CnP (n = 473)
No. Patients with any AE	471 (99.6%)	471 (99.6%)
No. Patients with treatment-related AE	455 (96.2%)	354 (74.8%)
Blood and lymphatic system disorders	354 (74.8%)	100 (21.1%)
Cardiac disorders	17 (3.6%)	14 (3.0%)
Congenital, familial and genetic disorders	0	0
Ear and labyrinth disorders	12 (2.5%)	5 (1.1%)
Endocrine disorders	61 (12.9%)	60 (12.7%)
Eye disorders	19 (4.0%)	12 (2.5%)
Gastrointestinal disorders	325 (68.7%)	191 (40.4%)
General disorders and administration site conditions	277 (58.6%)	191 (40.4%)

Summary of AEs by System Organ Class	All Study Treatment Related IMpower 130 Atezo+CnP (n = 473)	Atezolizumab Only Related IMpower 130 Atezo+CnP (n = 473)
Hepatobiliary disorders	7 (1.5%)	5 (1.1%)
Immune system disorders	1 (0.2%)	1 (0.2%)
Infections and infestations	64 (13.5%)	35 (7.4%)
Injury, poisoning and procedural complications	13 (2.7%)	11 (2.3%)
Investigations	196 (41.4%)	100 (21.1%)
Metabolism and nutrition disorders	177 (37.4%)	93 (19.7%)
Musculoskeletal and connective tissue disorders	88 (18.6%)	75 (15.9%)
Neoplasms benign, malignant and unspecified (incl cysts and polyps)	0	0
Nervous system disorders	196 (41.4%)	77 (16.3%)
Pregnancy, puerperium and perinatal conditions	0	0
Psychiatric disorders	15 (3.2%)	9 (1.9%)
Renal and urinary disorders	39 (8.2%)	29 (6.1%)
Reproductive system and breast disorders	4 (0.8%)	4 (0.8%)
Respiratory, thoracic and mediastinal disorders	104 (22.0%)	68 (14.4%)
Skin and subcutaneous tissue disorders	207 (43.8%)	123 (26.0%)
Social circumstances	0	0
Surgical and medical procedures	0	0
Vascular disorders	25 (5.3%)	13 (2.7%)

AE=adverse event; Atezo=Atezolizumab; CCOD=clinical cutoff date; CP=Carboplatin+Paclitaxel.

Only events reported in the Adverse Events Form are included. Investigator text for AEs encoded using MedDRA v21.0. Percentages are based on N in the column headings. Multiple occurrences of the same AE in one individual are counted only once except for "Total number of events" row in which multiple occurrences of the same AE are counted separately. AEs collected after first treatment dose are included

Adverse drug reactions

The table below reflects the ADRs from the pooled safety data for atezolizumab including study IMpower133.

Atezolizumab monotherapy (n=3178)		System Organ Class ADR	Atezolizumab in combination therapy* (n=2759)	
Frequency (All Grades)	Incidence % (All Grades)		Frequency (All Grades)	Incidence % (All Grades)
Infections and infestations				
very common	368 (11.6%)	Urinary tract infection ^a	-	-
-	-	Lung infection ^b	very common	377 (13.7%)
Blood and Lymphatic System Disorders				
-	-	Anaemia	very common	916 (33.2%)
-	-	Neutropenia ^d	very common	930 (33.7%)
common	116 (3.7%)	Thrombocytopenia ^c	very common	586 (21.2%)
-	-	Leukopenia ^e	very common	322 (11.7%)
-	-	Lymphocyte count decreased	common	55 (2.0%)
Immune System Disorders				
common	51 (1.6%)	Infusion related reaction ^f	-	-
Endocrine Disorders				
uncommon	11 (0.3%)	Adrenal insufficiency ^j	-	-
uncommon	10 (0.3%)	Diabetes mellitus ⁱ	-	-
uncommon	30 (0.9%)	Hyperthyroidism ^h	-	-
rare	2 (<0.1%)	Hypophysitis ^k	-	-
common	164 (5.2%)	Hypothyroidism ^g	very common	420 (15.2%)
Metabolism and nutrition disorders				
very common	810 (25.5%)	Decreased appetite	very common	678 (24.6%)
common	138 (4.3%)	Hypokalemia	common	202 (7.3%)
-	-	Hypomagnesaemia	common	259 (9.4%)
common	169 (5.3%)	Hyponatremia	common	145 (5.3%)
common	103 (3.2%)	Hyperglycaemia	-	-

Nervous System Disorders				
-	-	Dizziness	very common	292 (10.6%)
-	-	Syncope	common	46 (1.7%)
uncommon	5 (0.2%)	Guillain-Barré syndrome m	-	-
uncommon	14 (0.4%)	Meningoencephalitis n	-	-
rare	1 (<0.1%)	Myasthenic syndrome o	-	-
-	-	Peripheral neuropathy l	very common	740 (26.8%)
-	-	Headache	very common	469 (17.0%)
Cardiac Disorders				
rare	2 (<0.1%)	Myocarditis ^p	-	-
Vascular Disorders				
common	102 (3.2%)	Hypotension	-	-
Respiratory, Thoracic, and Mediastinal Disorders				
very common	660 (20.8%)	Cough	very common	554 (20.1%)
very common	651 (20.5%)	Dyspnoea	very common	481 (17.4%)
common	73 (2.3%)	Hypoxia	-	-
common	101 (3.2%)	Nasal congestion	-	-
common	87 (2.7%)	Pneumonitis ^q	-	-
common	141 (4.4%)	Nasopharyngitis	-	-
-	-	Dysphonia	common	155 (5.6%)
Gastrointestinal Disorders				
common	268 (8.4%)	Abdominal pain	-	-
common	34 (1.1%)	Colitis ^s	-	-
-	-	Constipation	very common	745 (27.0%)
very common	626 (19.7%)	Diarrhoea ^r	very common	814 (29.5%)
common	82 (2.6%)	Dysphagia	-	-
very common	747 (23.5%)	Nausea	very common	1031 (37.4%)
uncommon	18 (0.6%)	Pancreatitis ^u	-	-
-	-	Stomatitis	common	259 (9.4%)
very common	477 (15.0%)	Vomiting	very common	527 (19.1%)
common	131 (4.1%)	Oropharyngeal pain ^t	-	-
-	-	Dysgeusia	common	199 (7.2%)
Hepatobiliary Disorders				
common	167 (5.3%)	ALT increased	common	219 (7.9%)
common	180 (5.7%)	AST increased	common	203 (7.4%)
common	62 (2.0%)	Hepatitis ^v	-	-
Skin and Subcutaneous Tissue Disorders				
very common	400 (12.6%)	Pruritus	very common	363 (13.2%)
very common	619 (19.5%)	Rash ^w	very common	785 (28.5%)
Musculoskeletal and Connective Tissue Disorders				
very common	441 (13.9%)	Arthralgia	very common	535 (19.4%)
very common	487 (15.3%)	Back pain	very common	373 (13.5%)
very common	414 (13.0%)	Musculoskeletal pain ^x	very common	510 (18.5%)
Uncommon	13 (0.4%)	Myositis ^y	-	-
Renal Disorders				
-	-	Proteinuria ^z	common	215 (7.8%)
rare	3 (<0.1%)	Nephritis ^{aa}	-	-
General Disorders and Administration				
very common	461 (14.5%)	Asthenia	very common	487 (17.7%)
common	207 (6.5%)	Chills	-	-
very common	1142 (35.9%)	Fatigue	very common	1003 (36.4%)
common	186 (5.9%)	Influenza like illness	-	-
very common	638 (20.1%)	Pyrexia	very common	473 (17.1%)

^a Includes reports of urinary tract infection, cystitis, pyelonephritis, escherichia urinary tract infection, urinary tract infection bacterial, kidney infection, pyelonephritis acute, urinary tract infection fungal, urinary tract infection pseudomonal.

^b Includes reports of pneumonia, bronchitis, lung infection, lower respiratory tract infection, infective exacerbation of COPD, infectious pleural effusion, tracheobronchitis, atypical pneumonia, lung abscess, pyopneumothorax.

^c Includes reports of thrombocytopenia and platelet count decreased.

^d Includes reports of neutropenia, neutrophil count decreased, febrile neutropenia, neutropenic sepsis, granulocytopenia.

^e Includes reports of white blood cell count decreased and leukopenia.

^f Includes reports of cytokine release syndrome, hypersensitivity, anaphylaxis.

^g Includes reports of autoimmune hypothyroidism, autoimmune thyroiditis, blood thyroid stimulating hormone abnormal, blood thyroid stimulating hormone decreased, blood thyroid stimulating hormone increased, euthyroid sick syndrome, goitre, hypothyroidism, myxoedema, thyroid disorder, thyroid function test abnormal, thyroiditis, thyroiditis acute, thyroxine decreased, thyroxine free decreased, thyroxine free increased, thyroxine increased, tri-iodothyronine decreased, tri-iodothyronine free abnormal, tri-iodothyronine free decreased, tri-iodothyronine free increased.

^h Includes reports of hyperthyroidism, Basedow's disease, endocrine ophthalmopathy, exophthalmos.

ⁱ Includes reports of diabetes mellitus, type 1 diabetes mellitus, diabetic ketoacidosis, ketoacidosis.

^j Includes reports of adrenal insufficiency and primary adrenal insufficiency.

^{ak} Includes reports of hypophysitis and temperature regulation disorder.

^l Includes reports of neuropathy peripheral, autoimmune neuropathy, peripheral sensory neuropathy, polyneuropathy, herpes zoster, peripheral motor neuropathy, neuralgic amyotrophy, peripheral sensorimotor neuropathy, toxic neuropathy, axonal neuropathy, lumbosacral plexopathy, neuropathic arthropathy, peripheral nerve infection.

^m Includes reports of Guillain-Barré syndrome and demyelinating polyneuropathy.

ⁿ Includes reports of encephalitis, meningitis, photophobia.

^{ao} Includes reports of myasthenia gravis.

^p Reported in studies outside the pooled dataset. The frequency is based on the program wide exposure.

^q Includes reports of pneumonitis, lung infiltration, bronchiolitis, interstitial lung disease, radiation pneumonitis.

^r Includes reports of diarrhoea, defaecation urgency, frequent bowel movements, and gastrointestinal hypermotility.

^s Includes reports of colitis, autoimmune colitis, colitis ischaemic, colitis microscopic, colitis ulcerative.

^t Includes reports of oropharyngeal pain, oropharyngeal discomfort and throat irritation.

^u Includes reports of autoimmune pancreatitis, pancreatitis, pancreatitis acute, lipase increased, amylase increased.

^v Includes reports of ascites, autoimmune hepatitis, hepatocellular injury, hepatitis, hepatitis acute, hepatotoxicity, liver disorder, drug-induced liver injury, hepatic failure, hepatic steatosis, hepatic lesion, oesophageal varices haemorrhage, varices oesophageal.

^w Includes reports of acne, acne pustular, dermatitis, dermatitis acneiform, dermatitis allergic, dermatitis bullous, dermatitis exfoliative generalised, drug eruption, eczema, eczema infected, erythema, erythema multiforme, erythema of eyelid, exfoliative rash, eyelid rash, fixed eruption, folliculitis, furuncle, generalised erythema, palmar-plantar erythrodysesthesia syndrome, rash, rash erythematous, rash generalised, rash macular, rash maculo-papular, rash papular, rash papulosquamous, rash pruritic, rash pustular, rash vesicular, seborrhoeic dermatitis, skin exfoliation, skin toxicity, skin ulcer, toxic epidermal necrolysis, toxic skin eruption.

^x Includes reports of musculoskeletal pain and myalgia.

^y Includes reports of myositis, rhabdomyolysis, polymyalgia rheumatica, dermatomyositis, muscle abscess, myoglobin urine present.

^z Includes reports of proteinuria, protein urine present, haemoglobinuria, nephrotic syndrome.

^{aa} Includes report of nephritis, Henoch-Schonlein Purpura nephritis.

*includes studies GO29436, GO29438, GO29537, GO30081, WO29074, WO29637, WO29522.

Serious adverse event/deaths/other significant events

SAE

Table 61: Serious Adverse Events, with Onset Date Before Cross Over by Medical Concept, Preferred Term and Highest NCI CTCAE Grade - Study IMpower130

Serious Adverse Events, with Onset Date Before Cross Over by Medical Concept, Preferred Term and Highest NCI CTCAE Grade
Safety-Evaluable Patients
Protocol: GO29537 (Data Cutoff: 15MAR2018)

AE of Special Interest Medical Concept MedDRA Preferred Term	Grade	CnP (Actual) (N=232)	Atezo+CnP (Actual) (N=473)	All Patients (N=705)
- Any adverse events -	- Any Grade -	5 (2.2%)	30 (6.3%)	35 (5.0%)
	1	0	2 (0.4%)	2 (0.3%)
	2	1 (0.4%)	8 (1.7%)	9 (1.3%)
	3	2 (0.9%)	14 (3.0%)	16 (2.3%)
	4	1 (0.4%)	3 (0.6%)	4 (0.6%)
	5	1 (0.4%)	3 (0.6%)	4 (0.6%)

Atezo=Atezolizumab, CnP=Carboplatin+Nab-Paclitaxel.

Investigator text for AEs encoded using MedDRA v21.0. Multiple occurrences of the same AE in one individual are counted once at the highest grade for this patient. To the SOC (System Organ Class) Overall row counts, a patient contributes only with the AE occurring with the highest grade within the SOC. Percentages are based on N in the column headings. AEs collected after first treatment dose are included.

Table 62: Serious adverse events with an incidence of at least 1% in either arm/population (safety evaluable population) - Study IMpower130 and pooled safety datasets

MedDRA Preferred Term	IMpower130				
	CnP (N=232)	Atezo+CnP (N=473)	Atezo+Chemo Combo (N=2421)	Atezo Mono NSCLC (N=1636)	Atezo Mono (N=3178)
Total number of patients with at least one adverse event	88 (37.9%)	240 (50.7%)	1073 (44.3%)	627 (38.3%)	1309 (41.2%)
Total number of events	138	489	2023	1043	2267
PNEUMONIA	14 (6.0%)	28 (5.9%)	144 (5.9%)	75 (4.6%)	98 (3.1%)
DIARRHOEA	2 (0.9%)	14 (3.0%)	52 (2.1%)	10 (0.6%)	21 (0.7%)
LUNG INFECTION	1 (0.4%)	14 (3.0%)	28 (1.2%)	9 (0.6%)	15 (0.5%)
NEUTROPENIA	2 (0.9%)	14 (3.0%)	34 (1.4%)	0	0
PULMONARY EMBOLISM	5 (2.2%)	14 (3.0%)	40 (1.7%)	20 (1.2%)	42 (1.3%)
ANAEMIA	8 (3.4%)	13 (2.7%)	49 (2.0%)	10 (0.6%)	29 (0.9%)
CHRONIC OBSTRUCTIVE PULMONARY DISEASE	3 (1.3%)	12 (2.5%)	35 (1.4%)	9 (0.6%)	10 (0.3%)
DYSPNOEA	1 (0.4%)	11 (2.3%)	32 (1.3%)	51 (3.1%)	89 (2.8%)
FEBRILE NEUTROPENIA	5 (2.2%)	9 (1.9%)	93 (3.8%)	1 (<0.1%)	7 (0.2%)
PLEURAL EFFUSION	0	8 (1.7%)	16 (0.7%)	26 (1.6%)	40 (1.3%)
PYREXIA	2 (0.9%)	8 (1.7%)	45 (1.9%)	32 (2.0%)	79 (2.5%)
PNEUMONITIS	3 (1.3%)	7 (1.5%)	51 (2.1%)	24 (1.5%)	35 (1.1%)
NEUTROPHIL COUNT DECREASED	0	6 (1.3%)	10 (0.4%)	0	1 (<0.1%)
PERICARDIAL EFFUSION	0	6 (1.3%)	12 (0.5%)	10 (0.6%)	14 (0.4%)
SEPSIS	2 (0.9%)	6 (1.3%)	27 (1.1%)	11 (0.7%)	41 (1.3%)
THROMBOCYTOPENIA	0	6 (1.3%)	31 (1.3%)	2 (0.1%)	6 (0.2%)
VOMITING	4 (1.7%)	6 (1.3%)	27 (1.1%)	11 (0.7%)	19 (0.6%)
BRONCHITIS	1 (0.4%)	5 (1.1%)	19 (0.8%)	5 (0.3%)	7 (0.2%)
CHEST PAIN	0	5 (1.1%)	11 (0.5%)	10 (0.6%)	13 (0.4%)
HAEMOPTYSIS	3 (1.3%)	5 (1.1%)	26 (1.2%)	18 (1.1%)	20 (0.6%)
INFLUENZA	1 (0.4%)	5 (1.1%)	11 (0.5%)	3 (0.2%)	5 (0.2%)
NAUSEA	4 (1.7%)	5 (1.1%)	17 (0.7%)	12 (0.7%)	19 (0.6%)
RENAL FAILURE	2 (0.9%)	5 (1.1%)	8 (0.3%)	2 (0.1%)	18 (0.6%)
SEPTIC SHOCK	2 (0.9%)	5 (1.1%)	13 (0.5%)	5 (0.3%)	11 (0.3%)
ACUTE KIDNEY INJURY	0	4 (0.8%)	20 (0.8%)	6 (0.4%)	37 (1.2%)
URINARY TRACT INFECTION	1 (0.4%)	4 (0.8%)	20 (0.8%)	4 (0.2%)	61 (1.9%)
BACK PAIN	0	3 (0.6%)	11 (0.5%)	14 (0.9%)	35 (1.1%)
DEATH	4 (1.7%)	1 (0.2%)	15 (0.6%)	9 (0.6%)	19 (0.6%)

Grade 5 AEs due to PD are excluded for studies G027831 and G028625.
Investigator text for AEs encoded using MedDRA v21.0. Percentages are based on N in the column headings. For frequency counts by preferred term, multiple occurrences of the same RE in an individual are counted only once. For frequency counts of "Total number of events" rows, the multiple occurrences of the same RE in an individual are counted separately. All treatment emergent AEs are included. Sort order is by decreasing frequency in Atezo+CnP treatment arm.
Atezo=Atezolizumab, CnP=Carboplatin+Nab-Paclitaxel. Atezo+Chemo Combo: G029537 (IMPOWER130) Arm A + G029437 (IMPOWER131) Arm A+B + G029436 (IMPOWER150) Arm A+B + G029438 (IMPOWER132) Arm A + G030081 (IMPOWER133) Arm A. Atezo Mono NSCLC: G028915 (OAK) + G028753 (POPLAR) + G028754 (BIRCH) + G028625 (FIR) + G027831 (PCD4989g - NSCLC Cohort). Atezo Mono: G028915 (OAK) + G028753 (POPLAR) + G028754 (BIRCH) + G028625 (FIR) + G027831 (PCD4989g - All Cohorts) + G029293 (IMVIGOR210) + G029294 (IMVIGOR211) + W029074 (IMMOTION150 Arm B). Clinical cut-off dates: G027831:31MAR2016, G028625:07JAN2015, G028753:01DEC2015, G028754:01DEC2015, G029293:04JUL2016, G028915:07JUL2016, G029294:13MAR2017, G029436:22JAN2018, G029437:20APR2018, G029438:18MAY2018, G029537:15MAR2018, G030081:24APR2018, W029074:17OCT2016.

Death

Table 63: Cases of deaths (safety evaluable population) - Study IMpower130 and pooled safety datasets

	IMpower130				
	CnP (N=232)	Atezo+CnP (N=473)	Atezo+Chemo Combo (N=2421)	Atezo Mono NSCLC (N=1636)	Atezo Mono (N=3178)
All death					
n	134 (57.8%)	240 (50.7%)	1277 (52.7%)	907 (55.4%)	1884 (59.3%)
<=30 days from last study drug administration	24 (10.3%)	41 (8.7%)	235 (9.7%)	167 (10.2%)	352 (11.1%)
>30 days from last study drug administration	110 (47.4%)	199 (42.1%)	1042 (43.0%)	740 (45.2%)	1532 (48.2%)
Primary cause of death					
n	134 (57.8%)	240 (50.7%)	1277 (52.7%)	907 (55.4%)	1884 (59.3%)
ADVERSE EVENT	15 (6.5%)	25 (5.3%)	143 (5.9%)	68 (4.2%)	120 (3.8%)
PROGRESSIVE DISEASE	114 (49.1%)	200 (42.3%)	1062 (43.9%)	774 (47.3%)	1463 (46.0%)
OTHER	5 (2.2%)	15 (3.2%)	72 (3.0%)	65 (4.0%)	301 (9.5%)

Deaths due to other includes unrelated adverse events outside of reporting window. Other Cause of Death is displayed verbatim.
Patient G029293-268804-1310 in Atezo Mono group has missing data for Other Cause of Death.
Atezo=Atezolizumab, CnP=Carboplatin+Nab-Paclitaxel. Atezo+Chemo Combo: G029537 (IMPOWER130) Arm A + G029437 (IMPOWER131) Arm A+B + G029436 (IMPOWER150) Arm A+B + G029438 (IMPOWER132) Arm A + G030081 (IMPOWER133) Arm A. Atezo Mono NSCLC: G028915 (OAK) + G028753 (POPLAR) + G028754 (BIRCH) + G028625 (FIR) + G027831 (PCD4989g - NSCLC Cohort). Atezo Mono: G028915 (OAK) + G028753 (POPLAR) + G028754 (BIRCH) + G028625 (FIR) + G027831 (PCD4989g - All Cohorts) + G029293 (IMVIGOR210) + G029294 (IMVIGOR211) + W029074 (IMMOTION150 Arm B). Clinical cut-off dates: G027831:31MAR2016, G028625:07JAN2015, G028753:01DEC2015, G028754:01DEC2015, G029293:04JUL2016, G028915:07JUL2016, G029294:13MAR2017, G029436:22JAN2018, G029437:20APR2018, G029438:18MAY2018, G029537:15MAR2018, G030081:24APR2018, W029074:17OCT2016.

Table 64: Adverse events leading to death by system organ class and preferred term (safety evaluable population) - Study IMpower130

MedDRA System Organ Class MedDRA Preferred Term	CnP (Actual) (N=232)	Atezo+CnP (Actual) (N=473)
Total number of patients with at least one adverse event	13 (5.6%)	25 (5.3%)
Overall total number of events	13	25
INFECTIONS AND INFESTATIONS		
Total number of patients with at least one adverse event	5 (2.2%)	8 (1.7%)
PNEUMONIA	2 (0.9%)	5 (1.1%)
SEPSIS	2 (0.9%)	1 (0.2%)
SEPTIC SHOCK	1 (0.4%)	1 (0.2%)
STAPHYLOCOCCAL SEPSIS	0	1 (0.2%)
RESPIRATORY, THORACIC AND MEDIASTINAL DISORDERS		
Total number of patients with at least one adverse event	3 (1.3%)	8 (1.7%)
PULMONARY EMBOLISM	1 (0.4%)	4 (0.8%)
PNEUMONITIS	1 (0.4%)	2 (0.4%)
ASPIRATION	0	1 (0.2%)
RESPIRATORY DISTRESS	0	1 (0.2%)
CHRONIC OBSTRUCTIVE PULMONARY DISEASE	1 (0.4%)	0
CARDIAC DISORDERS		
Total number of patients with at least one adverse event	0	7 (1.5%)
MYOCARDIAL INFARCTION	0	3 (0.6%)
CARDIAC ARREST	0	2 (0.4%)
CARDIO-RESPIRATORY ARREST	0	1 (0.2%)
VENTRICULAR TACHYCARDIA	0	1 (0.2%)
GENERAL DISORDERS AND ADMINISTRATION SITE CONDITIONS		
Total number of patients with at least one adverse event	5 (2.2%)	1 (0.2%)
DEATH	4 (1.7%)	1 (0.2%)
SUDDEN DEATH	1 (0.4%)	0
HEPATOBIILIARY DISORDERS		
Total number of patients with at least one adverse event	0	1 (0.2%)
HEPATIC CIRRHOSIS	0	1 (0.2%)

Atezo=Atezolizumab, CnP=Carboplatin+Nab-Paclitaxel.

Investigator test for AEs encoded using MedDRA v21.0. Percentages are based on N in the column headings. For frequency counts by preferred term, multiple occurrences of the same AE in an individual are counted only once. For frequency counts of "Total number of events" rows, the multiple occurrences of the same AE in an individual are counted separately. AEs collected after first treatment dose are included.

Data Cutoff: 15MAR2018; RAVE Data Extracted: 18 May 2018.

AESI for atezolizumab

Table 65: Summary of adverse events of special interest for atezolizumab (safety evaluable population) - Study IMpower130 and pooled safety datasets

	IMpower130				
	CnP (N=232)	Atezo+CnP (N=473)	Atezo+Chemo Combo (N=2421)	Atezo Mono NSCLC (N=1636)	Atezo Mono (N=3178)
Total number of patients with at least one AESI	58 (25.0%)	213 (45.0%)	1164 (48.1%)	532 (32.5%)	1098 (34.6%)
Total number of AESI events	91	436	2395	1016	2186
Total number of patients with at least one					
Treatment-related AESI	34 (14.7%)	168 (35.5%)	975 (40.3%)	382 (23.3%)	794 (25.0%)
Atezo-related AESI	2 (0.9%)	156 (33.0%)	880 (36.3%)	382 (23.3%)	794 (25.0%)
Grade 3-4 AESI	11 (4.7%)	33 (7.0%)	250 (10.3%)	105 (6.4%)	248 (7.8%)
Treatment-related Grade 3-4 AESI	6 (2.6%)	24 (5.1%)	213 (8.8%)	78 (4.8%)	173 (5.4%)
Atezo-related Grade 3-4 AESI	1 (0.4%)	22 (4.7%)	194 (8.0%)	78 (4.8%)	173 (5.4%)
Grade 5 AESI	1 (0.4%)	3 (0.6%)	11 (0.5%)	2 (0.1%)	4 (0.1%)
Treatment-related Grade 5 AESI	0	3 (0.6%)	11 (0.5%)	0	2 (<0.1%)
Atezo-related Grade 5 AESI	0	2 (0.4%)	10 (0.4%)	0	2 (<0.1%)
Serious AESI	5 (2.2%)	30 (6.3%)	173 (7.1%)	77 (4.7%)	151 (4.8%)
Treatment-related serious AESI	2 (0.9%)	26 (5.5%)	155 (6.4%)	64 (3.9%)	127 (4.0%)
Atezo-related serious AESI	0	25 (5.3%)	149 (6.2%)	64 (3.9%)	127 (4.0%)
AESI leading to any Treatment withdrawal	5 (2.2%)	13 (2.7%)	134 (5.5%)	35 (2.1%)	58 (1.8%)
AESI leading to Atezo withdrawal	0	12 (2.5%)	122 (5.0%)	35 (2.1%)	58 (1.8%)
AESI leading to Treatment modification/interruption	6 (2.6%)	52 (11.0%)	299 (12.4%)	108 (6.6%)	210 (6.6%)
AESI Treated with Systemic Corticosteroids	13 (5.6%)	70 (14.8%)	373 (15.4%)	115 (7.0%)	247 (7.8%)
AE of Special Interest Medical Concepts: patients with at least one					
Immune-Related Rash	35 (15.1%)	114 (24.1%)	616 (25.4%)	303 (18.5%)	620 (19.5%)
Immune-Related Hypothyroidism	1 (0.4%)	70 (14.8%)	277 (11.4%)	76 (4.6%)	164 (5.2%)
Immune-Related Hepatitis (Diagnosis and Lab Abnormalities)	19 (8.2%)	46 (9.7%)	345 (14.3%)	152 (9.3%)	343 (10.8%)
Immune-Related Hepatitis (Lab Abnormalities)	19 (8.2%)	39 (8.2%)	315 (13.0%)	137 (8.4%)	315 (9.9%)
Immune-Related Pneumonitis	3 (1.3%)	31 (6.6%)	134 (5.5%)	56 (3.4%)	87 (2.7%)
Immune-Related Hyperthyroidism	1 (0.4%)	23 (4.9%)	91 (3.8%)	15 (0.9%)	30 (0.9%)
Infusion-Related Reactions	3 (1.3%)	10 (2.1%)	70 (2.9%)	19 (1.2%)	34 (1.1%)
Immune-Related Hepatitis (Diagnosis)	3 (1.3%)	8 (1.7%)	44 (1.8%)	27 (1.7%)	62 (2.0%)
Immune-Related Adrenal Insufficiency	0	7 (1.5%)	18 (0.7%)	6 (0.4%)	12 (0.4%)
Immune-Related Colitis	1 (0.4%)	5 (1.1%)	38 (1.6%)	15 (0.9%)	34 (1.1%)
Autoimmune Hemolytic Anemia	0	4 (0.8%)	6 (0.2%)	3 (0.2%)	4 (0.1%)
Immune-Related Diabetes Mellitus	2 (0.9%)	4 (0.8%)	16 (0.7%)	5 (0.3%)	11 (0.3%)
Immune-Related Nephritis	0	4 (0.8%)	12 (0.5%)	1 (<0.1%)	3 (<0.1%)
Immune-Related Meningoencephalitis	0	3 (0.6%)	10 (0.4%)	9 (0.6%)	13 (0.4%)
Immune-Related Severe Cutaneous Reactions	0	3 (0.6%)	19 (0.8%)	12 (0.7%)	22 (0.7%)

AE of Special Interest Medical Concepts: patients with at least one					
Immune-Related Hypophysitis	0	2 (0.4%)	7 (0.3%)	1 (<0.1%)	2 (<0.1%)
Immune-Related Meningitis	0	2 (0.4%)	7 (0.3%)	7 (0.4%)	11 (0.3%)
Immune-Related Myositis	0	2 (0.4%)	6 (0.2%)	3 (0.2%)	8 (0.3%)
Immune-Related Pancreatitis	1 (0.4%)	2 (0.4%)	17 (0.7%)	8 (0.5%)	18 (0.6%)
Immune-Related Encephalitis	0	1 (0.2%)	3 (0.1%)	2 (0.1%)	2 (<0.1%)
Immune-Related Ocular Inflammatory Toxicity	0	1 (0.2%)	7 (0.3%)	9 (0.6%)	16 (0.5%)
Immune-Related Vasculitis	0	1 (0.2%)	6 (0.2%)	4 (0.2%)	7 (0.2%)
Rhabdomyolysis	0	1 (0.2%)	5 (0.2%)	2 (0.1%)	5 (0.2%)
Immune-Related Guillain-Barre Syndrome	0	0	2 (<0.1%)	5 (0.3%)	5 (0.2%)
Immune-Related Myasthenia Gravis	0	0	0	0	1 (<0.1%)
Immune-Related Myocarditis	0	0	1 (<0.1%)	0	0
Systemic Immune Activation	0	0	1 (<0.1%)	1 (<0.1%)	2 (<0.1%)

Grade 5 AEs due to PD are excluded for studies G027831 and G028625. Investigator text for AEs encoded using MedDRA v21.0. Percentages are based on N in the column headings. Multiple occurrences of the same AE in one individual are counted only once except for "Total number of AESI events" row in which multiple occurrences of the same AE are counted separately. All treatment emergent AEs are included. Sort order is by decreasing frequency in Atezo+CnP treatment arm.
Atezo=Atezolizumab, CnP=Carboplatin+Nab-Paclitaxel. Atezo+Chemo Combo: G029537 (IMPOWER130) Arm A + G029437 (IMPOWER131) Arm A+B + G029436 (IMPOWER150) Arm A+B + G029438 (IMPOWER132) Arm A + G030081 (IMPOWER133) Arm A. Atezo Mono NSCLC: G028915 (OAK) + G028753 (POPLAR) + G028754 (BIRCH) + G028625 (FIR) + G027831 (PCD4989g - NSCLC Cohort). Atezo Mono: G028915 (OAK) + G028753 (POPLAR) + G028754 (BIRCH) + G028625 (FIR) + G027831 (PCD4989g - All Cohorts) + G029293 (IMVIGOR210) + G029294 (IMVIGOR211) + W029074 (IMMOTION150 Arm B). Clinical cut-off dates: G027831:31MAR2016, G028625:07JAN2015, G028753:01DEC2015, G028754:01DEC2015, G029293:04JUL2016, G028915:07JUL2016, G029294:13MAR2017, G029436:22JAN2018, G029437:20APR2018, G029438:18MAY2018, G029537:15MAR2018, G030081:24APR2018, W029074:17OCT2016.

Important AESI for Atezolizumab (IMpower130 Atezo+CnP)

AESIs for atezolizumab by medical concepts grouped under important AESIs correspond to important identified risks.

Table 66: Summary of safety information for important adverse events of special interest for atezolizumab (IMpower130 Atezo+CnP, safety-evaluable population) - Study IMpower130

Important AESI N=213 (45.0%)	No. of Patients (%)				Months (Range)		No. of Patients (%)	
	Severity			Resolved (All Grades) ^a	Median Time to Onset (All Grades)	Median Duration (All Grades)	AESIs Leading to Atezolizumab Withdrawal (All Grades)	AESIs Requiring Corticosteroids (imAEs, All Grades)
	All Grades	Grade 3-4	Grade 5					
	Atezo+CnP	Atezo+CnP	Atezo+CnP		Atezo+CnP	Atezo+CnP	Atezo+CnP	Atezo+CnP
Immune-related:								
Hypothyroidism	70 (14.8)	3 (0.6%)	0	33 (47.1%)	5.29 (0.1-21.5)	12.4 (0*-24*)	0	8 (1.7%)
Hepatitis (diagnosis and laboratory abnormality)	46 (9.7%)	14 (3.0%)	1 (0.2%)	32 (69.6%)	1.38 (0.0-17.9)	1.3 (0-20*)	4 (0.8%)	18 (3.8%)
Hepatitis (diagnosis)	8 (1.7%)	2 (0.4%)	1 (0.2%)	5 (62.5%)	1.38 (0.3-6.4)	1.5 (0*-20*)	4 (0.8%)	7 (1.5%)
Hepatitis (laboratory abnormality)	39 (8.2%)	12 (2.5%)	0	27 (69.2%)	1.38 (0.0-17.9)	1.2 (0-20*)	0	11 (2.3%)
Hyperthyroidism	23 (4.9%)	1 (0.2%)	0	15 (65.2%)	4.57 (0.3-22.8)	3.1 (1-12*)	1 (0.2%)	1 (0.2%)
Infusion-related reactions	10 (2.1%)	1 (0.2%)	0	10 (100%)	1.08 (0.0-14.9)	0.0 (0-0)	0	1 (0.2%)
Pneumonitis	31 (6.6%)	0	2 (0.4%)	22 (71.0%)	3.52 (0.3-24.3)	2.1 (0-22*)	3 (0.6%)	21 (4.4%)
Colitis	5 (1.1%)	5 (1.1%)	0	4 (80.0%)	1.84 (0.4-17.6)	0.5 (0-1*)	2 (0.4%)	5 (1.1%)
Guillain-Barré syndrome	0	0	0	0	NE	NE	0	0
Pancreatitis	2 (0.4%)	0	0	1 (50.0%)	5.90 (0.5-11.3)	NE (0-6*)	0	0
Diabetes mellitus	4 (0.8%)	2 (0.4%)	0	1 (25.0%)	7.80 (1.4-17.9)	NE (0*-12*)	0	2 (0.4%)
Hypophysitis	2 (0.4%)	0	0	0	5.19 (5.1-5.3)	NE (13*-18*)	1 (0.2%)	2 (0.4%)
Nephritis	4 (0.8%)	3 (0.6%)	0	3 (75.0%)	3.37 (2.3-4.2)	0.9 (0-23*)	1 (0.2%)	4 (0.8%)
Adrenal insufficiency	7 (1.5%)	0	0	2 (28.6%)	7.13 (4.7-19.6)	NE (1-14*)	0	4 (0.8%)
Meningoencephalitis	3 (0.6%)	1 (0.2%)	0	2 (66.7%)	4.76 (1.7-6.2)	2.9 (1-3)	0	1 (0.2%)
Myocarditis	0	0	0	0	NE	NE	0	0
Myasthenia gravis	0	0	0	0	NE	NE	0	0

AESI=adverse event of special interest; Atezo=atezolizumab; CnP=carboplatin+nab-paclitaxel; imAE=immune-mediated adverse event; NE=not estimable.

* Censored.

^a Only the first event within each important AESI medical concept occurring for each patient is counted for resolution.

Hypophysitis occurred in 0.4% (2/473) of patients who received atezolizumab in combination with nab-paclitaxel and carboplatin. The median time to onset was 5.2 months (range: 5.1 to 5.3 months). Both patients required the use of corticosteroids.

Important AESI for Atezolizumab (Atezo+Chemo Combo)

Table 67: Summary of Safety Information for Important Adverse Events of Special Interest for Atezolizumab (Atezo+Chemo Combo, Safety-Evaluable Population) – pooled safety dataset

Important AE/SAE N=1164 (48.1%)	No. of Patients (%)				Months (Range)		No. of Patients (%)	
	Severity			Resolved (All Grades) *	Median Time to Onset (All Grades)	Median Duration (All Grades)	AE/SAEs Leading to Atezolizumab Withdrawal (All Grades)	AE/SAEs Requiring Corticosteroids (Immune, All Grades)
	All Grades	Grade 3-4	Grade 5					
Immune-related:								
Hypothyroidism	277 (11.4%)	6 (0.3%)	0	109 (39.4%)	5.32 (0.1-24.9)	NE (0*-26*)	3 (0.1%)	32 (1.3%)
Hepatitis (diagnosis and laboratory abnormality)	345 (14.3%)	100 (4.1%)	4 (0.2%)	270 (78.3%)	1.45 (0.0-26.9)	1.0 (0-24*)	33 (1.4%)	117 (4.8%)
Hepatitis (diagnosis)	44 (1.8%)	17 (0.7%)	3 (0.1%)	31 (70.5%)	2.60 (0.3-16.9)	1.9 (0-23*)	16 (0.7%)	24 (1.0%)
Hepatitis (laboratory abnormality)	315 (13.0%)	86 (3.6%)	1 (<0.1%)	246 (78.1%)	1.41 (0.0-28.9)	1.0 (0-24*)	17 (0.7%)	97 (4.0%)
Hyperthyroidism	91 (3.8%)	4 (0.2%)	0	67 (73.6%)	3.52 (0.0-22.8)	2.8 (0*-21*)	2 (<0.1%)	12 (0.5%)
Infusion-related reactions	70 (2.9%)	12 (0.5%)	0	70 (100%)	0.74 (0.0-14.9)	0.0 (0-0)	7 (0.3%)	8 (0.3%)
Pneumonitis	134 (5.5%)	33 (1.4%)	6 (0.2%)	85 (63.4%)	4.22 (0.3-27.4)	2.2 (0-22*)	43 (1.8%)	102 (4.2%)
Colitis	38 (1.6%)	24 (1.0%)	0	29 (76.3%)	2.30 (0.2-17.6)	1.0 (0-23*)	6 (0.2%)	26 (1.1%)
Gillain Barré syndrome	2 (<0.1%)	1 (<0.1%)	1 (<0.1%)	1 (50.0%)	3.14 (2.6-3.6)	NE (1-4*)	1 (<0.1%)	1 (<0.1%)
Pancreatitis	17 (0.7%)	6 (0.3%)	0	12 (70.6%)	4.37 (0.1-17.1)	0.7 (0-21*)	1 (<0.1%)	3 (0.1%)
Diabetes mellitus	16 (0.7%)	8 (0.3%)	0	2 (12.5%)	6.00 (0.8-17.9)	NE	0	5 (0.2%)
Hypophysitis	7 (0.3%)	2 (<0.1%)	0	2 (28.6%)	6.37 (5.0-17.5)	NE (0*-18*)	2 (<0.1%)	4 (0.2%)
Nephritis	12 (0.5%)	8 (0.3%)	0	7 (58.3%)	8.16 (2.3-17.9)	7.2 (0-23*)	6 (0.2%)	9 (0.4%)
Adrenal insufficiency	18 (0.7%)	3 (0.1%)	0	6 (33.3%)	6.90 (0.6-19.6)	NE (0-22*)	1 (<0.1%)	15 (0.6%)
Meningoencephalitis	10 (0.4%)	4 (0.2%)	0	9 (90.0%)	1.31 (0.5-6.2)	1.5 (0-7)	1 (<0.1%)	4 (0.2%)
Myocarditis	1 (<0.1%)	1 (<0.1%)	0	0	1.36 (1.4-1.4)	NE (1*-1*)	0	1 (<0.1%)
Myasthenia gravis	0	0	0	0	NE	NE	0	0

AESi=adverse event of special interest; Atezo=atezolizumab; ImAE=immune-mediated adverse event; NE=not estimable

* Censored.

* Only the first event within each important AESI medical concept occurring for each patient is counted for resolution.

Other AESI for Atezolizumab

Other AESIs include immune-related AEs (e.g., rash), potential risks, and class effects. These events are not considered important identified risks for atezolizumab as either the event is not considered severe (rash) or there is not enough evidence to confirm a causal relationship between the event and treatment with atezolizumab.

Table 68: Summary of Immune-Related skin reactions - Study IMpower130 and pooled safety datasets

IMpower130						
AE of Special Interest Medical Concept MedDRA Preferred Term	Grade	CnP (N=232)	Atezo+CnP (N=473)	Atezo+Chemo Combo (N=2421)	Atezo Mono NSCLC (N=1636)	Atezo Mono (N=3176)
IMMUNE-RELATED RASH						
- Overall -	- Any Grade -	35 (15.1%)	114 (24.1%)	616 (25.4%)	303 (18.5%)	620 (19.5%)
	1 or 2	33 (14.2%)	109 (23.0%)	563 (23.3%)	288 (17.6%)	585 (18.4%)
	3 or 4	2 (0.9%)	5 (1.1%)	53 (2.2%)	15 (0.9%)	34 (1.1%)
	5	0	0	0	0	1 (<0.1%)
RASH	- Any Grade -	16 (6.9%)	66 (14.0%)	335 (13.8%)	169 (10.3%)	358 (11.3%)
	1 or 2	15 (6.5%)	64 (13.5%)	314 (13.0%)	161 (9.8%)	344 (10.8%)
	3 or 4	1 (0.4%)	2 (0.4%)	21 (0.9%)	7 (0.4%)	14 (0.4%)
RASH MACULO-PAPULAR	- Any Grade -	7 (3.0%)	18 (3.8%)	75 (3.1%)	32 (2.0%)	83 (2.6%)
	1 or 2	7 (3.0%)	17 (3.6%)	65 (2.7%)	29 (1.8%)	75 (2.4%)
	3 or 4	0	1 (0.2%)	10 (0.4%)	3 (0.2%)	8 (0.3%)
ERYTHEMA	- Any Grade -	0	10 (2.1%)	67 (2.8%)	29 (1.8%)	69 (2.2%)
	1 or 2	0	10 (2.1%)	65 (2.7%)	29 (1.8%)	68 (2.1%)
	3 or 4	0	0	2 (<0.1%)	0	1 (<0.1%)
DERMATITIS ACNEIFORM	- Any Grade -	1 (0.4%)	6 (1.3%)	30 (1.2%)	12 (0.7%)	24 (0.8%)
	1 or 2	1 (0.4%)	6 (1.3%)	29 (1.2%)	12 (0.7%)	24 (0.8%)
	3 or 4	0	0	1 (<0.1%)	0	0
ECZEMA	- Any Grade -	0	5 (1.1%)	23 (1.0%)	13 (0.8%)	24 (0.8%)
	1 or 2	0	5 (1.1%)	23 (1.0%)	13 (0.8%)	24 (0.8%)
RASH ERYTHEMATOUS	- Any Grade -	0	4 (0.8%)	20 (0.8%)	6 (0.4%)	17 (0.5%)
	1 or 2	0	4 (0.8%)	19 (0.8%)	6 (0.4%)	17 (0.5%)
	3 or 4	0	0	1 (<0.1%)	0	0
SKIN ULCER	- Any Grade -	3 (1.3%)	4 (0.8%)	15 (0.6%)	7 (0.4%)	15 (0.5%)
	1 or 2	3 (1.3%)	4 (0.8%)	14 (0.6%)	7 (0.4%)	14 (0.4%)
	3 or 4	0	0	1 (<0.1%)	0	1 (<0.1%)
RASH PRURITIC	- Any Grade -	2 (0.9%)	3 (0.6%)	17 (0.7%)	14 (0.9%)	28 (0.9%)
	1 or 2	2 (0.9%)	3 (0.6%)	16 (0.7%)	13 (0.8%)	26 (0.8%)
	3 or 4	0	0	1 (<0.1%)	1 (<0.1%)	2 (<0.1%)
DERMATITIS	- Any Grade -	1 (0.4%)	0 (0.4%)	22 (0.9%)	13 (0.8%)	21 (0.7%)
	1 or 2	1 (0.4%)	0 (0.4%)	18 (0.7%)	12 (0.7%)	20 (0.6%)
	3 or 4	0	0	4 (0.2%)	1 (<0.1%)	1 (<0.1%)
ERYTHEMA MULTIIFORME	- Any Grade -	0	0 (0.4%)	12 (0.5%)	4 (0.2%)	9 (0.3%)
	1 or 2	0	0 (0.4%)	6 (0.2%)	3 (0.2%)	8 (0.3%)
	3 or 4	0	0	6 (0.2%)	1 (<0.1%)	1 (<0.1%)
FOLLICULITIS	- Any Grade -	1 (0.4%)	2 (0.4%)	13 (0.5%)	9 (0.3%)	12 (0.4%)
	1 or 2	1 (0.4%)	2 (0.4%)	13 (0.5%)	9 (0.3%)	12 (0.4%)
TOXIC EPIDERMAL NECROLYSIS						
- Any Grade -	- Any Grade -	0	0	0	0	2 (<0.1%)
3 or 4	3 or 4	0	0	0	0	1 (<0.1%)
5	5	0	0	0	0	1 (<0.1%)

IMMUNE-RELATED SEVERE CUTANEOUS REACTIONS						
- Overall -	- Any Grade -	0	3 (0.6%)	19 (0.8%)	12 (0.7%)	22 (0.7%)
	1 or 2	0	3 (0.6%)	13 (0.5%)	10 (0.6%)	18 (0.6%)
	3 or 4	0	0	6 (0.2%)	2 (0.1%)	3 (<0.1%)
	5	0	0	0	0	1 (<0.1%)
ERYTHEMA MULTIFORME	- Any Grade -	0	2 (0.4%)	12 (0.5%)	4 (0.2%)	9 (0.3%)
	1 or 2	0	2 (0.4%)	6 (0.2%)	3 (0.2%)	8 (0.3%)
	3 or 4	0	0	6 (0.2%)	1 (<0.1%)	1 (<0.1%)
	5	0	0	0	0	1 (<0.1%)
TOXIC SKIN ERUPTION	- Any Grade -	0	1 (0.2%)	3 (0.1%)	1 (<0.1%)	2 (<0.1%)
	1 or 2	0	1 (0.2%)	3 (0.1%)	1 (<0.1%)	2 (<0.1%)
DERMATITIS BULLOUS	- Any Grade -	0	0	2 (<0.1%)	5 (0.3%)	7 (0.2%)
	1 or 2	0	0	2 (<0.1%)	4 (0.2%)	6 (0.2%)
	3 or 4	0	0	0	1 (<0.1%)	1 (<0.1%)
	5	0	0	0	0	1 (<0.1%)
DERMATITIS EXFOLIATIVE GENERALISED	- Any Grade -	0	0	0	1 (<0.1%)	1 (<0.1%)
	1 or 2	0	0	0	1 (<0.1%)	1 (<0.1%)
EXFOLIATIVE RASH	- Any Grade -	0	0	2 (<0.1%)	1 (<0.1%)	1 (<0.1%)
	1 or 2	0	0	2 (<0.1%)	1 (<0.1%)	1 (<0.1%)
TOXIC EPIDERMAL NECROLYSIS	- Any Grade -	0	0	0	0	2 (<0.1%)
	3 or 4	0	0	0	0	1 (<0.1%)
	5	0	0	0	0	1 (<0.1%)

Laboratory findings

Table 69: Summary of clinically relevant shifts from baseline in laboratory parameters from Grade 0-2 at baseline to Grade 3-4 post-baseline (safety-evaluable population) - Study IMpower130

Laboratory Test Category	Grade Direction	CnP (Actual) (N=232)	Ateso+CnP (Actual) (N=473)
Chemistry			
Albumin (g/L)	Low	0/226	0/455
Alkaline Phosphatase (U/L)	High	2/227 (0.9%)	11/461 (2.4%)
Bilirubin (umol/L)	High	2/229 (0.9%)	2/459 (0.4%)
Calcium (mmol/L)	High	0/226	0/458
	Low	0/227	0/458
Creatinine (umol/L)	High	1/229 (0.4%)	14/463 (3.0%)
Glucose (mmol/L)	Low	0/225	0/460
Magnesium (mmol/L)	High	3/218 (1.4%)	11/445 (2.5%)
	Low	0/218	0/445
Phosphorus (mmol/L)	Low	0/218	0/435
Potassium (mmol/L)	High	1/229 (0.4%)	8/461 (1.7%)
	Low	0/229	0/461
SGOT/AST (U/L)	High	4/227 (1.8%)	9/458 (2.0%)
SGPT/ALT (U/L)	High	8/229 (3.5%)	11/462 (2.4%)
Sodium (mmol/L)	High	0/229	2/462 (0.4%)
	Low	0/221	0/457
Coagulation			
Activated Partial Thromboplastin Time (sec)	High	0/78	0/106
International Normalized Ratio (FRACTION)	High	1/91 (1.1%)	3/148 (2.0%)
Hematology			
Hemoglobin (g/L)	High	3/229 (1.3%)	5/465 (1.1%)
	Low	0/229	0/465
Lymphocytes Abs (10 ⁹ /L)	High	0/228	0/464
	Low	0/220	0/457
Neutrophils, Total, Abs (10 ⁹ /L)	Low	0/228	0/464
Platelet (10 ⁹ /L)	Low	0/229	0/466
White Blood Cell Count (10 ⁹ /L)	High	0/229	0/465
	Low	0/228	0/465

Ateso=Atesolisumab, CnP=Carboplatin+Nab-Paclitaxel.
 NCI-CTCAE=National Cancer Institute Common Terminology Criteria for Adverse Events. Extended
 NCI-CTCAE v4 was used.
 For each laboratory test, patients with at least one post-baseline assessment were included in the
 analysis. For each cell, the denominator is the number of patients with baseline values of NCI-
 CTCAE Grade 0-2 in the specified direction of abnormality or with missing baseline values.
 Data Cutoff: 15MAR2018; RAVE Data Extracted: 18 May 2018.

Hy's Law Analysis

Hy's law cases were defined as elevated ALT or AST ($>3 \times \text{ULN}$) in combination with either an elevated total bilirubin ($>2 \times \text{ULN}$) or clinical jaundice in the absence of cholestasis or other causes of hyperbilirubinaemia. A total of 4 patients were identified as having at least one elevated total bilirubin ($>2 \times \text{ULN}$) within 7 days after the latest elevated ALT or AST ($>3 \times \text{ULN}$), including 5 patients in the Atezo-CnP arm and 1 patient in the CnP arm.

Of these 4 patients, two patients were diagnosed with immune mediated hepatitis; both patients recovered and their liver enzymes returned to within normal levels. One patient was reported to have not recovered. This patient had a confirmed tumoural obstruction of the biliary tract. One patient was reported to have been recovering from the hepatic event, however this patient later died due to disease progression.

Table 70: Summary of clinically significant changes in electrocardiogram (safety-evaluable population) - Study IMpower130

Visit Name	CnP (Actual) (N=232)	Atezo+CnP (Actual) (N=473)	All Patients (N=705)
Screening			
n	202	408	610
Abnormal, not clinically significant	98 (48.5%)	166 (40.7%)	264 (43.3%)
Abnormal, clinically significant	0	4 (1.0%)	4 (0.7%)
Normal	103 (51.0%)	237 (58.1%)	340 (55.7%)
Unable to Evaluate	1 (0.5%)	1 (0.2%)	2 (0.3%)
Post-baseline			
n	57	83	140
Abnormal, not clinically significant	23 (40.4%)	27 (32.5%)	50 (35.7%)
Abnormal, clinically significant	0	3 (3.6%)	3 (2.1%)
Normal	34 (59.6%)	53 (63.9%)	87 (62.1%)

Atezo=Atezolizumab, CnP=Carboplatin+Nab-Paclitaxel.
Data Cutoff: 15MAR2018; RAVE Data Extracted: 18 May 2018.

Table 71: Summary of vital signs among subjects without abnormality at baseline (safety-evaluable population) - Study IMpower130

Parameter Description	Direction of Abnormality	CnP (Actual) (N=232)	Atezo+CnP (Actual) (N=473)
Diastolic Blood Pressure			
	High	86/162 (53.1%)	181/337 (53.7%)
	Low	50/162 (30.9%)	135/337 (40.1%)
Pulse Rate			
	High	59/194 (30.4%)	132/408 (32.4%)
	Low	30/194 (15.5%)	96/408 (23.5%)
Respiratory Rate			
	High	61/169 (36.1%)	175/342 (51.2%)
	Low	2/169 (1.2%)	12/342 (3.5%)
Systolic Blood Pressure			
	High	83/102 (81.4%)	166/208 (79.8%)
	Low	11/102 (10.8%)	21/208 (10.1%)
Temperature			
	High	20/129 (15.5%)	49/262 (18.7%)
	Low	86/129 (66.7%)	206/262 (78.6%)

Atezo=Atezolizumab, CnP=Carboplatin+Nab-Paclitaxel.
Abnormalities are based on normal ranges: Diastolic Blood Pressure 60-80 mmHg; Pulse Rate 60-100 beats/min; Respiratory Rate: 12-18 breaths/min; Systolic Blood Pressure: 90-120 mmHg; Temperature: 36.5-37.3 C ;
Table entries provide the number of patients with an on-treatment assessment abnormality in the direction specified among patients without abnormality at baseline or with missing baseline values. Baseline is the patient's last observation prior to initiation of study drug.
Data Cutoff: 15MAR2018; RAVE Data Extracted: 18 May 2018.

Safety in special populations

Intrinsic factors

Age

Table 72: Overview of safety by age (safety-evaluable population) – pooled safety dataset

	Atezo+Chemo Combo (N=2421)				
	<65 (N=1251)	≥65 (N=1170)	65 – 74 (N=927)	75 – 84 (N=234)	≥85 (N=9)
Total number of patients with at least one AE	1237 (98.9%)	1151 (98.4%)	909 (98.1%)	233 (99.6%)	9 (100%)
Total number of Events	17852	17236	13144	3977	115
Total number of patients with at least one					
Treatment-related AE	1169 (93.4%)	1110 (94.9%)	875 (94.4%)	226 (96.6%)	9 (100%)
Atezo-related AE	899 (71.9%)	840 (71.8%)	650 (70.1%)	184 (78.6%)	6 (66.7%)
Grade 3-4 AE	802 (64.1%)	798 (68.2%)	617 (66.6%)	174 (74.4%)	7 (77.8%)
Treatment-related Grade 3-4 AE	673 (53.8%)	707 (60.4%)	542 (58.5%)	159 (67.9%)	6 (66.7%)
Atezo-related Grade 3-4 AE	330 (26.4%)	328 (28.0%)	255 (27.5%)	70 (29.9%)	3 (33.3%)
Grade 5 AE	61 (4.9%)	83 (7.1%)	67 (7.2%)	14 (6.0%)	2 (22.2%)
Treatment-related Grade 5 AE	23 (1.8%)	27 (2.3%)	23 (2.5%)	3 (1.3%)	1 (11.1%)
Atezo-related Grade 5 AE	16 (1.3%)	15 (1.3%)	12 (1.3%)	3 (1.3%)	0
Serious AE	507 (40.5%)	566 (48.4%)	439 (47.4%)	122 (52.1%)	5 (55.6%)
Treatment-related serious AE	281 (22.5%)	298 (25.5%)	238 (25.7%)	57 (24.4%)	3 (33.3%)
Atezo-related serious AE	184 (14.7%)	176 (15.0%)	141 (15.2%)	34 (14.5%)	1 (11.1%)
AE leading to Treatment withdrawal	264 (21.1%)	324 (27.7%)	254 (27.4%)	67 (28.6%)	3 (33.3%)
AE leading to Atezo withdrawal	144 (11.5%)	175 (15.0%)	139 (15.0%)	33 (14.1%)	3 (33.3%)
AE leading to Treatment modification/interruption	806 (64.4%)	802 (68.5%)	623 (67.2%)	173 (73.9%)	6 (66.7%)

Investigator text for AEs encoded using MedDRA v21.0. Percentages are based on N in the column headings. Multiple occurrences of the same AE in one individual are counted only once except for "Total number of events" row in which multiple occurrences of the same AE are counted separately. All treatment emergent AEs are included.
 Atezo+Atezolizumab, Atezo+Chemo Combo: G029537 (IMPOWER130) Arm A + G029437 (IMPOWER131) Arm A+B + G029436 (IMPOWER150) Arm A+B + G029438 (IMPOWER132) Arm A + G030081 (IMPOWER133) Arm A. Clinical cut-off dates: G029436:22JAN2018, G029437:20APR2018, G029438:18MAY2018, G029537:18MAR2018, G030081:24APR2018.

Gender

Table 73: Overview of safety by gender (safety-evaluable population) - pooled safety dataset

	Atezo+Chemo Combo (N=2421)	
	Female (N=812)	Male (N=1609)
Total number of patients with at least one AE	802 (98.8%)	1586 (98.6%)
Total number of Events	13605	21483
Total number of patients with at least one		
Treatment-related AE	779 (95.9%)	1500 (93.2%)
Atezo-related AE	603 (74.3%)	1136 (70.6%)
Grade 3-4 AE	573 (70.6%)	1027 (63.8%)
Treatment-related Grade 3-4 AE	508 (62.6%)	872 (54.2%)
Atezo-related Grade 3-4 AE	249 (30.7%)	409 (25.4%)
Grade 5 AE	33 (4.1%)	111 (6.9%)
Treatment-related Grade 5 AE	15 (1.8%)	35 (2.2%)
Atezo-related Grade 5 AE	10 (1.2%)	21 (1.3%)
Serious AE	378 (46.6%)	695 (43.2%)
Treatment-related serious AE	208 (25.6%)	371 (23.1%)
Atezo-related serious AE	128 (15.8%)	232 (14.4%)
AE leading to Treatment withdrawal	186 (22.9%)	402 (25.0%)
AE leading to Atezo withdrawal	90 (11.1%)	229 (14.2%)
AE leading to Treatment modification/interruption	582 (71.7%)	1026 (63.8%)

Investigator text for AEs encoded using MedDRA v21.0. Percentages are based on N in the column headings. Multiple occurrences of the same AE in one individual are counted only once except for "Total number of events" row in which multiple occurrences of the same AE are counted separately. All treatment emergent AEs are included.
 Atezo+Atezolizumab, Atezo+Chemo Combo: G029537 (IMPOWER130) Arm A + G029437 (IMPOWER131) Arm A+B + G029436 (IMPOWER150) Arm A+B + G029438 (IMPOWER132) Arm A + G030081 (IMPOWER133) Arm A. Clinical cut-off dates: G029436:22JAN2018, G029437:20APR2018, G029438:18MAY2018, G029537:18MAR2018, G030081:24APR2018.

Race

Table 74: Overview of safety by race (safety-evaluable population) - pooled safety dataset

	Atezo+Chemo Combo (N=2421)			
	White (N=1986)	Black (N=39)	Asian (N=293)	Other (N=103)
Total number of patients with at least one AE	1956 (98.5%)	39 (100%)	291 (99.3%)	102 (99.0%)
Total number of Events	28044	817	4694	1533
Total number of patients with at least one				
Treatment-related AE	1863 (93.8%)	37 (94.9%)	284 (96.9%)	95 (92.2%)
Atezo-related AE	1355 (69.7%)	30 (76.9%)	248 (84.6%)	76 (73.8%)
Grade 3-4 AE	1271 (64.0%)	26 (66.7%)	232 (79.2%)	71 (68.9%)
Treatment-related Grade 3-4 AE	1061 (54.4%)	23 (59.0%)	218 (74.4%)	58 (56.3%)
Atezo-related Grade 3-4 AE	493 (24.8%)	11 (28.2%)	126 (43.0%)	28 (27.2%)
Grade 5 AE	126 (6.3%)	3 (7.7%)	10 (3.4%)	5 (4.9%)
Treatment-related Grade 5 AE	40 (2.0%)	1 (2.6%)	8 (2.7%)	1 (1.0%)
Atezo-related Grade 5 AE	23 (1.2%)	1 (2.6%)	6 (2.0%)	1 (1.0%)
Serious AE	873 (44.0%)	20 (51.3%)	136 (46.4%)	44 (42.7%)
Treatment-related serious AE	455 (22.9%)	10 (25.6%)	89 (30.4%)	25 (24.3%)
Atezo-related serious AE	266 (13.4%)	7 (17.9%)	70 (23.9%)	17 (16.5%)
AE leading to Treatment withdrawal	474 (23.9%)	8 (20.5%)	81 (27.6%)	25 (24.3%)
AE leading to Atezo withdrawal	253 (12.7%)	5 (12.8%)	49 (16.7%)	12 (11.7%)
AE leading to Treatment modification/interruption	1302 (65.6%)	28 (71.8%)	211 (72.0%)	67 (65.0%)

Investigator text for AEs encoded using MedDRA v21.0. Percentages are based on N in the column headings. Multiple occurrences of the same AE in one individual are counted only once except for "Total number of events" row in which multiple occurrences of the same AE are counted separately. All treatment emergent AEs are included.
Atezo+Pembrolizumab, Atezo+Chemo Combo: G029537 (IMPOWER130) Arm A + G029437 (IMPOWER131) Arm A+B + G029436 (IMPOWER150) Arm A+B + G029438 (IMPOWER132) Arm A + G030081 (IMPOWER133) Arm A. Clinical cut-off dates: G029436:22JAN2018, G029437:20APR2018, G029438:18MAY2018, G029537:15MAR2018, G030081:24APR2018.

Extrinsic factors

Region

Table 75: Overview of safety by region (safety-evaluable population) - pooled safety dataset

	Atezo+Chemo Combo (N=2421)				
	Asia-Pacific (N=264)	Australia (N=102)	Central and South America (N=118)	Europe and Middle East (N=1315)	North America (N=622)
Total number of patients with at least one AE	263 (99.6%)	101 (99.0%)	117 (99.2%)	1288 (97.9%)	619 (99.5%)
Total number of Events	4087	1574	1588	16250	11589
Total number of patients with at least one					
Treatment-related AE	258 (97.7%)	98 (96.1%)	110 (93.2%)	1213 (92.2%)	600 (96.5%)
Atezo-related AE	227 (86.0%)	69 (67.6%)	70 (59.3%)	887 (67.5%)	486 (78.1%)
Grade 3-4 AE	209 (79.2%)	67 (65.7%)	65 (55.1%)	808 (61.4%)	481 (77.5%)
Treatment-related Grade 3-4 AE	199 (75.4%)	55 (53.9%)	53 (44.9%)	685 (52.1%)	388 (62.4%)
Atezo-related Grade 3-4 AE	116 (43.9%)	21 (20.6%)	17 (14.4%)	336 (25.6%)	168 (27.0%)
Grade 5 AE	11 (4.2%)	5 (4.9%)	17 (14.4%)	73 (5.6%)	38 (6.1%)
Treatment-related Grade 5 AE	8 (3.0%)	0	6 (5.1%)	29 (2.2%)	7 (1.1%)
Atezo-related Grade 5 AE	6 (2.3%)	0	4 (3.4%)	17 (1.3%)	4 (0.6%)
Serious AE	116 (43.9%)	60 (58.8%)	44 (37.3%)	552 (42.0%)	301 (48.4%)
Treatment-related serious AE	81 (30.7%)	31 (30.4%)	21 (17.8%)	331 (25.2%)	115 (18.5%)
Atezo-related serious AE	66 (25.0%)	19 (18.6%)	10 (8.5%)	235 (17.9%)	60 (9.6%)
AE leading to Treatment withdrawal	73 (27.7%)	24 (23.5%)	24 (20.3%)	312 (23.7%)	155 (24.9%)
AE leading to Atezo withdrawal	45 (17.0%)	13 (12.7%)	16 (13.6%)	167 (12.7%)	78 (12.5%)
AE leading to Treatment modification/interruption	190 (72.0%)	69 (67.6%)	70 (59.3%)	825 (62.7%)	454 (73.0%)

Investigator text for AEs encoded using MedDRA v21.0. Percentages are based on N in the column headings. Multiple occurrences of the same AE in one individual are counted only once except for "Total number of events" row in which multiple occurrences of the same AE are counted separately. All treatment emergent AEs are included.
Atezo+Pembrolizumab, Atezo+Chemo Combo: G029537 (IMPOWER130) Arm A + G029437 (IMPOWER131) Arm A+B + G029436 (IMPOWER150) Arm A+B + G029438 (IMPOWER132) Arm A + G030081 (IMPOWER133) Arm A. Clinical cut-off dates: G029436:22JAN2018, G029437:20APR2018, G029438:18MAY2018, G029537:15MAR2018, G030081:24APR2018.

Tobacco use history

Table 76: Overview of safety by tobacco use history (safety-evaluable population) - pooled safety dataset

	Atezo+Chemo Combo (N=2421)		
	Never (N=328)	Current (N=613)	Previous (N=1480)
Total number of patients with at least one AE	317 (96.6%)	604 (98.5%)	1467 (99.1%)
Total number of Events	4673	8012	22403
Total number of patients with at least one			
Treatment-related AE	299 (91.2%)	572 (93.3%)	1408 (95.1%)
Atezo-related AE	233 (71.0%)	421 (68.7%)	1085 (73.3%)
Grade 3-4 AE	197 (60.1%)	397 (64.8%)	1006 (68.0%)
Treatment-related Grade 3-4 AE	171 (52.1%)	327 (53.3%)	882 (59.6%)
Atezo-related Grade 3-4 AE	73 (22.3%)	151 (24.6%)	434 (29.3%)
Grade 5 AE	11 (3.4%)	30 (4.9%)	103 (7.0%)
Treatment-related Grade 5 AE	3 (0.9%)	12 (2.0%)	35 (2.4%)
Atezo-related Grade 5 AE	3 (0.9%)	3 (0.5%)	25 (1.7%)
Serious AE	111 (33.8%)	255 (41.6%)	707 (47.8%)
Treatment-related serious AE	63 (19.2%)	128 (20.9%)	388 (26.2%)
Atezo-related serious AE	42 (12.8%)	72 (11.7%)	246 (16.6%)
AE leading to Treatment withdrawal	63 (19.2%)	134 (21.9%)	391 (26.4%)
AE leading to Atezo withdrawal	26 (7.9%)	67 (10.9%)	226 (15.3%)
AE leading to Treatment modification/interruption	198 (60.4%)	403 (65.7%)	1007 (68.0%)

Investigator text for AEs encoded using MedDRA v21.0. Percentages are based on N in the column headings. Multiple occurrences of the same AE in one individual are counted only once except for "Total number of events" row in which multiple occurrences of the same AE are counted separately. All treatment emergent AEs are included.
Atezo=Atezolizumab. Atezo+Chemo Combo: G029537 (IMPOWER130) Arm A + G029437 (IMPOWER131) Arm A+B + G029436 (IMPOWER150) Arm A+B + G029438 (IMPOWER132) Arm A + G030081 (IMPOWER133) Arm A. Clinical cut-off dates: G029436:22JAN2018, G029437:20APR2018, G029438:18MAY2018, G029537:15MAR2018, G030081:24APR2018.

Safety related to drug-drug interactions and other interactions

No drug-drug interaction studies have been submitted with this application.

Discontinuation due to adverse events

AEs leading to withdrawal from study

Overall study period

Table 77: Adverse events leading to treatment withdrawal in at least 1% of patients in any treatment arm by system organ class and preferred term (safety-evaluable population) - Study IMpower130

MedDRA System Organ Class MedDRA Preferred Term	CnP (Actual) (N=232)	Ateso+CnP (Actual) (N=473)
Total number of patients with at least one adverse event	51 (22.0%)	125 (26.4%)
Overall total number of events	63	172
BLOOD AND LYMPHATIC SYSTEM DISORDERS		
Total number of patients with at least one adverse event	11 (4.7%)	33 (7.0%)
THROMBOCYTOPENIA	4 (1.7%)	17 (3.6%)
NEUTROPENIA	6 (2.6%)	14 (3.0%)
ANAEMIA	2 (0.9%)	8 (1.7%)
NERVOUS SYSTEM DISORDERS		
Total number of patients with at least one adverse event	9 (3.9%)	20 (4.2%)
PERIPHERAL SENSORY NEUROPATHY	4 (1.7%)	3 (0.6%)
GENERAL DISORDERS AND ADMINISTRATION SITE CONDITIONS		
Total number of patients with at least one adverse event	6 (2.6%)	20 (4.2%)
FATIGUE	4 (1.7%)	7 (1.5%)
INFECTIONS AND INFESTATIONS		
Total number of patients with at least one adverse event	8 (3.4%)	13 (2.7%)
PNEUMONIA	3 (1.3%)	6 (1.3%)
RESPIRATORY, THORACIC AND MEDIASTINAL DISORDERS		
Total number of patients with at least one adverse event	5 (2.2%)	15 (3.2%)
DYSPNOEA	0	6 (1.3%)
INVESTIGATIONS		
Total number of patients with at least one adverse event	3 (1.3%)	16 (3.4%)
NEUTROPHIL COUNT DECREASED	1 (0.4%)	8 (1.7%)
PLATELET COUNT DECREASED	2 (0.9%)	8 (1.7%)

Ateso=Atezolizumab, CnP=Carboplatin+Nab-Paclitaxel.

Investigator text for AEs encoded using MedDRA v21.0. Percentages are based on N in the column headings. For frequency counts by preferred term, multiple occurrences of the same AE in an individual are counted only once. For frequency counts of "Total number of events" rows, the multiple occurrences of the same AE in an individual are counted separately. AEs collected after first treatment dose are included.

Data Cutoff: 15MAR2018; RAVE Data Extracted: 18 May 2018.

Induction phase

Table 78: Adverse events leading to treatment withdrawal during the induction phase in at least 1% of patients in any treatment arm by system organ class and preferred term (safety-evaluable population) Study IMpower130

MedDRA System Organ Class MedDRA Preferred Term	CnP (Actual) (N=232)	Ateso+CnP (Actual) (N=473)
Total number of patients with at least one adverse event	33 (14.2%)	83 (17.5%)
Overall total number of events	38	106
BLOOD AND LYMPHATIC SYSTEM DISORDERS		
Total number of patients with at least one adverse event	9 (3.9%)	25 (5.3%)
THROMBOCYTOPENIA	3 (1.3%)	11 (2.3%)
NEUTROPENIA	4 (1.7%)	10 (2.1%)
ANAEMIA	1 (0.4%)	7 (1.5%)
GENERAL DISORDERS AND ADMINISTRATION SITE CONDITIONS		
Total number of patients with at least one adverse event	4 (1.7%)	11 (2.3%)
FATIGUE	3 (1.3%)	5 (1.1%)
INVESTIGATIONS		
Total number of patients with at least one adverse event	1 (0.4%)	10 (2.1%)
PLATELET COUNT DECREASED	1 (0.4%)	6 (1.3%)
NEUTROPHIL COUNT DECREASED	0	5 (1.1%)

Ateso=Atezolizumab, CnP=Carboplatin+Nab-Paclitaxel.

Investigator text for AEs encoded using MedDRA v21.0. Percentages are based on N in the column headings. For frequency counts by preferred term, multiple occurrences of the same AE in an individual are counted only once. For frequency counts of "Total number of events" rows, the multiple occurrences of the same AE in an individual are counted separately. AEs collected after first treatment dose are included.

Data Cutoff: 15MAR2018; RAVE Data Extracted: 18 May 2018.

Maintenance phase

Adverse Events Leading to Withdrawal of Atezolizumab Treatment

A total of 24 patients (6.8%) in the Atezo-CnP arm experienced AEs leading to withdrawal of atezolizumab during the maintenance phase. The most common AE (>2 patients) leading to atezolizumab treatment withdrawal during maintenance was dyspnoea.

Adverse Events Leading to Withdrawal of Pemetrexed Treatment (Pemetrexed Switch Maintenance Phase)

A total of 6 patients (12.2%) in the CnP arm experienced AEs leading to withdrawal of pemetrexed. The most common AEs (>1% of patients) causing withdrawal from pemetrexed were fatigue, death (PT), pleural effusion, neutropenia, sepsis, creatine renal clearance increased, acute kidney injury, and drug eruption, each leading to pemetrexed withdrawal in 2.0% of patients in the CnP arm.

AEs leading to dose modification/interruption

Table 79: Adverse events leading to dose modification/interruption in at least 2% of patients in any treatment arm (onset date before cross-over) by system organ class and preferred term (safety-evaluable population) Study IMpower130

MedDRA System Organ Class MedDRA Preferred Term	CnP (Actual) (N=232)	Atezo+CnP (Actual) (N=473)
Total number of patients with at least one adverse event	186 (80.2%)	402 (85.0%)
Overall total number of events	544	1472
BLOOD AND LYMPHATIC SYSTEM DISORDERS		
Total number of patients with at least one adverse event	128 (55.2%)	255 (53.9%)
NEUTROPENIA	83 (35.8%)	182 (38.5%)
THROMBOCYTOPENIA	46 (19.8%)	93 (19.7%)
ANAEMIA	31 (13.4%)	77 (16.3%)
LEUKOPENIA	13 (5.6%)	27 (5.7%)
FEBRILE NEUTROPENIA	5 (2.2%)	7 (1.5%)
INVESTIGATIONS		
Total number of patients with at least one adverse event	57 (24.6%)	137 (29.0%)
Total number of events	109	285
PLATELET COUNT DECREASED	26 (11.2%)	71 (15.0%)
NEUTROPHIL COUNT DECREASED	31 (13.4%)	68 (14.4%)
WHITE BLOOD CELL COUNT DECREASED	9 (3.9%)	27 (5.7%)
BLOOD CREATININE INCREASED	1 (0.4%)	11 (2.3%)
GENERAL DISORDERS AND ADMINISTRATION SITE CONDITIONS		
Total number of patients with at least one adverse event	33 (14.2%)	85 (18.0%)
FATIGUE	13 (5.6%)	35 (7.4%)
PYREXIA	3 (1.3%)	23 (4.9%)
ASTHENIA	11 (4.7%)	10 (2.1%)
GASTROINTESTINAL DISORDERS		
Total number of patients with at least one adverse event	24 (10.3%)	80 (16.9%)
Total number of events	38	128
DIARRHOEA	13 (5.6%)	43 (9.1%)
NAUSEA	4 (1.7%)	21 (4.4%)
VOMITING	3 (1.3%)	15 (3.2%)
INFECTIONS AND INFESTATIONS		
Total number of patients with at least one adverse event	30 (12.9%)	72 (15.2%)
PNEUMONIA	11 (4.7%)	18 (3.8%)
RESPIRATORY, THORACIC AND MEDIASTINAL DISORDERS		
Total number of patients with at least one adverse event	17 (7.3%)	69 (14.6%)
DYSPNOEA	5 (2.2%)	17 (3.6%)
PNEUMONITIS	1 (0.4%)	13 (2.7%)
METABOLISM AND NUTRITION DISORDERS		
Total number of patients with at least one adverse event	10 (4.3%)	33 (7.0%)
DEHYDRATION	1 (0.4%)	12 (2.5%)
DECREASED APPETITE	4 (1.7%)	11 (2.3%)

Atezo=Atezolizumab, CnP=Carboplatin+Nab-Paclitaxel.

Investigator report for AEs encoded using MedDRA v21.0. Percentages are based on N in the column headings. For frequency counts by preferred term, multiple occurrences of the same AE in an individual are counted only once. For frequency counts of "Total number of events" rows, the multiple occurrences of the same AE in an individual are counted separately. AEs collected after first treatment dose are included.

Data Cutoff: 15MAR2018; RAVE Data Extracted: 18 May 2018.

Immunogenicity

Table 80: Exposure to atezolizumab by atezolizumab ADA status (ADA-evaluable population) - Study IMpower130

	ADA- (N=346)	ADA+ (N=100)	All ADA Evaluable Patients (N=446)
Treatment duration (M)			
n	346	100	446
Mean (SD)	9.8 (7.2)	8.3 (6.5)	9.4 (7.1)
Median	7.8	6.1	7.6
Min - Max	0 - 32	0 - 26	0 - 32
Number of doses			
n	346	100	446
Mean (SD)	13.9 (9.9)	12.0 (8.9)	13.5 (9.7)
Median	11.0	9.0	10.5
Min - Max	1 - 45	1 - 38	1 - 45

Atezo=Atezolizumab, CnP=Carboplatin+Nab-Paclitaxel.

ADA = Anti-Therapeutic Antibodies; ADA- = Without TX Enhanced/Induced; ADA+ = With TX Enhanced/Induced.

M=Months. Treatment duration is the date of the last dose of study medication minus the date of the first dose plus one day.

Dose intensity is the number of doses actually received divided by the expected number of doses.

Data Cutoff: 15MAR2018; RAVE Data Extracted: 18 May 2018.

Table 81: Overview of adverse events by atezolizumab ADA status (ADA-evaluable population) - Study IMpower130

	ADA- (N=346)	ADA+ (N=100)
Total number of patients with at least one adverse event	346 (100.0%)	100 (100.0%)
Total number of events	6781	1760
Total number of patients with at least one		
Treatment-related AE	334 (96.5%)	100 (100.0%)
Grade 3-4 AE	285 (82.4%)	81 (81.0%)
Treatment-related Grade 3-4 AE	259 (74.9%)	74 (74.0%)
Grade 5 AE	15 (4.3%)	3 (3.0%)
Treatment-related Grade 5 AE	4 (1.2%)	1 (1.0%)
Serious Adverse Event	166 (48.0%)	54 (54.0%)
Treatment-Related Serious Adverse Event	78 (22.5%)	23 (23.0%)
AE leading to withdrawal from any treatment	83 (24.0%)	31 (31.0%)
-MPDL3280A	33 (9.5%)	17 (17.0%)
-Nab-Paclitaxel	64 (18.5%)	22 (22.0%)
-Carboplatin	57 (16.5%)	17 (17.0%)
-Pemetrexed	0	1 (1.0%)
AE leading to any dose modification/interruption	295 (85.3%)	93 (93.0%)
-MPDL3280A	215 (62.1%)	72 (72.0%)
-Nab-Paclitaxel	278 (80.3%)	86 (86.0%)
-Carboplatin	242 (69.9%)	75 (75.0%)
-Pemetrexed	6 (1.7%)	1 (1.0%)

Atezo=Atezolizumab, CnP=Carboplatin+Nab-Paclitaxel.

ADA = Anti-Therapeutic Antibodies; ADA- = Without TX Enhanced/Induced; ADA+ = With TX Enhanced/Induced.

Only events reported in the Adverse Events Form are included.

Investigator text for AEs encoded using MedDRA v21.0. Percentages are based on N in the column headings. Multiple occurrences of the same AE in one individual are counted only once except for "Total number of events" row in which multiple occurrences of the same AE are counted separately.

Counts in "Grade 3-4 AE" are number of patients whose highest grades of AE are 3 or 4.

Data Cutoff: 15MAR2018; RAVE Data Extracted: 18 May 2018.

Table 82: Adverse events with at least 2% higher incidence in ADA-positive patients by preferred term and atezolizumab ADA status (ADA-evaluable population) - Study IMpower130

System Organ Class	Preferred Term	Atezo+CnP, No. of Patients (%)	
		Atezo ADA-Negative (N=346)	Atezo ADA-Positive (N=100)
Gastrointestinal Disorders	Nausea	168 (48.6%)	54 (54.0%)
	Dyspepsia	19 (5.5%)	10 (10.0%)
	Abdominal pain upper	12 (3.5%)	9 (9.0%)
	Dry mouth	15 (4.3%)	6 (6.0%)
General Disorders and Administration Site Conditions	Asthenia	59 (17.1%)	23 (23.0%)
	Pyrexia	56 (16.2%)	23 (23.0%)
	Oedema peripheral	42 (12.1%)	18 (18.0%)
	Chills	14 (4.0%)	8 (8.0%)
	Face oedema	2 (0.6%)	3 (3.0%)
Blood and Lymphatic System Disorders	Neutropenia	162 (46.8%)	54 (54.0%)
	Thrombocytopenia	93 (26.9%)	34 (34.0%)
	Leukopenia	37 (10.7%)	14 (14.0%)
Respiratory, Thoracic and Mediastinal Disorders	Dyspnoea	97 (28.0%)	33 (33.0%)
	Haemoptysis	18 (5.2%)	10 (10.0%)
	Pleural effusion	6 (1.7%)	8 (8.0%)
Nervous System Disorders	Dysgeusia	41 (11.8%)	15 (15.0%)
	Neurotoxicity	3 (0.9%)	5 (5.0%)
Metabolism and Nutrition Disorders	Hypokalaemia	51 (14.7%)	18 (18.0%)
	Dehydration	35 (10.1%)	12 (12.0%)
	Hyponatraemia	20 (5.8%)	8 (8.0%)
	Hypoalbuminaemia	8 (2.3%)	5 (5.0%)
Infections and Infestations	Bronchitis	21 (6.1%)	9 (9.0%)
	Herpes zoster	3 (0.9%)	3 (3.0%)
Musculoskeletal and Connective Tissue Disorders	Back pain	60 (17.3%)	22 (22.0%)
	Pain in extremity	36 (10.4%)	16 (16.0%)
	Bone pain	15 (4.3%)	7 (7.0%)
Cardiac Disorders	Palpitations	7 (2.0%)	4 (4.0%)
Injury, Poisoning and Procedural Complications	Infusion-related reaction	5 (1.4%)	6 (6.0%)

ADA=anti-drug antibody; AE=adverse event; Atezo=atezolizumab; CnP=carboplatin+nab-paclitaxel.

Note: Investigator text for AEs encoded using MedDRA v21.0. Percentages were based on N in the column headings. For frequency counts by Preferred Term, multiple occurrences of the same AE in an individual were counted only once. AEs collected after first treatment dose were included.

Table 83: Adverse events of special interest with at least 2% difference between ADA-negative and ADA-positive patients by preferred term and atezolizumab ADA status (ADA-evaluable population) - Study IMpower130

MedDRA Preferred Term	Atezo+CnP, No. of Patients (%)	
	Atezo ADA-Negative (N=346)	Atezo ADA-Positive (N=100)
Rash maculopapular	12 (3.5%)	6 (6.0%)
Hypothyroidism	46 (13.3%)	7 (7.0%)
Pneumonitis	20 (5.8%)	3 (3.0%)
Hyperthyroidism	20 (5.8%)	3 (3.0%)
Infusion-related reaction	5 (1.4%)	5 (5.0%)

ADA=anti-drug antibody; AE=adverse event; Atezo=atezolizumab;
CnP=carboplatin+nab-paclitaxel.

Table 84: Safety by Treatment-Emergent ADA Status in Atezolizumab + Chemo Combination Population (Pooled Safety Population)

	Atezo+Chemo Combo (N=2267)	
	ADA- (N=1543)	ADA+ (N=724)
Total number of patients with at least one AE	1532 (99.3%)	718 (99.2%)
Total number of Events	23982	10204
Total number of patients with at least one		
Treatment-related AE	1476 (95.7%)	696 (96.1%)
Atezo-related AE	1125 (72.9%)	547 (75.6%)
Grade 3-4 AE	1051 (68.1%)	476 (65.7%)
Treatment-related Grade 3-4 AE	905 (58.7%)	414 (57.2%)
Atezo-related Grade 3-4 AE	418 (27.1%)	207 (28.6%)
Grade 5 AE	63 (4.1%)	32 (4.4%)
Treatment-related Grade 5 AE	19 (1.2%)	11 (1.5%)
Atezo-related Grade 5 AE	13 (0.8%)	9 (1.2%)
Serious AE	651 (42.2%)	321 (44.3%)
Treatment-related serious AE	334 (21.6%)	188 (26.0%)
Atezo-related serious AE	211 (13.7%)	118 (16.3%)
AE leading to Treatment withdrawal	352 (22.8%)	181 (25.0%)
AE leading to Atezo withdrawal	173 (11.2%)	101 (14.0%)
AE leading to Treatment modification/interruption	1067 (69.2%)	496 (68.5%)

Investigator text for AEs encoded using MedDRA v21.0. Percentages are based on N in the column headings. Multiple occurrences of the same AE in one individual are counted only once except for "Total number of events" row in which multiple occurrences of the same AE are counted separately. All treatment emergent AEs are included.

Atezo=Atezolizumab. Atezo+Chemo Combo: GO29537 (IMPOWER130) Arm A + GO29437 (IMPOWER131) Arm A+B + GO29436 (IMPOWER150) Arm A+B + GO29438 (IMPOWER132) Arm A + GO30081 (IMPOWER133) Arm A. Clinical cut-off dates: GO29436:22JAN2018, GO29437:20APR2018, GO29438:22MAY2018, GO29537:15MAR2018, GO30081:24APR2018.

Post marketing experience

Since the international birth date (18 May 2016) through 17 May 2018, an estimated cumulative total of 20,783 patients have received atezolizumab from marketing experience (United States, n=18,470; European Union, n=987; Japan, n=181; rest of the world, n=1145). No new or unexpected safety findings were identified in the post-marketing setting for atezolizumab used as a monotherapy.

The combination regimen of atezolizumab with carboplatin and nab-paclitaxel administered in IMpower130 is not approved yet.

2.5.1. Discussion on clinical safety

Overall, Atezo+CnP caused more gastrointestinal (diarrhoea and vomiting), general disorders (pyrexia), respiratory disorders (dyspnoea and cough), nervous disorders (headache, dizziness), infections (urinary

tract infection and pneumonia), skin disorders (rash and pruritus), decrease in platelet and neutrophil counts, musculoskeletal disorders (pain in muscles and joints) and hypothyroidism.

There were clearly more grade ≥ 3 AEs clearly reported in the Atezo+CnP arm vs the comparator arm, 85.8% vs. 76.3%. Main differences were seen with regards to neutropenia, thrombocytopenia and anaemia.

There were also clearly more infections in the Atezo+CnP arm as a consequence of the more frequently observed neutropenia. There was almost a doubling in the rate of Grade ≥ 3 infections, 17.5% vs. 9.5%.

While dizziness was observed in 10.8% in the CnP arm and 16.1% in the Atezo+CnP (see Table 51) no syncope was observed in the CnP arm. However, 13 (2.7%) syncope were reported in the Atezo+CnP arm out of 26 (1.1%) in the Atezo+Chemo combo, in other words half of the reported syncope in the Atezo+Chemo combo originated from the IMpower130. Syncope and Dizziness were added as ADRs in section 4.8 of the SmPC.

The proportion of patients experiencing SAEs was higher in the Atezo+CnP arm (50.7%) compared with the CnP arm (37.9%). Lung infection, neutropenia, diarrhoea were reported with $\geq 2\%$ difference. Deaths as an AE (Table 64) were reported in 4 patients in the CnP arm (1.7%) and 1 patient (0.2%) in the Atezo+CnP arm.

Based on the safety data submitted, a causal relationship with atezolizumab in combination with chemotherapy is established for the following ADRs which have been included in section 4.8 of the SmPC: lung infection, lymphocyte count decreased, vomiting, dysgeusia, AST/ALT increased, back pain and asthenia.

Looking at the AEs leading to death, there were no clinically relevant differences between the two treatment arms.

Atezo in combination with chemotherapy seems to lead to an increase in AESI compared to the use of atezo as monotherapy, although the number of Grade 3-4 AESI was similar, when comparing the different arms including atezo. However the number of AESI leading to treatment modification/interruption or use of systemic use of corticosteroids was clearly higher in the atezo+chemo/CnP group compared to the monotherapy arms.

Looking at the individual AESIs, the main differences between atezo+CnP and the monotherapy arms, were seen with regards to rash, hypothyroidism, pneumonitis and hyperthyroidism. Important AESIs unique to patients in the Atezo+CnP arm included immune-related adrenal insufficiency, immune-related nephritis, immune-related meningoencephalitis, and immune-related hypophysitis. There were no cases of immune-related Guillain-Barre syndrome, immune-related myocarditis, or immune-related myasthenia gravis in either arm.

It is also noted that that there were 4 Grade 5 AESI (2 cases of hepatitis and 2 cases of pneumonitis).

With regards to laboratory parameters, clinically relevant shifts in ALT, creatinine, magnesium and potassium are observed in the atezo+CnP. The majority resolved. Four patients met the criteria for Hy's Law. Two patients had immune related hepatitis. Their liver function returned to normal. One patient had a confirmed tumoural obstruction of the biliary tract. The last patient was recovering from the hepatitis event, but died due to PD.

Overall, there were slightly more AEs leading to withdrawal in the Atezo+CnP arm compared to the CnP arm, 26.4% vs. 22.0%. The main difference was in the "blood and lymphatic system disorder" SOC. In general, the withdrawal rate is considered high, however, having in mind the clinical setting, it is acceptable.

The majority of patients, 80.2% and 85.0% in the CnP and atezo+CnP arms respectively experienced an AE leading to a dose modification/interruption. There is no clear pattern, but merely a slightly higher number of AEs in all SOCs. Nonetheless, it seems that the majority of patients were kept on treatment with dose modifications/interruptions.

From the safety database all the adverse reactions reported in clinical trials have been included in the Summary of Product Characteristics. No changes to the RMP are needed as a result of the new safety data submitted as part of the application.

2.5.2. Conclusions on clinical safety

Overall, the safety profile of Atezo in combination with CnP is considered acceptable. However, there are more AEs and AESI in the Atezo+CnP arm, but no new safety concerns or major concerns were identified.

The current RMP is adequate to manage the risks associated with Tecentriq in this new indication.

2.5.3. PSUR cycle

The requirements for submission of periodic safety update reports for this medicinal product are set out in the list of Union reference dates (EURD list) provided for under Article 107c(7) of Directive 2001/83/EC and any subsequent updates published on the European medicines web-portal.

2.6. Risk management plan

The CHMP received the following PRAC Advice on the submitted Risk Management Plan:

The PRAC considered that the risk management plan version 9.1 is acceptable.

The CHMP endorsed the Risk Management Plan version 9.1 with the following content:

Safety concerns

Summary of safety concerns	
Important identified risks	Immune-related hepatitis Immune-related pneumonitis Immune-related colitis Immune-related pancreatitis Immune-related endocrinopathies (diabetes mellitus, hypothyroidism, hyperthyroidism, adrenal insufficiency and hypophysitis) Immune-related neuropathies (Guillain-Barré syndrome, and myasthenic syndrome / myasthenia gravis) Immune-related meningoencephalitis Infusion-related reactions Immune-related myocarditis Immune-related nephritis Immune-related myositis
Important potential risks	Anti-drug antibodies Embryo-fetal toxicity
Missing information	Concomitant use with other immuno-modulatory drugs Long term use Concomitant or sequential use of atezolizumab with intra-vesical Bacillus Calmette-Guérin vaccine for the treatment of urothelial carcinoma

No changes to the list of safety concerns were made as a result of this extension of indication.

Pharmacovigilance plan

Study Status	Summary of Objectives	Safety concerns addressed	Milestones	Due dates
Category 1 - Imposed mandatory additional pharmacovigilance activities which are conditions of the marketing authorization				
There are no Imposed mandatory additional pharmacovigilance activities which are conditions of the marketing authorization				
Category 2 – Imposed mandatory additional pharmacovigilance activities which are Specific Obligations in the context of a conditional marketing authorization or a marketing authorization under exceptional circumstances				
There are no Imposed mandatory additional pharmacovigilance activities which are Specific Obligations in the context of a conditional marketing authorization or a marketing authorization under exceptional circumstances				
Category 3 - Required additional pharmacovigilance activities				
GO28915 (OAK) A Phase III, Open-Label, Multicenter, Randomized Study to Investigate the Efficacy and Safety of Atezolizumab (Anti-PD-L1 Antibody) Compared with Docetaxel in Patients with Non–Small Cell Lung Cancer After Failure with Platinum-Containing Chemotherapy Ongoing	To determine if atezolizumab treatment results in an improved OS compared with docetaxel To evaluate safety and tolerability of atezolizumab compared with docetaxel To evaluate incidence of ADAs against atezolizumab and to explore the potential relationship of the immunogenicity response with pharmacokinetics, safety, and efficacy	Anti-drug antibodies	Final CSR	December 2019
GO29322: A Phase IB Study of the Safety and Pharmacology of atezolizumab Administered with Ipilimumab or Interferon-Alpha in Patients with Locally Advanced or Metastatic Solid Tumors Ongoing	To evaluate the safety and tolerability of atezolizumab and ipilimumab in combination in patients with advanced or metastatic NSCLC or melanoma. To evaluate the safety and tolerability of atezolizumab and interferon alfa-2b in combination in patients with advanced or metastatic RCC or melanoma	Concomitant use with other immunomodulatory drugs	Final CSR	March 2020
WO29635: A Phase IB/II, Open-Label Study of the Safety and Pharmacology of Atezolizumab Administered with or without Bacille Calmette-Guérin in Patients with High Risk Non Muscle-Invasive Bladder Cancer	To evaluate the safety and tolerability of atezolizumab as a single agent and in combination with BCG. To identify the DLTs and to determine the MTD or tolerability at the MAD of BCG in combination with atezolizumab	Concomitant or sequential use of atezolizumab with intra-vesical BCG vaccine for the treatment of urothelial carcinoma	Final CSR	June 2022

Ongoing				
MO39171 (TAIL): Single-Arm Long-Term Safety and Efficacy Study of atezolizumab in previously treated NSCLC Patients Ongoing	To evaluate the long-term safety of atezolizumab on the bases of the following endpoints: The incidence of all serious adverse events (SAEs) related to atezolizumab treatment and the incidence of immune-related adverse events (irAEs) related to atezolizumab treatment	Long-term use	Final CSR	May 2022
MO29983: An Open-Label, Single Arm, Multicenter, Safety Study of atezolizumab in Locally Advanced or Metastatic Urothelial or Non-Urothelial Carcinoma of the Urinary Tract Ongoing	To evaluate the safety of atezolizumab based on the following endpoints: Nature, severity, duration, frequency and timing of adverse events (AEs) and changes in vital signs, physical findings, and clinical laboratory results during and following atezolizumab administration.	Long-term use	Final CSR	Q1 2023
WO40486 (Observational Study) Evaluation of the effectiveness of HCP educational materials which aims to facilitate early recognition and intervention of the following important immune-related risks: Pneumonitis, hepatitis, colitis, pancreatitis, endocrinopathies, neuropathies, meningoencephalitis, myocarditis, nephritis, and infusion-related reactions Ongoing	The overall objective is to evaluate the effectiveness of the HCP brochure designed to mitigate important immune-related risks in patients receiving atezolizumab in the European Union. Data from HCP surveys and reporting rates for the important identified immune related risks will be collected and analyzed to evaluate effectiveness of the HCP brochure	Immune-related hepatitis Immune-related pneumonitis Immune-related colitis Immune-related pancreatitis Immune-related endocrinopathies (diabetes mellitus, hypothyroidism, hyperthyroidism, adrenal insufficiency, and hypophysitis) Immune-related neuropathies (Guillain-Barré syndrome, and myasthenic syndrome / myasthenia gravis) Immune related meningoencephalitis Infusion-related reactions Immune-related myocarditis Immune-related nephritis	Protocol submission Interim report Final Report	February 2018 December 2020 December 2022

No new studies were added to the pharmacovigilance plan as a result of this extension of indication.

Risk minimisation measures

Safety concern	Risk minimization measures	Pharmacovigilance activities
Immune-Related Hepatitis	Routine risk minimization measures: Proposed measures are described in the E.U. SmPC under the following sections: Section 4.2 Posology and method of administration Section 4.4 Special Warnings and Precautions for Use Section 4.8 Undesirable effects Additional risk minimization measures: • Educational materials for HCPs • Patient alert cards	Routine pharmacovigilance activities beyond adverse reactions reporting and signal detection: None Additional pharmacovigilance activities: WO40486 (Observational Study) Evaluation of the effectiveness of HCP educational materials which aims to facilitate early recognition and intervention of the following important immune-related risks: Pneumonitis, hepatitis, colitis, pancreatitis, endocrinopathies, neuropathies, meningoencephalitis, myocarditis, nephritis, and infusion-related reactions.
Immune-Related Pneumonitis	Routine risk minimization measures: Proposed measures are described in the E.U. SmPC under the following sections: Section 4.2 Posology and method of administration Section 4.4 Special Warnings and Precautions for Use Section 4.8 Undesirable effects Additional risk minimization measures: • Educational materials for HCPs • Patient alert cards	Routine pharmacovigilance activities beyond adverse reactions reporting and signal detection: None Additional pharmacovigilance activities: WO40486 (Observational Study) Evaluation of the effectiveness of HCP educational materials which aims to facilitate early recognition and intervention of the following important immune-related risks: Pneumonitis, hepatitis, colitis, pancreatitis, endocrinopathies, neuropathies, meningoencephalitis, myocarditis, nephritis, and infusion-related reactions.

Safety concern	Risk minimization measures	Pharmacovigilance activities
Immune-Related Colitis	Routine risk minimization measures: Proposed measures are described in the E.U. SmPC under the following sections: Section 4.2 Posology and method of administration Section 4.4 Special Warnings and Precautions for Use Section 4.8 Undesirable effects Additional risk minimization measures: • Educational materials for HCPs • Patient alert cards	Routine pharmacovigilance activities beyond adverse reactions reporting and signal detection: None Additional pharmacovigilance activities: WO40486 (Observational Study) Evaluation of the effectiveness of HCP educational materials which aims to facilitate early recognition and intervention of the following important immune-related risks: Pneumonitis, hepatitis, colitis, pancreatitis, endocrinopathies, neuropathies, meningoencephalitis, myocarditis, nephritis, and infusion-related reactions.
Immune-Related Pancreatitis	Routine risk minimization measures: Proposed measures are described in the E.U. SmPC under the following sections: Section 4.2 Posology and method of administration Section 4.4 Special Warnings and Precautions for Use Section 4.8 Undesirable effects Additional risk minimization measures: • Educational materials for HCPs • Patient alert cards	Routine pharmacovigilance activities beyond adverse reactions reporting and signal detection: None Additional pharmacovigilance activities: WO40486 (Observational Study) Evaluation of the effectiveness of HCP educational materials which aims to facilitate early recognition and intervention of the following important immune-related risks: Pneumonitis, hepatitis, colitis, pancreatitis, endocrinopathies, neuropathies, meningoencephalitis, myocarditis, nephritis, and infusion-related reactions.

Safety concern	Risk minimization measures	Pharmacovigilance activities
Immune-Related Endocrinopathies (Diabetes Mellitus, Hypothyroidism, Hyperthyroidism, Adrenal Insufficiency, and Hypophysitis)	Routine risk minimization measures: Proposed measures are described in the E.U. SmPC under the following sections: Section 4.2 Posology and method of administration Section 4.4 Special Warnings and Precautions for Use Section 4.8 Undesirable effects Additional risk minimization measures: • Educational materials for HCPs • Patient alert cards	Routine pharmacovigilance activities beyond adverse reactions reporting and signal detection: None Additional pharmacovigilance activities: WO40486 (Observational Study) Evaluation of the effectiveness of HCP educational materials which aims to facilitate early recognition and intervention of the following important immune-related risks: Pneumonitis, hepatitis, colitis, pancreatitis, endocrinopathies, neuropathies, meningoencephalitis, myocarditis, nephritis and infusion-related reactions.
Immune-Related Neuropathies (Guillain-Barre Syndrome and Myasthenia Gravis)	Routine risk minimization measures: Proposed measures are described in the E.U. SmPC under the following sections: Section 4.2 Posology and method of administration Section 4.4 Special Warnings and Precautions for Use Section 4.8 Undesirable effects Additional risk minimization measures: • Educational materials for HCPs • Patient alert cards	Routine pharmacovigilance activities beyond adverse reactions reporting and signal detection: None Additional pharmacovigilance activities: WO40486 (Observational Study) Evaluation of the effectiveness of HCP educational materials which aims to facilitate early recognition and intervention of the following important immune-related risks: Pneumonitis, hepatitis, colitis, pancreatitis, endocrinopathies, neuropathies, meningoencephalitis, myocarditis, nephritis, and infusion-related reactions.

Safety concern	Risk minimization measures	Pharmacovigilance activities
Immune-Related Meningoencephalitis	Routine risk minimization measures: Proposed measures are described in the E.U. SmPC under the following sections: Section 4.2 Posology and method of administration Section 4.4 Special Warnings and Precautions for Use Section 4.8 Undesirable effects Additional risk minimization measures: • Educational materials for HCPs • Patient alert cards	Routine pharmacovigilance activities beyond adverse reactions reporting and signal detection: None Additional pharmacovigilance activities: WO40486 (Observational Study) Evaluation of the effectiveness of HCP educational materials which aims to facilitate early recognition and intervention of the following important immune-related risks: Pneumonitis, hepatitis, colitis, pancreatitis, endocrinopathies, neuropathies, meningoencephalitis, myocarditis, nephritis and infusion-related reactions.
Infusion-Related Reactions	Routine risk minimization measures: Proposed measures are described in the E.U. SmPC under the following sections: Section 4.2 Posology and method of administration Section 4.4 Special Warnings and Precautions for Use Section 4.8 Undesirable effects Additional risk minimization measures: • Educational materials for HCPs • Patient alert cards	Routine pharmacovigilance activities beyond adverse reactions reporting and signal detection: None Additional pharmacovigilance activities: WO40486 (Observational Study) Evaluation of the effectiveness of HCP educational materials which aims to facilitate early recognition and intervention of the following important immune-related risks: Pneumonitis, hepatitis, colitis, pancreatitis, endocrinopathies, neuropathies, meningoencephalitis, myocarditis, nephritis, and infusion-related reactions.

Safety concern	Risk minimization measures	Pharmacovigilance activities
Immune-Related Myocarditis	Routine risk minimization measures: Proposed measures are described in the E.U. SmPC under the following sections: Section 4.2 Posology and method of administration Section 4.4 Special Warnings and Precautions for Use Section 4.8 Undesirable effects Additional risk minimization measures: • Educational materials for HCPs • Patient alert cards	Routine pharmacovigilance activities beyond adverse reactions reporting and signal detection: None Additional pharmacovigilance activities: WO40486 (Observational Study) Evaluation of the effectiveness of HCP educational materials which aims to facilitate early recognition of and intervention in the following important immune-related risks: Pneumonitis, hepatitis, colitis, pancreatitis, endocrinopathies, neuropathies, meningoencephalitis, myocarditis, nephritis, and infusion-related reactions.
Immune-related nephritis	Routine risk minimization measures: Proposed measures are described in the E.U. SmPC under the following sections: Section 4.2 Posology and method of administration Section 4.4 Special Warnings and Precautions for Use Section 4.8 –Undesirable effects Additional risk minimization measures: • Educational materials for HCPs • Patient alert cards	Routine pharmacovigilance activities beyond adverse reactions reporting and signal detection: None Additional pharmacovigilance activities: WO40486 (Observational Study) Evaluation of the effectiveness of HCP educational materials which aims to facilitate early recognition of and intervention in the following important immune-related risks: Pneumonitis, hepatitis, colitis, pancreatitis, endocrinopathies, neuropathies, meningoencephalitis, myocarditis, nephritis, and infusion-related reactions.

Safety concern	Risk minimization measures	Pharmacovigilance activities
Immune-related myositis	Routine risk minimization measures: Proposed measures are described in the E.U. SmPC under the following sections: Section 4.2 Posology and method of administration Section 4.4 Special Warnings and Precautions for Use Section 4.8 Undesirable effects Additional risk minimization measures: • Educational materials for HCPs • Patient alert cards	Routine pharmacovigilance activities beyond adverse reactions reporting and signal detection: None Additional pharmacovigilance activities: None
Anti-drug Antibodies	Routine risk minimization measures: Proposed measures are described in the E.U. SmPC under the following sections: Section 4.8 Undesirable effects No additional risk minimization measures	Routine pharmacovigilance activities beyond adverse reactions reporting and signal detection: None Additional pharmacovigilance activities: Study GO28915 (OAK)
Embryo-fetal toxicity	Routine risk minimization measures: Proposed measures are described in the E.U. SmPC under the following sections: Section 4.6 Fertility, pregnancy and lactation Section 5.3 Preclinical safety data No additional risk minimization measures	Routine pharmacovigilance activities beyond adverse reactions reporting and signal detection: None Additional pharmacovigilance activities: None

Safety concern	Risk minimization measures	Pharmacovigilance activities
Concomitant use with other immuno-modulatory agents	Routine risk minimization measures: This safety concern considered as missing information is mentioned as one of the exclusion criteria within the Warnings and Precautions and description of studies included in the E.U. SmPC. No Additional risk minimization measures	Routine pharmacovigilance activities beyond adverse reactions reporting and signal detection: None Additional pharmacovigilance activities: Study GO29322
Long-term use	Routine risk minimization measures: Proposed text in E.U. SmPC: None No Additional risk minimization measures	Routine pharmacovigilance activities beyond adverse reactions reporting and signal detection: None Additional pharmacovigilance activities: Studies: • MO29983 • MO39171
Concomitant or sequential use of atezolizumab with intra-vesical Bacillus Calmette-Guérin vaccine for the treatment of urothelial carcinoma.	Routine risk minimization measures: Proposed measures are described in the E.U. SmPC under the following sections: Section 4.4 Special Warnings and Precautions for Use: Includes language that patients who were administered a live attenuated vaccine with 28 days prior to enrolment were excluded from clinical trials No Additional risk minimization measures	Routine pharmacovigilance activities beyond adverse reactions reporting and signal detection: None Additional pharmacovigilance activities: Study WO29635

The risk minimisations measures remain unchanged as a result of this extension of indication.

2.7. Update of the Product information

As a consequence of this new indication, sections 4.1, 4.2, 4.4, 4.8 and 5.1 of the SmPC have been updated. The Package Leaflet has been updated accordingly.

The new indication has only been reflected in the SmPC for the 1,200 mg strength, however the safety sections have been aligned between the 840 mg and 1,200 mg strengths.

2.7.1. User consultation

A justification for not performing a full user consultation with target patient groups on the package leaflet has been submitted by the MAH and has been found acceptable for the following reasons:

- No significant changes impacting the readability of the package leaflet are made. The new additions follow the same structure and use similar descriptions and terminology as used in the approved package leaflet.
- The target group of users will be similar between the approved indication (locally advanced or metastatic NSCLC previously treated with chemotherapy) and the applied indication (first-line treatment of metastatic non-squamous NSCLC), with no significant age difference.
- The posology proposed in this application is the same as for the currently approved indication.

3. Benefit-Risk Balance

3.1. Therapeutic Context

3.1.1. Disease or condition

The proposed new therapeutic indication is for atezolizumab in combination with nab paclitaxel and carboplatin, for the first line treatment of adult patients with metastatic non-squamous NSCLC who do not have EGFR mutant or ALK positive NSCLC.

Lung cancer remains the leading cause of cancer death worldwide, estimated to be responsible for nearly one in five cancer deaths globally (1.76 million deaths, 18.4% of the total; Globocan 2018). There were estimated to be 2.1 million new cases in 2018 (11.6% of the total), 58% of which occurred in less developed regions. This disease is the most common cancer in men worldwide (1.4 million [15% of cancers in men]) and accounts for the highest absolute number of cancer deaths globally (1.2 million deaths [24% of cancer deaths in men]; Fitzmaurice et al. 2015). NSCLC is the predominant subtype, accounting for approximately 85% of all cases (Molina et al. 2008; Howlader et al. 2014).

NSCLC can be divided into two major histologic types: non-squamous and squamous cell carcinoma (Travis et al. 2011). Non-squamous histology accounts for more than half of all NSCLC, whereas squamous histology accounts for approximately 30% (Langer et al. 2010).

Genetic changes that have prognostic and/or predictive significance in NSCLC include EGFR mutations in exons 18 through 21 and rearrangements in the ALK gene. EGFR mutations are rare in squamous NSCLC but are seen in 10%-40% of patients with non-squamous NSCLC (Herbst et al. 2008). ALK rearrangements have been reported in 2%-6% of NSCLC patients, predominantly in non-squamous NSCLC (Kwak et al. 2010; Crinò et al. 2011).

3.1.2. Available therapies and unmet medical need

For patients who have metastatic non-squamous NSCLC without an oncogenic driver mutation (such as *EGFR* mutations or *ALK* translocations), the standard-of-care options in first-line are anti-PD-1 monotherapy (only for high PD-L1 expression) or platinum-based chemotherapy in combination with anti-PD-1/PD-L1 agents.

3.1.3. Main clinical studies

The MAH has provided results from the IMpower 130 study, which is a relatively large (> 700 patients) phase III, open label, randomised study conducted to evaluate the efficacy and safety of atezolizumab in combination with nab paclitaxel and carboplatin, in chemotherapy naïve patients with metastatic non-squamous NSCLC.

3.2. Favourable effects

The primary analysis was conducted in all patients, excluding those with *EGFR* mutations or *ALK* rearrangements, defined as ITT-WT population (n=679). Patients had a median survival follow up time of 18.6 months. The study met one of its co-primary endpoints, PFS in ITT-WT. Results showed a HR of 0.64 (95%CI: 0.54, 0.77; p<0.0001). The median duration of PFS was 7.0 months in the atezo+CnP arm. With regards to OS in ITT-WT, median OS was 18.6 months in the atezo+CnP vs 13.9 months in the comparator arm (HR=0.79 (95%CI 0.64, 0.98).

Updated OS data with median follow-up of 24.1 months showed a HR = 0.82 (95% CI: 0.67, 1.01) and the p-value = 0.057. As OS data has matured, the difference between the two treatment arms is no longer statistically significant, but the study is considered positive as at least one of the co-primary endpoints is statistically significant. The updated results are clearly reflected in section 5.1.

Concerning the secondary endpoints, results were overall in line with the primary analyses in the ITT-WT population.

Finally, the secondary analyses of OS and PFS in the ITT population show similar results to the ITT-WT population. Breaking down the treatment effect per PD-L1 expression, it is clearly seen that the majority of patients are TC0/IC0, however, a similar treatment effect is observed across all the different PD-L1 subgroups. These results are in line with the primary estimate.

3.3. Uncertainties and limitations about favourable effects

There are no uncertainties regarding the favourable effects of the Atezo+CnP combination.

3.4. Unfavourable effects

Overall incidences of treatment-related AEs are numerically higher in the Atezo+CnP arm (96.2 % vs 92.7% in the CnP arm).

Incidence of Grade 3-4 adverse events was higher in the Atezo+CnP arm, especially grade 4 AEs (27.7% vs 15.9%) were reported numerically higher in the Atezo+CnP (85.0%) than in the CnP arm (74.6%).

The proportion of patients experiencing SAEs was higher in the Atezo+CnP arm (50.7%) compared with the CnP arm (37.9%).

Lung infection, neutropenia, diarrhoea were reported with $\geq 2\%$ difference.

Deaths due to AEs were reported in 4 patients in the CnP arm (1.7%) and 1 patient (0.2%) in the Atezo+CnP arm.

3.5. Uncertainties and limitations about unfavourable effects

There are no uncertainties or limitations about the unfavourable effects.

3.6. Effects Table

Table 85: Effects Table for atezolizumab in combination with nab paclitaxel and carboplatin, in 1L metastatic non-squamous NSCLC (data cut-off: 15 March 2018)

Effect	Short description	Unit	Treatment	Control	Uncertainties / Strength of evidence	References
Favourable Effects						
PFS ITT-WT	Progression free survival	mont hs	7.0	5.5	HR=0.64 (95%CI: 0.54, 0.77; p<0.0001)	
OS ITT-WT	Overall survival	mont hs	18.6	13.9	HR=0.79 (95%CI: 0.64, 0.98) , p = 0.033	
Unfavourable Effects						
Grade 3/4 infections and infestations		%	17.5	9.5		
Grade 3/4 neutropenia		%	32.3	28.4		
AESI		%	45	25.0		
Grade 3-5 AEs		%	80.5	70.7		
Discontinuation		%	26.4	22.0		
SAE		%	50.7	37.9		

Abbreviations: PFS: progression free survival; OS: overall survival; AESI: adverse events of special interest; AEs: Adverse events; SAEs: serious adverse events. HR: hazard ratio

3.7. Benefit-risk assessment and discussion

3.7.1. Importance of favourable and unfavourable effects

In the intended indication, there is a need for further treatment options, especially for patients with PD-L1 negative tumours or with low PD-L1 expression. Study IMpower130 shows clinically meaningful results in both PD-L1 positive and PD-L1 negative patients. In the ITT-WT population the median OS was 18.6 months vs 13.9 months in the atezo+CnP and CnP arms respectively, an improvement of median OS by 4.7 months, which is considered clinically encouraging in this patient population with a dismal prognosis. Updated OS results no longer show a statistically significant result. PFS results in the ITT-WT population, showed a HR of 0.64 (95% CI: 0.54, 0.77; p<0.0001). The median duration of PFS was 7.0 months in the atezo+CnP arm. Overall, study IMpower130 is considered positive, since it met both of its co-primary endpoints in the ITT-WT population at the primary analysis.

There are more AEs, Grade 3/4 AEs, AESI and discontinuation in the atezo+CnP arm, but overall the safety profile of Atezo in combination with CnP is considered acceptable. The majority of AEs are clinically manageable with supportive therapy and/or dose reduction/interruptions.

The addition of atezolizumab to CnP is in the interest of the patient as shown by the difference in OS and PFS compared to the control arm.

3.7.2. Balance of benefits and risks

The B/R balance of Tecentriq in combination with nab paclitaxel and carboplatin, for the first line treatment of adult patients with metastatic non-squamous NSCLC who do not have EGFR mutant or ALK positive NSCLC is positive.

3.7.3. Additional considerations on the benefit-risk balance

N/A

3.8. Conclusions

The overall B/R of Tecentriq is positive.

4. Recommendations

Outcome

Based on the review of the submitted data, the CHMP considers the following variation acceptable and therefore recommends by consensus the variation to the terms of the Marketing Authorisation, concerning the following change:

Variation accepted		Type	Annexes affected
C.I.6.a	C.I.6.a - Change(s) to therapeutic indication(s) - Addition of a new therapeutic indication or modification of an approved one	Type II	I and IIIB

Extension of Indication to include Tecentriq, in combination with nab-paclitaxel and carboplatin, indicated for the first-line treatment of adult patients with metastatic non-squamous non-small cell lung cancer (NSCLC) who do not have EGFR mutant or ALK-positive NSCLC; as a consequence, sections 4.1, 4.2, 4.8 and 5.1 of the SmPC are updated. The Package Leaflet is updated in accordance. RMP version 9.1 has been accepted.

The variation leads to amendments to the Summary of Product Characteristics and Package Leaflet and to the Risk Management Plan (RMP).

5. EPAR changes

The EPAR will be updated following Commission Decision for this variation. In particular the EPAR module "*steps after the authorisation*" will be updated as follows:

Scope

Extension of Indication to include Tecentriq, in combination with nab-paclitaxel and carboplatin, indicated for the first-line treatment of adult patients with metastatic non-squamous non-small cell lung cancer (NSCLC) who do not have EGFR mutant or ALK-positive NSCLC; as a consequence, sections 4.1, 4.2, 4.8 and 5.1 of the SmPC are updated. The Package Leaflet is updated in accordance. RMP version 9.1 has been accepted.

Summary

Please refer to the Scientific Discussion Tecentriq-H-C-4143-II-19.